

International Climate News

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Abstract

We develop novel high-frequency indices that measure climate attention across a wide range of developed and emerging economies. By analyzing the text of over 23 million Tweets published by leading national newspapers, we find that a country experiencing more severe climate news shocks tends to see both an inflow of capital and an appreciation of its currency. In addition, brown stocks experience large and persistent negative returns after a global climate news shock if located in highly exposed countries. A risk-sharing model in which investors price climate news shocks and trade consumption and investment goods in global markets rationalizes these findings.

First Draft: June 28, 2023. This Draft: September 18, 2026.

Keywords: International Climate Indices, Text Analysis, Trade, Currencies.

JEL Classification: F3, F4, G1.

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1 Introduction

In recent years, there has been a growing focus on the interaction between climate dynamics and economic activities. Predicted changes in climate patterns are expected to result in a significant redistribution of economic operations across various locations, sectors, and corporations. At the same time, the valuation of stocks and bonds is increasingly reflecting climate risk factors. However, there has been limited attention given to the relationship between climate-related news shocks and international economic variables, such as currencies and global capital flows.

In this paper, we develop a new international general equilibrium model with climate news shocks to generate testable predictions for asset prices, exchange rates, and international trade. To empirically test the model, we construct a new index that captures high-frequency climate news shocks based on the content of Twitter feeds from major newspapers in both advanced and emerging economies. By adopting a high-frequency approach, we mitigate the challenges of studying the long-run effects of climate shocks in short samples and offer new insights for asset pricing.

We document three main empirical findings. First, countries with greater exposure to climate attention shocks experience a persistent currency appreciation. Second, brown firms located in highly exposed countries tend to experience large negative and persistent stock returns after adverse climate news shocks. To the best of our knowledge, we are the first to document no significant effect on high-emission firms located in countries with low exposure to climate news shocks. Third, international current accounts respond to climate news shocks in a manner consistent with a recursive risk-sharing mechanism. Specifically, countries that are highly sensitive to climate news shocks hedge through international financial markets and receive resources—reflected in a decline in net exports—when adverse news shocks materialize. This result highlights the significant impact of climate news on both international prices and quantities.

On the theory side, we consider a general equilibrium model in which: (i) investors price climate news shocks (Colacito et al. 2018); (ii) investment goods are used to increase either green or brown assets; (iii) there are both productivity and global climate news shocks; and (iv) countries have different climate news sensitivities.

To derive analytical expressions, we focus on a two-country, two-period model in which time-1 utility depends on both consumption and a component linked to global climate news shocks, mediated by the sensitivity parameter b . At time 2, production is endogenous and determined by investment decisions made at time 1. Time-2 utility is influenced by exogenous productivity shocks as well as disruptions caused by brown assets, which can be partially offset by green assets. Importantly, “disruption” is interpreted broadly to include both potential physical damages and transition risks. We assume complete markets, allowing the two countries to trade time-0 securities equities to hedge against both productivity and climate news shocks. We contribute to the climate finance literature by offering a general equilibrium analysis in which climate news is priced in a multi-country setting. While the model is intentionally stylized to highlight the core mechanism, it offers a valuable and tractable framework for understanding the role of these risks in an international context.

Specifically, our model predicts that a country more sensitive to a global climate news shock (i.e., with higher b) should experience both a currency appreciation and a decline in net exports when an adverse climate news shock materializes. A decline in net exports corresponds to an inflow of resources—consistent with the transfer prescribed by international risk-sharing. In addition, the model allows us to study the returns on both brown and green assets. An investment strategy that is long brown assets and short green assets delivers a negative excess return following an adverse global climate news shock. This loss is more pronounced in countries with greater sensitivity to climate news.

To test the predictions of our model, we apply textual analysis techniques to the full universe of tweets posted by major national newspapers across 25 countries between 2014 and 2022. We compute a cosine-similarity score with a climate-related reference corpus and use these scores to build a daily climate-attention measure for each country. These indices contain a common component, which we denote as the global climate attention index. Importantly, there is substantial heterogeneity in the country-level loadings on this global component. We denote these loadings as β s and

interpret them as the empirical counterpart to the sensitivity of country-level sentiment to global climate news shocks—that is, the model’s b coefficient. In both the data and the model, this dimension of cross-country heterogeneity is distinct from existing measures of climate vulnerability.

We identify days of high attention toward climate news and study the behavior of currencies after these days. We find that after negative climate-related news, the currencies of high- β countries appreciate. This result adds empirical support to the theoretical link between climate news shocks and currency values.

Additionally, in line with the empirical asset pricing literature, we estimate sensitivity coefficients of returns over various horizons and for more than 17,000 firms. We examine firms with varying emission intensities located in all the countries included in our study. Importantly, in our empirical investigation we use an interaction term comprising emissions and country-level exposure. The literature has already documented that climate attention shocks are followed by negative cumulative stock returns. We find that this effect is highly persistent and is concentrated among brown firms that are located in countries with high exposure. In contrast, brown firms in countries with low exposure exhibit no significant response in their equity returns.

Our climate attention indicators can be aggregated to the quarterly level, enabling us to combine them with lower-frequency international trade data. We find empirical support for the third prediction of our model—namely, that countries more exposed to climate news shocks receive resources through lower net exports.

In what follows, we provide a more detailed description of the novel contribution related to our climate news index. More specifically, as in [Engle et al. \(2020\)](#), we construct a climate attention index that measures the extent to which climate change is discussed in the news media. However, our method differs in that we focus on newspapers with a significant presence on Twitter and we span a large cross-section of countries. This approach allows us to compile a large dataset of over 23 million tweets, which we then aggregate at the country level across daily, weekly, monthly, and quarterly frequencies. We compare the aggregated text to a corpus of authoritative texts on climate change, similar to the method used by [Engle et al. \(2020\)](#). Our data coverage refers to a total of 25 countries that span a wide range of local languages, income levels, and geographical regions.

We construct climate news indices at the country level and aggregate them to create a global index by computing cross-country averages weighted equally, by GDP, and by Twitter volume (data available on our [website](#)). For each country in our dataset, we regress its country-specific climate index on the global index. The estimated slope, denoted as β , measures each country’s exposure to common or, equivalently, global news. Across countries, we find a substantial degree of variation in exposure. Importantly, our measure of sensitivity is weakly correlated with both the vulnerability score provided by the [Notre Dame Global Adaptation Initiative](#) and absolute latitude, implying that we are focusing on a distinct dimension of heterogeneity that is not directly related to vulnerability and that we interpret as climate-related attention.

Leveraging our granular dataset, we use high-frequency data to identify the impact of climate news shocks on asset prices. Specifically, we analyze the reaction of currencies and stocks in the days following a substantial increase in climate attention. To achieve this, we compute daily innovations in our global climate index and focus on the top 15% of these observations. This approach allows us to (i) reduce noise from non-climate-related news and (ii) better isolate days driven by “pure” climate news shocks. Such a methodology is commonly employed in the literature on identifying the effects of news related to policy announcements ([Leombroni et al. 2021](#)). Importantly, some days with substantial climate-related news might fall below the top 15% threshold due to the prominence of other topics in the news. Consequently, as is typical in this literature, we may have a downward bias, i.e., a reduced likelihood of detecting a significant effect of climate news on asset prices and currencies.

We further incorporate a sentiment analysis of climate-related tweets on days with high climate attention. Using the BERTopic algorithm, we cluster all tweets into topics and identify those closely related to climate. This subset is then divided into two groups based on whether the topic primarily pertains to climate-related physical or transition risk. Following this classification, each tweet on a given day is categorized as related to physical risk, transition risk, or non-climate topics. We apply multilingual sentiment classifiers to distinguish between positive and negative content. Our findings indicate that days with extremely high climate attention are predominantly characterized by negative climate-related news. The reaction of currencies and stock returns is mainly driven by negative news about transition risk.

Our analysis using daily-frequency data reveals that currencies of countries with

higher exposure to our global climate index tend to appreciate following climate news shocks, with this effect persisting over several days. Remarkably, this finding is observed not only in pairs of advanced economies but also in combinations involving at least one developing country. Moreover, the robustness of this finding is upheld regardless of (i) the method used for aggregating country-level climate indices into a single global index, and (ii) whether we use weekly or daily data.

Related literature. An important strand of recent literature employs textual analysis of earnings conference call transcripts to assess exposure to climate risk (see, for example, [Sautner et al. 2023a](#), [Hassan et al. 2019](#) and [Hassan et al. 2023](#)). We differ from this literature in that we focus on the public interest for climate, rather than on educated firm-level experts. Moreover, our access to social media outlets allows us to construct a high-frequency index for a large cross-section of countries. **Importantly, in our extensive Internet Appendix, we show that a ‘simple’ and free BERT model can perform very well when compared to current leading AI models.**

Our frictionless model builds on the tradition of international macro-finance models with recursive preferences and news shocks ([Colacito et al. 2018](#)). We present a unified framework to document that currencies, international capital flows, and investments in brown and green technologies are related to our climate-related news. We leave the study of settings with segmented markets (see, among others, [Sandulescu et al. 2020](#)) to future work.

A substantial body of literature has examined the behavior, predictability, and volatility of exchange rates ([Della Corte et al. 2009](#); [Della Corte et al. 2011](#); [Della Corte et al. 2016](#)). Our analysis contributes to this stream by focusing on the dynamic response of the foreign exchange market to news. In international asset pricing, extensive research has developed and analyzed models to understand asset prices, exchange rates, and macroeconomic interactions across countries ([Pavlova and Rigobon 2007, 2010, 2013](#); [Hassan 2013](#); [Hassan et al. 2015, 2016](#); [Zviadadze 2017](#)). We contribute to the broader understanding of international macroeconomic dynamics and asset valuation in response to global climate-related events.

Another line of research investigates the impact of political and economic risk on firms, asset markets, and international economic linkages ([Hassan et al. 2019, 2023](#)). This literature develops methodologies to quantify various risks and uncertainties

and assesses their economic and financial spillovers. Our study introduces a novel text-based measure of global attention to climate change as an indicator of perceived risk and uncertainty. Furthermore, we document the influence of our index on currency markets and international financial dynamics.

The literature on carry trade strategies has linked their risk premia to global risk factors (see, for example, [Lustig et al. 2011, 2014](#); [Mueller et al. 2017](#)). Our results suggest that climate-related global news is an important risk factor for both exchange rate movements and capital flows. [Jiang \(2024\)](#) and [Jiang et al. \(2024a\)](#) emphasize the importance of international demand for dollar-denominated safe assets for the global financial cycle. Our analysis complements their perspective by introducing climate news shocks as a distinct source of priced risk that affects exchange rates and capital flows when countries differ in their sensitivity to climate-related attention.

[Lee et al. \(2022\)](#) find heterogeneous impulse responses of monthly U.S. dollar (USD) real exchange rates of 76 countries to global temperature shocks. We differ from their work in several dimensions: (i) we have a broader cross-section of country pairs; (ii) our climate-related indices move with a broad set of climate-related news, not just realized temperature changes; and (iii) we link both currencies and international current accounts. [Javadi et al. \(2023\)](#) find no link between G10 currencies and the vulnerability of these countries to physical climate risk. Our broader indicator captures both physical and transition risk and suggests the presence of a relevant link at both the daily and the weekly frequency. Our paper is also complementary to [Hale \(2024\)](#), who investigates the effect of climate-related disasters on real exchange rates using a belief-formation framework with rigidities. In contrast to [Hale \(2024\)](#), we study high-frequency climate news shocks and their impact on international financial variables through the lens of a recursive risk-sharing model.

Our work is related to the evolving academic field of climate finance (see, among others, [Giglio et al. 2021a, b, 2023](#); [Starks 2023](#); [Hsu et al. 2023](#); [Bansal et al. 2019](#); [Pástor et al. 2021, 2022](#); [Colacito et al. 2019](#); [Acharya et al. 2022](#); [Bolton and Kacperczyk 2023, 2021](#); [Bolton et al. 2023](#); and [Kashyap et al. 2024](#)). Recent contributions in this area include [Kruttili et al. \(2025\)](#), who show that firm-level returns are sensitive to extreme weather uncertainty, and [Ivanov et al. \(2024\)](#), who study the effects of cap-and-trade regulation on credit supply. [Zhang \(2024\)](#) documents international variation in carbon return premia, while [Shi and Zhang \(2025\)](#) explore how oil price

fluctuations affect the greenium. On the financing side, [Albuquerque et al. \(2025\)](#) highlight the paradoxical effect of green lending on carbon emissions. Finally, [Jung et al. \(2023\)](#) develop measures of physical climate risk exposure using insurer data, illustrating the importance of sectoral sensitivity to climate shocks. The climate macro-finance literature has primarily focused on the link between climate risks, local assets and local macroeconomic dynamics. We differ from these studies for our attention to international quantities and currencies.

2 Climate News Shocks in General Equilibrium

We develop a two-period general equilibrium model to derive testable null hypotheses using our novel dataset. The model is designed to capture several key features observed in the data: (i) climate news shocks are priced, (ii) green and brown investments differ in their exposure to climate news shocks, and (iii) countries exhibit heterogeneous sensitivity to common (i.e., global) climate news shocks.

2.1 The International Integrated Assessment Model (I-IAM)

Environment and information timing. The main equilibrium conditions of our model are derived in [Appendix A](#). In what follows, we describe our economic environment and then derive the key predictions regarding stock returns, trade, and currencies. The model features two countries, home and foreign. Foreign variables are denoted by “*”. Due to the symmetry of our environment, many of our equations are presented for the home country, with the implicit understanding that they apply to the foreign country as well. Time is discrete and we assume that there are only two periods denoted by the following three dates $t = \{0, 1, 2\}$.

All shocks materialize at the end of the first period, i.e., at $t = 1$. Hence, at time $t = 0$ choices are made under uncertainty. At time $t = 2$, no additional shock materializes, or, equivalently, in the second period there is full resolution of uncertainty. Most importantly, all news received at time $t = 1$ affects the production frontier only at $t = 2$. Hence, from a time-1 perspective, these shocks represent pure news shocks.

Preferences. In each country, the representative household has the following recursive preferences:

$$u_t = (1 - \beta) \log \left(\tilde{C}_t \cdot \mathcal{G}_t \right) + \frac{\beta}{1 - \gamma} \log E_t [\exp \{ (1 - \gamma) u_{t+1} \}], \quad t = \{0, 1, 2\},$$

where $\mathcal{G}_t$ captures climate news shocks, and $\tilde{C}_t$ includes both a standard consumption bundle C_t and an endogenous disruption term D_t :

$$\tilde{C}_t = C_t \cdot D_t^{-1}.$$

The climate news process evolves as:

$$\mathcal{G}_t = e^{-b \cdot g_t},$$

where g_t is a zero-mean normally distributed global climate news shock. Since periods of heightened climate news coverage (positive g_t) typically reflect unfavorable information about climate risks, a positive realization of g_t reduces the agent's utility. The parameter b (and its foreign counterpart b^*) governs the exposure of each country to global climate news shocks. While this formulation accommodates any climate news shock, we interpret $\mathcal{G}_t$ as a climate news shock for two reasons: (i) agents in the economy can hedge this risk by investing in a green technology that mitigates future climate damage, and (ii) the economy is observationally equivalent to one in which $\mathcal{G}_t$ represents a news shock about climate disruptions occurring at date $t + 1$. Moreover, our model adopts a broad definition of climate news shocks, without distinguishing between shocks associated with physical risk and those linked to transition risk. In our empirical analysis, we leverage text analysis tools to separately identify these two channels.

When $\gamma = 1$, preferences reduce to expected-log utility. For $\gamma > 1$, households exhibit a preference for early resolution of uncertainty, which implies that their marginal utility adjusts immediately in response to new climate news. Specifically, agents care not only about the expected value of their future utility, but also about how their hedging decisions affect its conditional variance. To see this point, notice that when

continuation utilities are approximately log-normal, the following holds:

$$u_t \approx (1 - \beta) \log \left(\tilde{C}_t \cdot \mathcal{G}_t \right) + \beta E_t [u_{t+1}] + \beta \frac{(1 - \gamma)}{2} V_t [u_{t+1}].$$

This formulation shows that agents care not only about the expected value of future utility but also about its conditional volatility. From the perspective of time 0, countries with greater exposure to global climate news, i.e., higher b , face more uncertainty about their continuation utility. As a result, these agents have a stronger incentive to hedge against the realization of adverse climate news shocks at time 1. To hedge this risk, the more exposed country structures its international portfolio to receive higher returns in states where the climate news shock g is positive. When such a shock materializes, this country can import resources from abroad (resulting in negative net exports), and its currency appreciates.

In our setting, $\tilde{C}_0$ and g_0 are predetermined parameters that do not influence the allocation decisions at time $t = \{1, 2\}$. The same logic applies to D_1 and D_1^* , as they are independent of the economic choices made for time $t = 2$. Based on these considerations, we normalize $\tilde{C}_0$, D_1 , and D_1^* to one, and g_0 to zero. In addition, we normalize the foreign country's exposure to the global climate news shock to one ($b^* = 1$), so that a value of $b > 1$ indicates that the home country is more sensitive than the foreign country to the same climate news shock. By incorporating this process into the utility function, we capture agents' concerns about climate disruptions, while remaining agnostic as to whether those concerns stem from physical risks or transition risks.

Consumption bundles. We assume that each agent consumes a bundle composed of two consumption goods:

$$C_t = X_t^\lambda Y_t^{1-\lambda}, \quad \text{and} \quad C_t^* = X_t^{*(1-\lambda)} Y_t^{*\lambda},$$

where X_t is the home-produced good, and Y_t is the foreign-produced good. The parameter $\lambda > 1/2$ introduces home bias in consumption and ensures that the equilibrium exchange rate is not mechanically pinned to unity. The allocations at times 1 and 2 are determined endogenously through the recursive risk-sharing scheme described below.

Since the model features a finite horizon, we begin the analysis by solving the terminal period and proceed recursively backward.

Time-2 output and climate disruptions. At time $t = 2$, output in the home and foreign country is given by:

$$GDP_2 = e^\theta B, \quad \text{and} \quad GDP_2^* = e^{-\theta} B^*,$$

where $\theta \sim i.i.d.\mathcal{N}(0, \sigma_\theta^2)$ captures a country-specific productivity shock. A positive realization of θ implies that the home country is relatively more productive. The terms B and B^* represent the effective capital stock in the home and foreign country, respectively. These are Cobb-Douglas aggregates of domestic and foreign investment made at time 1:

$$B = I_{x,1}^{\lambda_I} I_{x,1}^{*1-\lambda_I}, \quad \text{and} \quad B^* = I_{y,1}^{1-\lambda_I} I_{y,1}^{*\lambda_I}.$$

We refer to these investments as “brown capital”, since they contribute to the severity of future climate disruptions. In contrast, investments in green technology, which help mitigate climate damage, are introduced later in the model.¹ Since the model ends at time 2, there is no further capital accumulation. Thus, all output at $t = 2$ is allocated to consumption goods:

$$GDP_2 = X_2 + X_2^*, \quad \text{and} \quad GDP_2^* = Y_2 + Y_2^*.$$

The adverse effect that economic activity has on climate conditions is measured by D_t . In our model, only D_2 and D_2^* are endogenous and depend on economic decisions made at time $t = 1$. Specifically, disruptions are modeled as increasing in *brown investments* (captured by GDP) and decreasing in *green investment*, i.e., investment in technologies that mitigate the climate impact of production. Let $I_{g,1}$ and $I_{g,1}^*$ denote green investments in the home and foreign countries, respectively. We assume that home and foreign “green capital” obtain from the following aggregators:

$$G = I_{g,1}^{\lambda_g} I_{g,1}^{*1-\lambda_g}, \quad \text{and} \quad G^* = I_{g,1}^{*\lambda_g} I_{g,1}^{1-\lambda_g}.$$

¹We let $\lambda_I \neq \lambda$ to allow for the possibility that consumption home bias differs from investments home bias. When $\lambda_I = \lambda$, the structure of the investment bundle mirrors that of the consumption bundle, as in [Backus et al. \(1994\)](#).

The disruption terms take the following form:

$$D_2 = \left(\frac{GDP_2^{\lambda_g} \cdot GDP_2^{*1-\lambda_g}}{G} \right)^a, \quad D_2^* = \left(\frac{GDP_2^{*\lambda_g} \cdot GDP_2^{1-\lambda_g}}{G^*} \right)^a,$$

where the parameter λ_g governs the relative importance of domestic versus foreign sources of disruption. When $\lambda_g = 0.5$, agents are equally concerned about domestic and foreign drivers of climate deterioration. When $\lambda_g > 0.5$, local disruptions lead to stronger deterioration of local utility. Disruptions should be interpreted broadly, as including both potential physical damages and distortions due to transition risk. The parameter a satisfies the restriction $0 < a < 1$, to capture the idea that investors care about climate damages, but place greater weight on the utility derived from consuming potentially polluting goods.

Time-1 constraints. At time $t = 1$, output is predetermined and normalized to 1 in each country for simplicity. The corresponding resource constraints reflect the possibility of international trade in both consumption and investment goods:

$$1 = X_1 + X_1^* + I_{x,1} + I_{y,1} + I_{g,1}, \quad 1 = Y_1 + Y_1^* + I_{y,1}^* + I_{x,1}^* + I_{g,1}^*.$$

From the home (foreign) country's perspective, $I_{x,1}$ ($I_{y,1}^*$) represents local investment, while $I_{y,1}$ ($I_{x,1}^*$) reflects investment abroad. Although capital stocks are country-specific, agents can trade both consumption and brown investment goods across borders without frictions in every state of the world, consistent with the setting in [Colacito et al. \(2018\)](#). Green investments are inherently local and cannot be traded internationally. For instance, an afforestation project undertaken in the home country cannot simply be relocated to a foreign country. Nonetheless, these investments have global implications: although undertaken locally, they enter symmetrically into both countries' disruption functions, D_2 and D_2^* , thus reflecting shared climate benefits.

2.2 Decentralized economy: household side

Financial market completeness. At time-0, both households have access to two country-level equity indices. We specify the payoff structure later; here we clarify

that the time-1 returns are state-contingent, i.e., they adjust with the news shocks. Households decide their portfolio positions at time-0 in order to hedge bad news. From time-1 onward, uncertainty is resolved and all assets pay the same risk-free return. As a result, at time-1 the cross-section of assets collapses to one safe asset and each household simply chooses how much to invest in it in order to support time-2 consumption. Because there are two shocks and two state-contingent claims, this structure is sufficient to span the relevant uncertainty and no additional securities are required.

In what follows, we also assume that both countries adopt both a local Emissions Trading System and a Carbon Border Adjustment Mechanism. We identify the implied optimal taxation that enables our decentralized economy to mimic the planner's equilibrium.

Household's problem. For the sake of focus, we state the problem of the household in the home country. The problem of the foreign household is similar and can be found in the Appendix. We start from the terminal period, taking wealth as given. The household solves the following problem,

$$\begin{aligned} \max_{X_2, Y_2} u_2 &\equiv \lambda \log X_2 + (1 - \lambda) \log Y_2 - \log D_2 \\ s.t. \quad X_2 + p_2 Y_2 &= W_2 \end{aligned}$$

by setting

$$X_2 = \lambda W_2, \quad Y_2 = (1 - \lambda) \frac{W_2}{p_2},$$

where the relative price clearing the time-2 goods market is $p_2 = \frac{1-\lambda}{\lambda} \frac{X_2}{Y_2}$.

At time 1, uncertainty is fully resolved and all assets pay the same risk-free return expressed in units of good X . Hence each household must simply decide how much to consume and how much to invest in a representative risk-free asset to support time-2 consumption. After dividing the objective by $1 - \beta$ and dropping terms independent

of the household's time-1 choices, the household solves:

$$\begin{aligned}
\max_{X_1, Y_1, Z_1} \quad & \log C_1 + \beta \log(C_2 D_2^{-1}) \\
= \quad & \lambda \log X_1 + (1 - \lambda) \log Y_1 + \beta \log W_2 + \text{const.}, \\
s.t. \quad & W_1 = X_1 + p_1 Y_1 + Z_1, \\
& W_2 = Z_1 R_{1|2}^f,
\end{aligned}$$

where Z_1 represents the amount of time-1 wealth saved to support consumption at time 2, and the second equation obtains after substituting the optimal consumption plan decided at time 2. At the optimum, the agent allocates the following constant shares of time-1 wealth to consumption and savings:

$$X_1 = \frac{\lambda}{1 + \beta} W_1, \quad Y_1 = \frac{1 - \lambda}{1 + \beta} \frac{W_1}{p_1}, \quad Z_1 = \frac{\beta}{1 + \beta} W_1.$$

Time-0 wealth is exogenous and normalized to 1 for both the home and foreign countries. The household invests a fraction Z_0^H in the local stock market, and $Z_0^F = 1 - Z_0^H$ in the foreign one. As a result, the home-country wealth at time 1 is:

$$W_1 = Z_0^H V_1^A + Z_0^F V_1^{A*}$$

where V_1^A (V_1^{A*}) is the time-1 cum-dividend aggregate stock value in the home (foreign) country. All values are {denominated in units of good X .

Since this decision is made before the realization of the news shocks, the portfolio allocation accounts for the fact that u_0 depends on the volatility of the continuation utility u_1 inherited via W_1 and p_1 . The latter term depends on the distribution of domestic and foreign wealth, W_1 and W_1^* , respectively. This dependence generates a fixed point problem that we solve in the Appendix. In what follows, we describe the production side of the economy and the implied returns.

2.3 Decentralized economy: firms' side

Regulation about emissions and offsets. In what follows, we adopt the following terminology. E_2 and E_2^* denote the emissions of the home and foreign brown firm at

time 2, respectively. Emissions are defined as follows:

$$E_2 = \left[GDP_2^{\lambda_g} GDP_2^{*1-\lambda_g} \right]^a \quad E_2^* = \left[GDP_2^{1-\lambda_g} GDP_2^{*\lambda_g} \right]^a.$$

The green firms produce carbon sequestration and offsets which are denoted as O_2 and O_2^* in the home and foreign countries, respectively. Offsets are computed as follows:

$$O_2 = \left[I_{g,1}^{\lambda_g} I_{g,1}^{*1-\lambda_g} \right]^a, \quad \text{and} \quad O_2^* = \left[I_{g,1}^{*\lambda_g} I_{g,1}^{1-\lambda_g} \right]^a.$$

Consistent with the specification of damages in Section 2.1, both emissions and offsets depend on activity in both countries: carbon is a global stock, so local climate outcomes reflect worldwide brown production and worldwide green investment, with λ_g governing the degree of home bias in their incidence. As a result, the disruption terms that affect the consumption bundle can be rewritten as:

$$D_2 = E_2/O_2, \quad D_2^* = E_2^*/O_2^*,$$

so that there is no disruption when countries are net-zero, i.e., when offsets and sequestration equal emissions, $E_2 = O_2$ ($E_2^* = O_2^*$), we have $D_2 = 1$ ($D_2^* = 1$).

We postulate that in each country the government taxes climate-related damage, as measured by $D_2 - 1$ in the home country ($D_2^* - 1$ in the foreign country). Each government sets the local tax on emissions (τ in the home country, and τ^* in the foreign one), or, equivalently, it sets the supply of emission allowances in the local ETS in order to target a specific price per ton of CO_{2e} .

Importantly, each government recognizes the relevance of offsets and sequestration and enables local offsets to count toward the assessment of the damage. Very much as in the EU ETS, we assume that international offsets are not allowed to count for the reduction of net emissions. Equivalently, taxation is applied to $D_2 - 1$. This restriction is an accounting one: foreign green investment still lowers domestic damage physically, through its presence in O_2 , but firms may not purchase foreign credits to reduce their own reported net emissions. The transfers described below compensate for this spillover instead. As in the EU, each country adopts a Carbon Border Adjustment Mechanism (CBAM), and taxes damages realized abroad. As a result, the home

country collects a total amount of taxes defined as

$$T := \tau (D_2 - 1) + p_2 \tau^* (D_2^* - 1),$$

where the last term is expressed in units of good X . We assume that all proceeds are given to the green firm as a lump-sum subsidy.

In addition, the international spillovers of green investment are rewarded by a system of international proportional transfers. The home green firm receives $p_2 p_o^{FH} I_{g,1}$ from abroad, and it rebates $p_o^{HF} I_{g,1}^*$ to the foreign green firm. We think of these transfers as consistent with the spirit of Article 6.4 of the Paris Agreement, which establishes a centralized mechanism through which private entities can trade emission reductions across borders. More precisely, these transfers are set among private firms, not via government-to-government transactions of the kind envisioned in Article 6.2. We derive p_o^{FH} and p_o^{HF} in what follows, as we describe the firms' problem in the home country. The derivations for the foreign firms can be found in the Appendix.

The brown firm. The brown firm must maximize the present value of its dividends:

$$V_1^B = \max_{I_{x,1}, I_{x,1}^*, O_2} \underbrace{(1 - I_{x,1} - p_1 I_{x,1}^*)}_{Div_1^B} + \frac{1}{R_{1|2}^f} \underbrace{[GDP_2 - p_o O_2 - T]}_{Div_2^B}, \quad (1)$$

where p_o denotes the price of offsets purchased from the green firm at time 2. Optimizing over O_2 yields the demand for offsets,

$$p_o = \tau \frac{E_2}{O_2^2},$$

implying that the offset value is anchored to the marginal reduction in the carbon tax. Optimal investment prescribes:

$$\lambda_I \frac{GDP_2}{I_{x,1}} \cdot \left(1 - a \lambda_g \frac{\tau E_2}{GDP_2 O_2} - a(1 - \lambda_g) \frac{\tau^* p_2 E_2^*}{GDP_2 O_2^*} \right) = R_{1|2}^f,$$

that is, the marginal product of investment net of taxation must equal the risk-free rate. A similar expression can be obtained for $I_{x,1}^*$. See the Appendix.

The green firm. The green firm maximizes the present value of its dividends:

$$V_1^G = \max_{I_{g,1}} \underbrace{-I_{g,1}}_{Div_1^G} + \frac{1}{R_{1|2}^f} \underbrace{[p_o O_2 - p_o^{HF} I_{g,1}^* + p_2 p_o^{FH} I_{g,1} + T]}_{Div_2^G}, \quad (2)$$

where $p_o O_2$ denotes the total sales of carbon credits in the home country; $p_2 p_o^{FH} I_{g,1} - p_o^{HF} I_{g,1}^*$ is the net international transfer that accounts for synergies across domestic and foreign green investments. The optimal green domestic investment condition,

$$1 = \frac{1}{R_{1|2}^f} \left(p_o \lambda_g a \frac{O_2}{I_{g,1}} + p_2 p_o^{FH} \right),$$

accounts for marginal benefits originating from both the domestic sale of offsets and the international compensation. The transfer from the home country to the foreign one is obtained from the optimal demand for foreign green investment:

$$p_o^{HF} = p_o (1 - \lambda_g) a \frac{O_2}{I_{g,1}^*},$$

and, by symmetry,

$$p_o^{FH} = p_o^* (1 - \lambda_g) a \frac{O_2^*}{I_{g,1}}.$$

Aggregate dividends and valuations. After having specified the dividends paid by each firm, we can define the aggregate dividends. At time 2, in the home country we have

$$Div_2^A := Div_2^G + Div_2^B = GDP_2 - p_o^{HF} I_{g,1}^* + p_2 p_o^{FH} I_{g,1}.$$

Carbon tax revenue and offset purchases are transfers between the domestic brown and green firms and therefore cancel in the aggregate, leaving output net of the international transfer. [At time 1, the cum-dividend valuation is

$$V_1^A = Div_1^A + \frac{1}{R_{1|2}^f} Div_2^A,$$

in which

$$Div_1^A = 1 - I_{g,1} - I_{x,1} - p_1 I_{x,1}^*.$$

Returns. Define time-1 gross returns as follows,

$$R_{0|1}^i = \frac{V_1^i}{V_0^i - Div_0^i}, \quad \forall i \in \{G, B, A\}, \quad (3)$$

where G , B , and A refer to green, brown, and aggregate returns, respectively. Because no shock is realized and no real-investment or consumption decision is made at $t = 0$, we assume $Div_0^i = 0$ and $V_0^i = E_0(V_1^i)$, $\forall i \in \{G, B, A\}$. We are interested in comparing the market value of assets before and after the realization of the shocks θ and g at time $t = 1$. The valuation V_0^i is predetermined in this economy, i.e., it can be taken as given. As a result, the return dynamics depend solely on V_1^i , i.e., the market value of cash flows from the perspective of time 1. Hence the return differential between brown and green capital is simply,

$$r_{0|1}^{BMG} := \log \left(\frac{R_{0|1}^B}{R_{0|1}^G} \right) = \log(V_1^B) - \log(V_1^G) + \text{const.},$$

and it reflects heterogeneous exposure to shocks across green and brown firms.

2.4 Optimal taxation and first-best

In order to replicate the first best, it is sufficient that governments arrange the taxation on net emissions as follows:

$$\tau = \frac{X_2 O_2}{\lambda E_2}, \quad \tau^* = \frac{Y_2^* O_2^*}{\lambda E_2^*}. \quad (4)$$

In this case, the marginal value of offsets becomes,

$$p_o = \frac{X_2}{\lambda O_2}, \quad p_2 p_o^* = \frac{1}{1 - \lambda} \frac{X_2^*}{O_2^*}, \quad (5)$$

as in the planner's problem. In the Appendix, we show that under this taxation scheme one can study the allocation via the planner's problem or the decentralized economy interchangeably.

Pareto weights, wealth, and portfolio weights. Let $\tilde{S}_1$ be the ratio of pseudo-Pareto weights after the news shocks are realized. In the Appendix, we show how to derive it using the methods developed in Colacito et al. (2018). Most importantly, we show that the time-1 allocation is increasing in $\tilde{S}_1$, that is, when the pseudo-Pareto weight of the home country increases relative to that of the foreign country, resources are shifted toward the home country. Define $\hat{\lambda} \equiv \lambda - a\lambda_g$. In the decentralized economy, we mimic the ratio of the pseudo-Pareto weights when time-1 wealth evolves as follows:

$$\begin{aligned} W_1 &= \frac{1 + \beta}{K(\tilde{S}_1)} \tilde{S}_1, \\ K(\tilde{S}_1) &:= K_0 + K_1 \tilde{S}_1 > 0, \\ K_0 &:= 1 - \lambda + \beta\lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} + \beta a(1 - \lambda_g) > 0, \\ K_1 &:= \lambda + \beta\lambda_I\hat{\lambda} + \beta(1 - \lambda_I)(1 - a - \hat{\lambda}) + \beta a\lambda_g > 0. \end{aligned} \tag{6}$$

In the Appendix, we show that time-1 wealth satisfies these conditions when the share of time-0 wealth allocated to the domestic stock market is:

$$Z_0^H = \frac{K_1 - \beta a(1 - \lambda_g)}{K_1 - K_0 - 2\beta a(1 - \lambda_g)} > 1. \tag{7}$$

This result implies that the home country holds a levered position in domestic equity, financed by shorting the foreign aggregate stock. In the next section, we clarify that when adverse global climate news shocks materialize and the home country is more sensitive to them, the foreign currency depreciates. Hence, by being short in a foreign currency-denominated stock, the home household experiences a wealth appreciation at time 1. This hedging strategy enables the home household to buy more goods and services from abroad at time 1 exactly in the states in which its marginal utility is higher.

2.5 Testable hypotheses

In what follows, we present three propositions that guide our empirical analysis. All proofs are provided in [Appendix A](#). The first two propositions concern the behavior of currencies and asset returns, and can be tested using high-frequency data. The

third proposition pertains to the dynamics of international current accounts, which we evaluate using quarterly data. In this section, we focus exclusively on the role of climate news shocks. However, our full model also incorporates an endogenous link between productivity news and climate disruption, consistent with standard integrated assessment frameworks. Detailed analytical derivations are provided in the Appendix.

We state our propositions in the most general way and we use results obtained under additional parametric restrictions in order to provide further intuitions. Without loss of generality, when presenting analytical formulas, we use the following assumption.

Assumption 1. *The following parametric restrictions hold: $a \in (0, 1)$, $\gamma > 1$, $\lambda = \lambda_I = \lambda_g \in (0.5, 1)$ and $\beta \in (0, 1)$.*

As anticipated in the previous section, the allocation at time $t = 1$ moves with an endogenous state variable called ratio of the pseudo-Pareto weights. At time $t = 1$, this variable is denoted as $\tilde{S}_1$. At time $t = 0$, this variable is normalized to 1, implying that the two countries have the same relative size before the realization of the shocks. In our simplified setting, the log-ratio of the Pareto weights at time $t = 1$ is approximately equal to:

$$\begin{aligned} \tilde{s}_1 = \bar{s} &+ (1 - \beta) \cdot (\gamma - 1) \cdot \left(\frac{b - 1}{1 + 2(\gamma - 1)\lambda_u^s} \right) \cdot g \\ &- \tilde{\beta} \cdot (\gamma - 1) \cdot \left(\frac{(1 - a)(2\lambda - 1)}{1 + 2(\gamma - 1)\lambda_u^s} \right) \cdot 2\theta, \end{aligned} \quad (8)$$

We define $\bar{s}$ as the deterministic expansion point of $\tilde{s}_1$. Under symmetry and the normalization $\tilde{S}_0 = 1$, this expansion point is $\bar{s} = 0$. The overbars in $\bar{\Delta e}$ and $\bar{r}_{0|1}^{BMG}$ likewise denote deterministic baseline constants. where $\tilde{\beta} = \beta(1 - \beta)$ and $\lambda_u^s > 0$ is a composite positive parameter.² If preferences are not time-additive ($\gamma > 1$), the log-ratio of the Pareto weights $\tilde{s}_1$ increases in response to an adverse climate news shock ($g > 0$) and it decreases in response to a positive productivity shock ($\theta > 0$). When the conditions in equation (4) apply, this is equivalent to an appreciation of the time-1 wealth of the home country.

²See Lemma 1 in the Appendix for detailed derivations of these expressions.

Proposition 1. *Let $\gamma > 1$, when $b > 1$ ($b < 1$), a global climate news shock $g > 0$ causes an appreciation of home (foreign) real exchange rate.*

In our model, the log-growth rate of the exchange rate at date $t = 1$ evolves as follows

$$\Delta e_1 = \overline{\Delta e} + \lambda_e^s \cdot \tilde{s}_1. \quad (9)$$

In our simplified calibration,

$$\lambda_e^s = (2\lambda - 1)^2 \cdot \left[1 - 2 \frac{\beta}{1 + \beta} (1 - \lambda)(1 - a) \right] > 0, \quad (10)$$

implying that the exchange rate appreciates in response to an adverse climate news shock ($g > 0$) and it depreciates in response to a positive productivity shock ($\theta > 0$). Intuitively, Proposition 1 implies that adverse global climate news shocks lead to an appreciation of the currencies of countries that are relatively more exposed to such news. This occurs because, when the shock materializes, the investors in these countries have a relatively higher marginal utility. Under complete markets, this translates into an increase in $\tilde{S}_1$ and a stronger real exchange rate.

Proposition 2. *By taking the first-order Taylor expansion on $\log(V_1^B) - \log(V_1^G)$ at $\tilde{s}_1 = 0$, we get the linearized form of $r_{0|1}^{BMG}$ as a function of the log pseudo-Pareto weight $\tilde{s}_1$:*

$$r_{0|1}^{BMG} = \overline{r_{0|1}^{BMG}} + \lambda_{BMG}^s \cdot \tilde{s}_1 \quad (11)$$

where

$$\lambda_{BMG}^s = \frac{1}{2} \left(\frac{a - 1}{1 - \Phi(a)} + \frac{2\lambda - 1 + 4\beta\lambda(\lambda - 1) + a\beta(2\lambda - 3)(2\lambda - 1)}{1 + \beta\Phi(a)} \right)$$

with

$$\Phi(a) = 2 \left(\frac{a}{(1 - a)\lambda^\lambda(1 - \lambda)^{1-\lambda}} \right)^a + a - 2.$$

When a is close to zero, the sensitivity λ_{BMG}^s is negative for $\lambda < 1$.

It follows that the excess return declines in response to a positive climate news shock ($g > 0$) and it increases in response to a positive productivity shock ($\theta > 0$). In addition, since the sensitivity of the ratio of the pseudo-Pareto weights is increasing in $b - 1$ (see equation (8)), the exposure of $r_{0|1}^{BMG}$ is relatively more negative when the

brown firm is located in highly exposed countries (higher b). This finding motivates the use of an *interaction* term between firm-level emissions (a common measure of firm's brown-ness) and country exposure in our regressions (equations (17) and (18)).

Given the home country's large exposure to climate news shocks, investing in domestic *green* technology becomes more valuable than investing in domestic *brown* technology, as it helps mitigate the rise in marginal utility in response to adverse climate news. This results in an appreciation of green assets, a depreciation of brown assets, and a corresponding decline in the spread. In the Appendix, we show that this proposition holds for all values of a implying empirically plausible damage-to-GDP ratios.

The next and last proposition focuses on international hedging via international trade.

Proposition 3. *Let $\gamma > 1$, when $b > 1$, a global climate news shock $g > 0$ decreases total net exports in the home country, implying that there is a reallocation of resources toward the country that ex-ante hedges the most.*

A positive climate news shock, $g > 0$, reveals adverse information about future global climate conditions and it leads to a decline in net exports for the country that is more exposed to such shocks, i.e., the country with a higher b coefficient. In this state of the world, without trade the more exposed country would experience a larger increase in marginal utility and would be therefore worse off. To restore equality in marginal utilities under optimal risk-sharing, the most exposed country hedges ex-ante and receives a transfer of resources from the less exposed one ex-post. In the decentralized economy, this transfer corresponds to higher imports and lower exports, leading to a decline in net exports of the home country. This pattern is a direct implication of the recursive risk-sharing equilibrium under complete markets as the net exports move together with $\tilde{S}_1$. In the data, we test this result by looking at country pairs sorted according to our estimated exposure to the global climate attention index.

3 Data

Data collection. In line with the methodology developed in [Engle et al. \(2020\)](#), we construct a climate attention index to assess the extent of discourse concerning climate change within the news media. Our approach differs from that of [Engle et al. \(2020\)](#) in that we have chosen to focus on news media outlets, particularly newspapers, that have a strong presence on the social media platform Twitter.³ Utilizing textual analysis, we generate country-level indices at both low and high frequencies. Our dataset encompasses a total of 25 countries, providing a comprehensive representation across a wide spectrum of local languages, income levels, and geographical regions. We have conveniently summarized our dataset in Table 1.

In what follows, we describe the construction of our climate attention index. In a second step, we show that in the data we can disentangle attention to topics related to physical risk and transition risk. Importantly, we document that spikes in climate attention are associated with news carrying negative sentiment. We return to this relationship directly in Section 4, where we test whether the FX and equity responses are stronger conditional on negative sentiment.

Climate attention index construction. We use the “universe” of Tweets from October 2014 until December 2022 from the Twitter accounts of official newspapers (the full list of newspapers is reported in table B.1 in the Appendix). We construct our index by comparing the news content in these tweets to a corpus of “[...] authoritative texts on the subject of climate change” as in [Engle et al. \(2020\)](#). Similar to [Engle et al. \(2020\)](#), we aggregate the authoritative text documents into a Climate Change Vocabulary (CCV), which amounts to the list of both unique terms and their associated frequency with which each one of them appears in the aggregated corpus. In the construction of our dataset, we use the text (if any) of the Twitter posts and not the respective articles. Before doing so, we preprocess both the corpus of authoritative texts and the newspaper’s tweets. The preprocess removes (1) any links, (2) any tags

³Our data collection ended in December 2022. The platform is currently identified as “X”. The list of newspapers included in our data set can be found in the Appendix. The selection of newspapers was determined through a two-step process: first, we verified whether the newspapers had been included in prior studies ([Baker et al. 2016](#)) and were categorized as primary information outlets. Second, we expanded upon this criterion by evaluating whether each newspaper’s Twitter handle had a substantial number of followers and maintained consistent posting activity.

TABLE 1. DATASET SUMMARY

Country	Tot. no. News Outlets	Tot. no. Tweets	Tot. no. Words	Tot. no. Terms	Language
AR	4	1,892,464	16,592,918	50,513	Spanish
AU	4	539,792	5,805,411	46,253	English
BR	3	1,091,173	9,157,179	46,841	Portuguese
CA	5	1,071,360	9,662,873	48,695	English
CH	4	312,897	2,477,651	40,932	German and French
CL	3	1,088,455	9,422,589	48,409	Spanish
CN	3	488,076	6,786,626	53,342	English
CN (HK)	2	305,490	2,982,160	28,143	English
CO	3	1,291,773	10,348,974	55,984	Spanish
DE	4	708,692	6,779,556	31,190	German
ES	4	1,484,502	13,662,696	61,176	Spanish
FR	4	1,004,595	8,950,751	54,021	French
IN	4	1,909,474	22,370,252	71,505	English
IT	3	1,268,190	10,209,216	42,084	Italian
JP	4	353,247	3,819,349	36,544	English
KR	4	302,946	2,814,363	28,923	English
MX	4	2,041,505	19,553,684	70,880	Spanish
NO	3	152,521	921,258	13,022	Norwegian
NZ	3	147,218	1,156,784	20,538	English
PT	3	802,011	6,102,651	40,929	Portuguese
SA	3	914,623	8,754,709	44,571	Arabic
SE	4	190,493	1,607,852	19,201	Swedish
UK	4	791,929	6,978,125	46,460	English
US	11	2,735,338	29,141,511	72,662	English
ZA	3	453,135	3,945,579	33,473	English
Total	96	23,341,899	220,004,717	1,106,291	

Notes: This table shows summary statistics for all news that we collect for a large cross section of countries. Our real-time data range from 1/10/2014 to 31/12/2022. For each country, we report the total number of Twitter accounts that we monitor (major newspapers and other media outlets), the total number of tweets collected, total number of words, total number of unique terms (accounting for unigrams and bigrams), and original language. We translated the corpus in [Engle et al. \(2020\)](#) to 8 different languages, in order to construct our index.

to Twitter accounts, (3) any special characters and/or numbers, and (4) stop words (e.g, “and” “or”). Furthermore, it stems each word (using the Snowball algorithm), and it keeps only words with more than two characters. The stemming procedure retains only the root of each terms; for example, terms like “work” and “working” are included as unique terms. See Figure [B.1](#) in the Appendix for a word cloud representation of the preprocessed corpus in English.

Second, we construct the analogous count from our preprocessed Twitter texts for every country. Our approach enables us to construct our measures both at high and low frequency. When studying the impact of climate attention shocks on currencies, we use a daily aggregation method. In what follows, we focus on monthly aggregation of tweets for illustrative purposes. We refer to the Twitter texts aggregated in a specific month for a specific country as “document”. At the monthly frequency, we have a total of 99 documents for each country. Next, we convert the term counts from each one of our documents into term frequency–inverse document frequency scores (in short, “tf-idf” scores). Finally, for each document, we compute the cosine similarity between its tf-idf scores and those of the entire corpus. We carry out the described procedure for every country within our sample.

In cases where a country uses a language other than English, we employ the Google Translate API to translate the original English corpus into the appropriate foreign language. These languages include German, Spanish, French, Norwegian, Swedish, Arabic, Portuguese, and Italian. This approach enables us to create Climate Change Vocabulary (CCV) sets for each of the available languages, facilitating the analysis of newspapers published in local languages. This approach mitigates the potential news bias that might arise from solely relying on English-language newspapers in non-English-speaking countries.⁴ For countries with multiple local languages, such as Switzerland, we generate an index for each language and calculate the average across these indices.

Figure 1 presents our weekly Climate Attention Index (CAI), which has been aggregated across countries. The index is constructed by taking the equally weighted average across all the countries in our sample. The calculation of the index at the country-level is necessary because we cannot mix documents written in different languages. Consolidating documents at the country-language-frequency level would yield highly comparable results.

Our index effectively reflects specific global events. For instance, the United Nations Climate Change Conference (COP26), held in early November 2021 with the aim of promoting commitments to combat climate change, garnered significant atten-

⁴For China, Japan and Korea, we retained the English-speaking newspapers posts due to the challenges in preprocessing languages with special characters. Our results are robust to the removal of these countries.

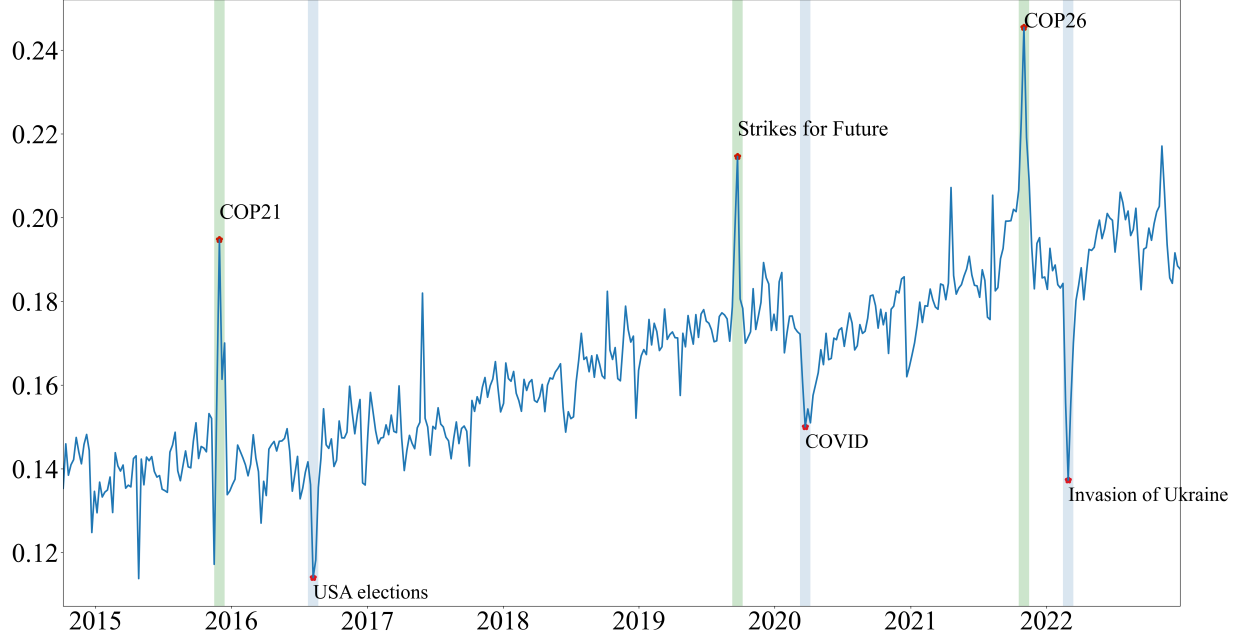


FIG. 1. WORLD CLIMATE ATTENTION INDEX (CAI)

Notes: This figure shows our Global Climate Attention Index. It is based on the methods described in section 3 applied to the countries listed in table 1 and a total of 8 languages.

tion worldwide, as evident in our index. Conversely, the COVID-19 pandemic and the invasion of Ukraine had the opposite effect on our index, with a notable decline attributed to a reduced proportion of climate-related discussions during these periods. Typically, significant events like elections, conflicts, and health crises tend to lead to a decrease in our index.

Heterogeneous sensitivity to global climate news. Figure 2 illustrates the same index across all of the countries included in our dataset. One of the novelties of our dataset is that it spans a large cross-section, which enables us to measure heterogeneity across countries. In the spirit of the international macro-finance literature, in what follows we focus primarily on heterogeneous exposures to global shocks (see Lustig et al. (2011)).

Specifically, let $CAI_{i,t}$ denote the index for country i at time t . The world climate

Climate Attention Index, monthly frequency, by country

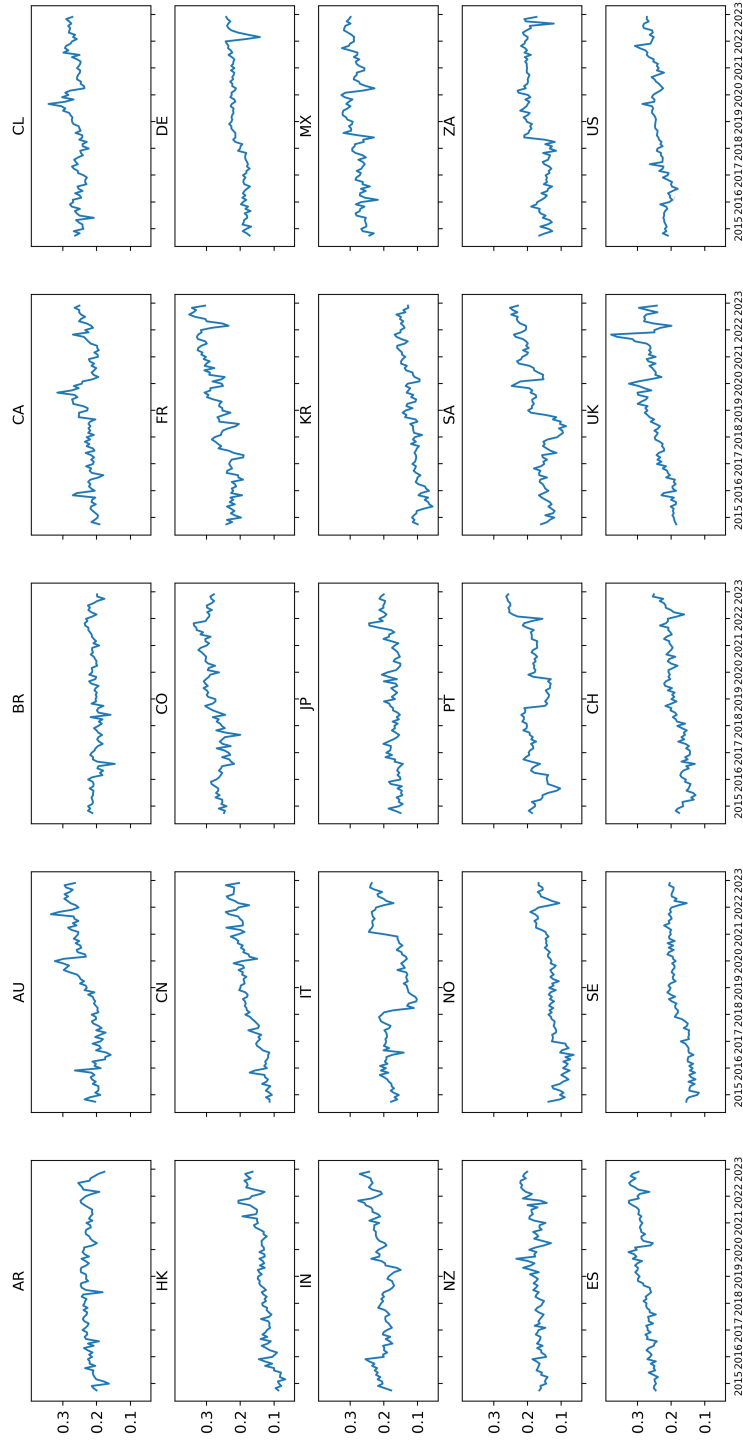


FIG. 2. WORLD CLIMATE ATTENTION INDEX (CAI) BY COUNTRY

Notes: This figure shows our Global Climate Attention Index. It is based on the methods described in section 3 applied to the countries listed in table 1 and 8 languages. For countries with multiple languages (e.g., Switzerland), the country-level index is the average of each language index.

attention index is defined as follows:

$$\overline{CAI}_t = \sum_i \omega_{i,t} CAI_{i,t}, \quad (12)$$

where $\omega_{i,t}$ is a country-specific weight. We provide results based on three alternative weighting schemes: (i) by equally weighting all countries, (ii) by weighting each country proportionally to its share of global GDP, and (iii) by weighting each country proportionally to its share of Twitter volume. Volume weights are based only on tweets with a similarity score with the corpus of authoritative text above median. Equivalently, we only include the volume of tweets that are the most relevant for the focus of our analysis. We then estimate the following regression:

$$\Delta CAI_{i,t} = \beta_i \cdot \Delta \overline{CAI}_t + u_{i,t}, \quad (13)$$

where Δ refers to the demeaned growth rate over time for each country. We regard β_i as a measure of exposure to a common, i.e., global, component, and $u_{i,t}$ as a pure local component. Equivalently, we interpret β_i as the empirical counterpart of b in the model, that is, the parameter that determines how much a country is sensitive to global climate news shocks.

Table 2 presents the estimated β parameters for each country in our sample under various weighting schemes for the global CAI index. These β values exhibit a strong positive correlation across different weighting schemes, with correlation coefficients exceeding 0.9. Using the equally weighted global CAI index as the reference, we observe that South Korea, Saudi Arabia, China (including both Hong Kong and Mainland), and Japan consistently possess the highest β values. Conversely, Switzerland, South Africa, Chile, Norway, and Germany consistently exhibit the lowest β values. The relevance of this heterogeneity is twofold. First, the model suggest that climate news shocks impact currencies and international trade when countries have heterogenous sensitivities ($b \neq b^*$). Our cross section features 138 country-pairs with exposures to the global CAI that are different from each other at the 10% confidence level.

Second, this cross section is *distinct* from common measures of vulnerability and exposure to physical risk. Specifically, we assess the correlation between our esti-

TABLE 2. ESTIMATED β FOR EACH COUNTRY

Country code	Equally weighted		GDP weighted		Volume weighted	
	β	s.e.	β	s.e.	β	s.e.
AR	0.929	0.209	0.574	0.148	1.047	0.246
AU	1.419	0.197	0.932	0.190	1.375	0.219
BR	0.616	0.293	0.470	0.218	0.717	0.294
CA	1.710	0.224	1.312	0.138	1.789	0.232
CL	0.241	0.285	0.113	0.175	0.262	0.291
CN (HK)	2.105	0.325	1.196	0.326	1.619	0.518
CN	2.014	0.344	1.787	0.142	1.887	0.450
CO	0.679	0.180	0.561	0.115	0.702	0.207
FR	0.662	0.247	0.320	0.243	0.712	0.274
DE	0.363	0.122	0.236	0.074	0.316	0.160
IN	1.410	0.367	1.144	0.184	1.547	0.330
IT	0.985	0.593	0.624	0.473	1.441	0.705
JP	1.895	0.271	1.472	0.236	1.750	0.233
KR	2.413	0.394	1.624	0.280	1.889	0.324
MX	0.613	0.243	0.376	0.141	0.847	0.244
NZ	0.811	0.293	0.428	0.223	0.767	0.318
NO	0.344	0.320	0.245	0.267	0.166	0.313
PT	0.469	0.663	0.083	0.432	0.130	0.614
SA	2.266	0.574	1.108	0.251	2.254	0.609
ZA	0.197	0.466	-0.029	0.466	0.063	0.612
ES	0.534	0.128	0.366	0.084	0.578	0.114
SE	0.592	0.251	0.255	0.116	0.394	0.208
CH	-0.028	0.414	-0.014	0.263	-0.168	0.335
UK	0.993	0.269	0.798	0.214	0.998	0.339
US	0.768	0.143	0.743	0.083	0.905	0.134

Notes: This table reports the estimated β_i reported in equation (13), that is, the country-specific exposure to the global climate attention shock. The global climate attention shock is constructed as the equally weighted, GDP-weighted, and Twitter volume-weighted cross-sectional average at each point in time. Volume- and GDP-based weights are lagged by one period. The sample ranges from 2015:Q1 to 2022:Q4. Standard errors are HAC-adjusted.

mated β values and alternative measures of climate exposure. We first use the vulnerability score provided by the [Notre Dame Global Adaptation Initiative](#). This index aims to encapsulate “a country’s exposure, sensitivity, and capacity to adapt to the negative effects of climate change.” We calculate the average of these scores over the longest available period (1995-2021). The upper panel of Figure 3 illustrates the relationship between our β estimates and the vulnerability score. We observe a pos-

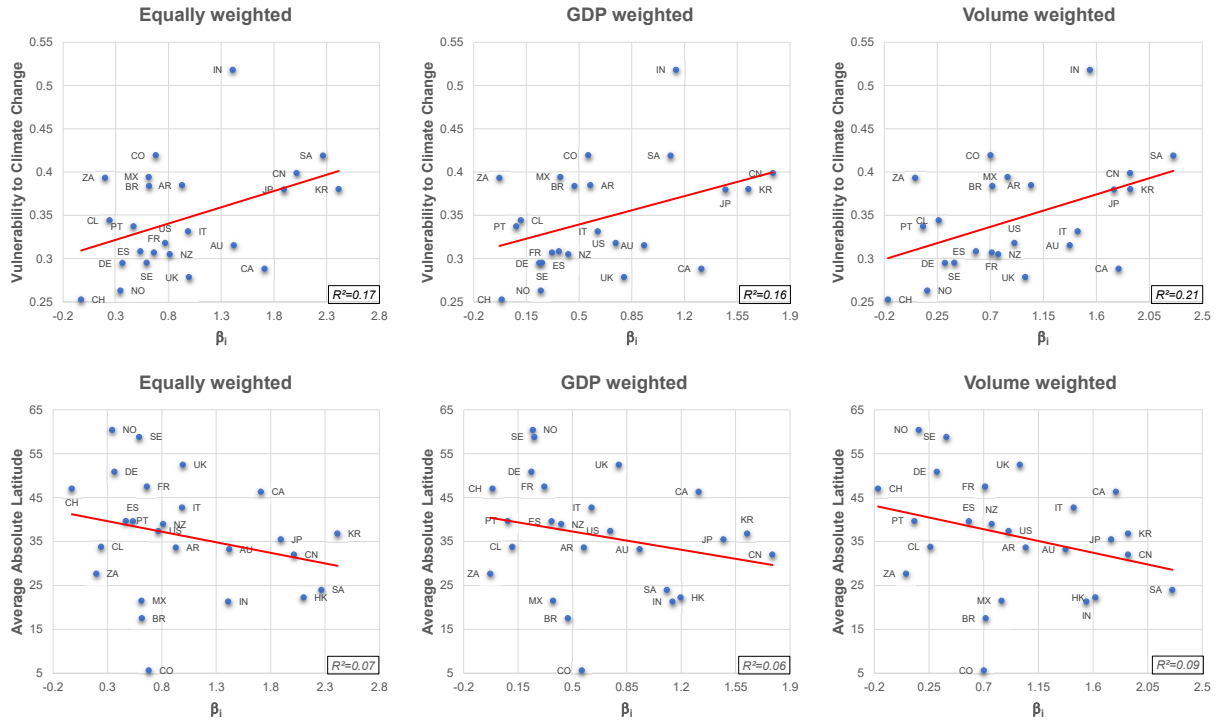


FIG. 3. β AGAINST VULNERABILITY TO CLIMATE CHANGE.

Notes: This figure shows the scatter plot of the estimated β s in equation (13), and measures of vulnerability to climate change across countries. The β s are estimated using the global climate attention shock constructed as either the equally-, GDP-, or Twitter volume-weighted cross-sectional average at each point in time. The upper panels consider each country's average vulnerability score from the [Notre Dame Global Adaptation Initiative](#). The lower panels refer to the average absolute latitude of each country, that is, a population-weighted average absolute latitude of all cities in each country. City-level latitudes are collected from the [World Cities Database](#).

itive and statistically significant correlation between our β estimates and this index, implying that countries with greater susceptibility to climate change usually exhibit higher β values, i.e, more attention exposure to global climate news. Importantly, a large share of the variance in our cross section is not explained by vulnerability.

Additionally, we relate our estimated β values to absolute latitude. Generally, countries situated in regions with higher absolute latitudes are usually considered less exposed to rising temperatures resulting from climate change. To assess this, we compile a dataset of city-level absolute latitudes for each country and compute the population-weighted average of these latitudes. Our analysis reveals a weak negative correlation between our β estimates and absolute latitude.

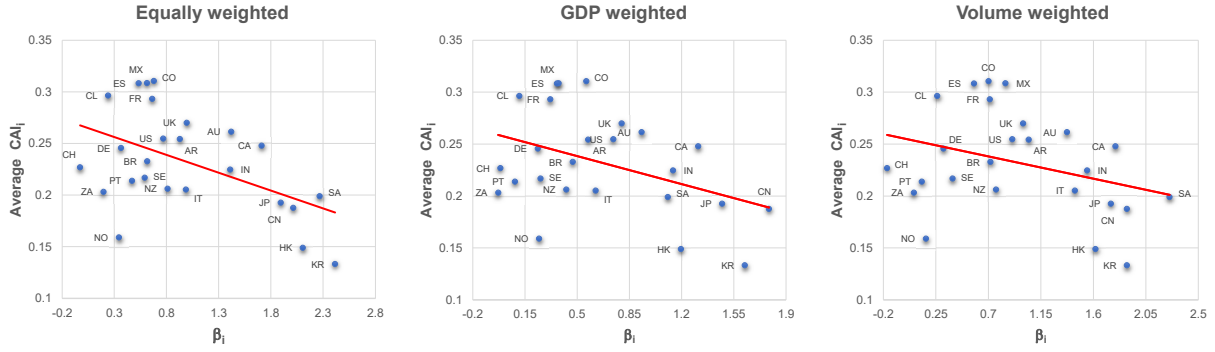


FIG. 4. β AGAINST AVERAGE ATTENTION TO CLIMATE CHANGE.

Notes: This figure shows the scatter plot of the estimated β s in equation (13), and the average attention to climate change across countries. The β s are estimated using the global climate attention shock constructed as either the equally-, GDP-, or Twitter volume-weighted cross-sectional average at each point in time.

In the context of our model, the vulnerability of the home country is captured by its disruption functions, D , while its sensitivity to global news shocks is governed by the country-specific parameter b . In the next section, we show that our estimated exposure coefficients provide information beyond what is captured by vulnerability alone. Equivalently, the coefficients β_i serve as empirical proxies for the model's sensitivity parameter b , and they highlight a novel finding: attention to global climate news can be high even in countries with low vulnerability, with significant implications for exchange rates, asset prices, and trade.

In figure 4, we plot the estimated β for each country against the country's average attention to climate news. We document a striking negative relationship, meaning that countries that are more attentive to climate news tend to have lower β values. We interpret this pattern as indicating that countries with lower sensitivity to climate attention shocks are those that maintain a more persistent and elevated level of climate awareness. This is consistent with the idea that in regions where climate considerations are deeply embedded in national policy agendas, marginal shifts in attention have less impact. Several of the low- β and high-average attention countries are in Europe (both EU and non-EU), a region long recognized as a leader in climate policy and sustainable finance. Through the lens of our model, this reduced sensitivity implies that these countries serve as natural providers of insurance in the global climate risk-sharing system.

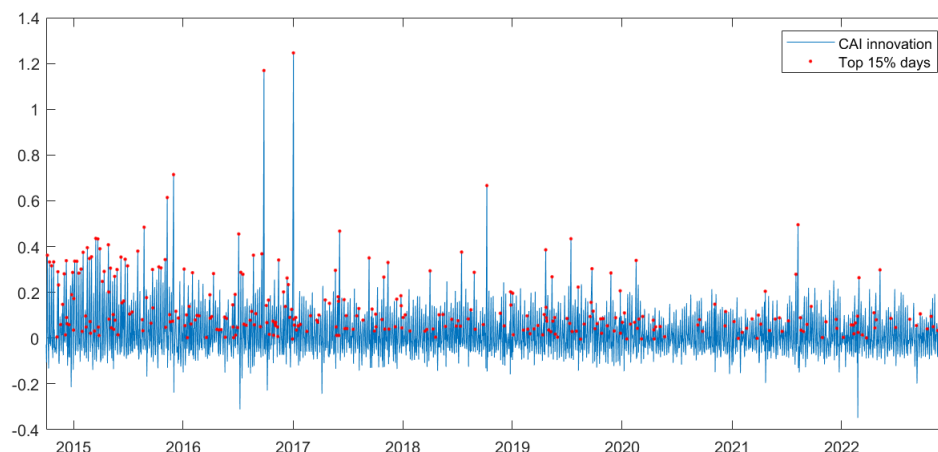


FIG. 5. INNOVATIONS TO WORLD CLIMATE ATTENTION INDEX (CAI)

Notes: This figure shows innovations to our equal-weights Global Climate Attention Index constructed as described in section 3. Our cross section of countries is reported in table 1. Country-level indices are aggregated using equal weights. Red dots are classified as climate-related news days (top-15% realizations). Weekends are excluded.

Top days. In the next section, we use high-frequency data to identify the impact of climate news shocks on exchange rates and stock returns. To identify innovations in our global index, we initially calculate the daily growth rate in the raw CAI index for each country and subsequently remove its mean. These demeaned indices at the country level are then winsorized at the 99.9% and aggregated to generate a global metric.⁵ The outcomes of this analysis are visualized in Figure 5.

Based on the distribution of innovations in our global climate index, we select the top 15% of these innovations.⁶ Specifically, we examine the reaction of currencies and stocks in the days following a substantial increase in climate attention. This approach serves two purposes: (i) to minimize noise from news unrelated to climate, and (ii) to more effectively identify days driven by “pure” climate news shocks. Such a methodology is commonly used in the literature investigating the effects of news related to policy announcements (see, for instance, [Leombroni et al. \(2021\)](#)). It is worth noting that some days with a high volume of climate-related news may fall

⁵When we compute our index at the daily frequency, it is possible that in some days and for some countries we could have extreme growth rates because our index gets close to zero. We remove both the top- and bottom-0.1% realizations from our sample.

⁶We analyze the distribution of news for each day of the week, which allows us to account for any patterns specific to particular days (e.g., Monday or Friday effects). This approach is standard in studies focusing on daily news.

below our top 15% threshold due to the dominance of other topics in the news. As a result, similar to previous studies, this procedure may introduce a downward bias, making it less likely to detect a significant effect of climate news on asset prices and currencies. In the Appendix, we report the news the we collected for selected top days. See Table B.3 and Figure B.2.

Volume for top-days. In principle, one could be concerned that our top-days represent days with no significant global non-climate events. That is, our “climate-focused” days may actually just be “news vacuums” days. In order to address this concern, we regress global weekday tweet-volume on a top-day dummy, as well as country, year, month, and weekday dummies. The implied coefficient in front of our top-day dummy is both small (the point estimate refers to 1 tweet) and insignificant with a p -value of 0.42 when clustering by date. When we use a two-way-clustered specification by both country and day, the p -value decreases to a marginal 0.07. Hence, after accounting for seasonality in the news cycle, our top-day dates are not slow news days, but rather days in which the news shift toward climate.

Measurement robustness. We assess the robustness of our measure along two dimensions and report the details in Appendices D.1 and D.2. First, we assess whether terms carrying a non-climate meaning contaminate the index by asking GPT-4o to audit the 813 English CCV stems that account for 90% of our index mass. For each stem, we record (i) the share of occurrences referring to broad environmental concepts, and (ii) the share referring specifically to climate change.

Purging the terms that combine a genuine climate meaning with predominantly non-climate usage (e.g., “target,” “protocol,” “transit,” and “commit”) leaves the series essentially unchanged. The innovations of the benchmark and purged indices have a correlation of 0.998. About 96% of our top climate-attention days continue to be classified as such, and the few days that change classification lie immediately below the top-15% cutoff, i.e., they are marginal cases.

Second, we ask whether a large language model would measure daily climate content more accurately than our benchmark methodology. We run simulations to produce a controlled setting in which each day’s true climate share is known. Under

the sentence-level design, the GPT-4o classification produces improvements that are economically negligible and typically statistically indistinguishable from ours.

We conclude that our index is robust to term ambiguity and that currently available LLM-based alternatives offer no improvement of practical value.

Topics for top days. After identifying top-attention days, we examine the content of the news shared during these periods. Following the text analysis literature (see, for example, [Hassan et al. \(2019\)](#)), we employ a pre-trained sentence embedding model to extract semantic representations of newspaper tweets.⁷ These embeddings are then clustered to identify a smaller set of key themes, or main topics. This dimensionality reduction enables the use of topic-level cosine similarity scores to classify tweets as climate-related and further distinguish between those primarily associated with physical risk and transition risk. The process involves several carefully designed steps aimed at minimizing discretion on our part, which we detail below.

First, we generate 100 climate risk-related sentences using ChatGPT: 50 focused on physical risks (e.g., extreme weather) and 50 on transition risks (e.g., regulatory changes). The full list of these sentences is provided in Table B.2 in the Appendix. Next, we analyze the identified top climate-attention days by preprocessing raw newspaper tweets, removing non-informative elements such as links and special characters. We then convert the cleaned tweets into high-dimensional vector representations (sentence embeddings), which serve as inputs for the BERTopic clustering algorithm (see [Grootendorst \(2022\)](#)).⁸

After identifying the key topics discussed in the media on these top-attention days, we compute cosine similarities between each topic embedding and each AI-generated sentence embedding. This approach allows us to assess how closely each topic aligns

⁷Sentence embeddings enter the analysis only at this stage. The CAI, and therefore the selection of top-attention days, is constructed entirely from the tf-idf cosine similarity described above; embeddings are used solely to classify tweets on days that have already been selected.

⁸BERTopic is a topic modeling technique that integrates transformer-based embeddings with clustering to extract coherent topics from large text datasets. We use the package provided at <https://maartengr.github.io/BERTopic/index.html>. We keep the following default options: (i) UMAP (Uniform Manifold Approximation and Projection) with five components, and (ii) HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) to group tweets into distinct topics. Since our data comprise tweets in different languages, we must use the embedding model “distiluse-base-multilingual-cased-v2” by [Reimers and Gurevych \(2019\)](#).

with climate-related themes. Specifically, a topic is classified as climate-related if its cosine similarity score with any AI-generated climate sentence falls within the top 0.1% of the score distribution (with a threshold of 0.546). Using the 99.9th percentile ensures a stringent definition of climate-related topics.

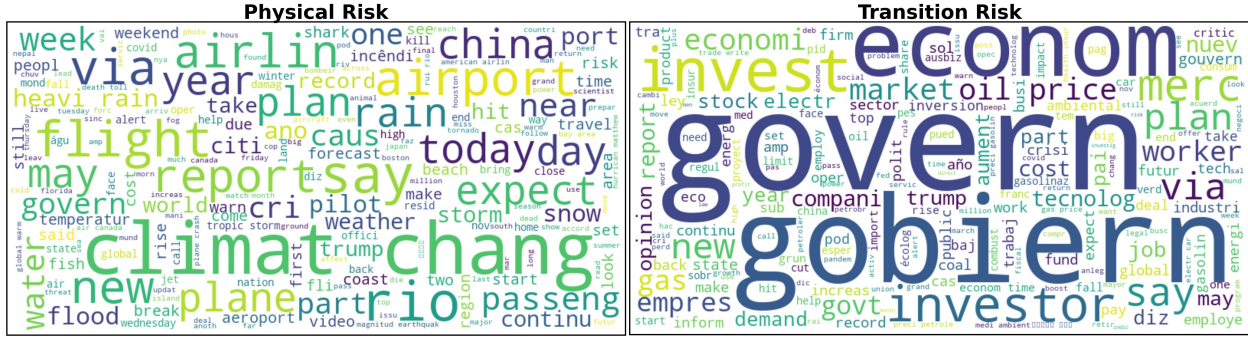
Finally, we classify each climate-related topic as either “Physical Risk” or “Transition Risk” based on its closest match to the sentences listed in Table B.2. If the most related sentence corresponds to physical (transition) risks, the topic is labeled accordingly. Figure 6 visualizes the most frequent words in these two groups of tweets, with the resulting word clouds reinforcing our methodology by clearly distinguishing between different sources of climate-related concerns. In the following analysis, we use these two groups of tweets in both our sentiment analysis and high-frequency asset pricing analysis.⁹

Sentiment of tweets. We analyze the sentiment of climate-related tweets during top days by using the Twitter-XLM-RoBERTa model from CardiffNLP.¹⁰ This is a transformer-based multilingual sentiment classifier suitable for text in multiple languages (Barbieri et al. 2022). Each one of our tweets is tokenized and passed through the pre-trained model in order to obtain a sentiment label (negative, neutral, or positive) based on the highest logit value.

Figure 7 presents the share of climate-related tweets in each risk category (physical and transition) that are classified as either positive or negative. In our sample, global news shocks related to climate typically have a negative tone. Equivalently, our top-attention days are usually bad-news days.

⁹ClimateBERT (see Webersinke et al. (2022)) is a natural candidate for analyzing climate-related text, given its domain-specific training. However, we chose not to use it for our analysis due to two key limitations. First, ClimateBERT is trained exclusively on English-language text, which poses challenges in our context where the dataset spans multiple countries and includes tweets in a variety of languages. Translating all non-English content into English could introduce distortions that bias the analysis, particularly when dealing with linguistic nuances or compressed phrasing typical of headline-style content shared by news outlets on social media. Second, ClimateBERT was originally pretrained on longer, formal texts such as news articles, research abstracts, and corporate disclosures, and was not explicitly optimized for short-form content like tweets or headlines. As a robustness check, we compared the share of physical risk tweets identified by our model to the share of tweets classified as such by both ClimateBERT and our model for U.S.-based content, and found a correlation of approximately 80%.

¹⁰Available at <https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base-sentiment>.



Notes: This figure shows the word cloud of root words extracted from climate-related Tweets in our sample of top days. Tweets are classified as climate-related if the cosine similarity of their associated topic exceeds the 99.9th percentile of the similarity scores calculated against a set of AI-generated reference sentences. Tweets are further classified as physical (transition) risk if their topic aligns most closely with a physical (transition) risk-related sentence (see table B.2). Our sample period starts in October 2014 and ends in December 2022. Transition risk generally refers to risks arising from changes in policies, technologies, and market preferences associated with climate action. Physical risk relates to the direct impact of climate change, such as extreme weather events and rising temperatures.

Backward- vs forward-looking tweets. We also examine whether climate-related tweets on top CAI days are predominantly backward- or forward-looking. A backward-looking tweet reports realized events, e.g., an extreme weather event that already occurred. A forward-looking tweet refers to future developments, e.g., pending legislation, targets, or commitments. Equivalently, we disentangle news about realized climate events from news about anticipated future risk. We perform this analysis across transition- and physical-risk related tweets.

Specifically, both risk type and temporal orientation are assigned jointly by GPT-4o-mini: we give each tweet’s text and posting date to GPT and ask it to: (i) reason step by step, anchoring strictly to the posting date; and (ii) classify the tweet’s risk type and its temporal orientation (past, present, or future).¹¹ For each tweet, the

¹¹To minimize token consumption, GPT classifications are run only on tweets that the BERT-based pipeline already identified as climate-related. Tweets flagged as non-climate by BERT are excluded from this analysis. We classify each tweet on top CAI days into one of six risk subcategories: (1) physical acute (discrete extreme weather events), (2) physical chronic (long-run physical shifts such as sea level rise or rising temperatures), (3) transition regulatory (carbon policy, emissions law, climate disclosure rules), (4) transition market (green/brown asset repricing, ESG investing, stranded assets), (5) transition reputational (corporate climate pledges, greenwashing, activist pressure), and (6) transition technology (renewables, clean tech, EVs, energy transition). We then map these GPT’s six subcategories into (1) physical risk (physical acute and physical chronic) and (2) transition risk (the four transition subcategories). For each classification, GPT also returns a confidence score $Confidence_i \in [0, 1]$.

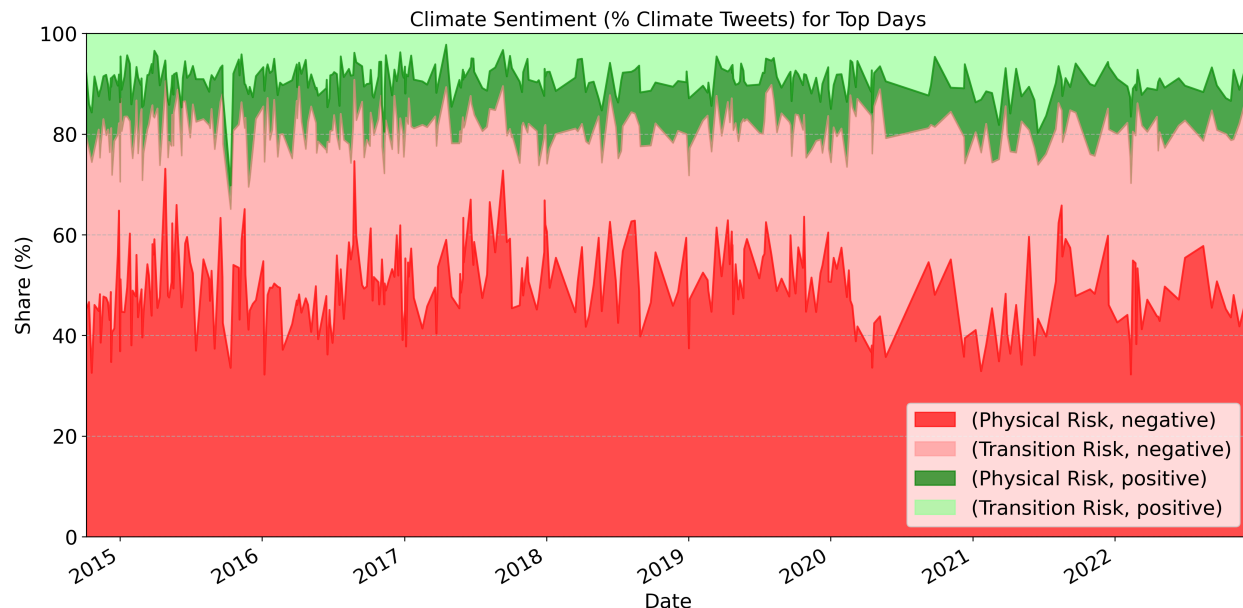


FIG. 7. SENTIMENT OF CLIMATE RELATED TWEETS DURING TOP DAYS.

Notes: This figure presents the share of climate-related tweets categorized by sentiment (positive or negative) and risk type (transition or physical) on days for which there is a positive spike (top-15%) in the global CAI innovations. The sentiment classification is based on a multilingual sentiment classifier [Barbieri et al. 2022](#)). The word clouds for transition and physical risk are reported in figure 6.

model receives the tweet text in its original language, and the posting date.

Figure 8 plots the resulting share of climate-related tweets on top CAI days by risk type $\times$ temporal orientation. Physical risk tweets are predominantly past-tense throughout the sample, consistent with physical risk news largely reporting realized events. Transition risk tweets are relatively more present or future-oriented than physical risk tweets. Pooled across the full sample, physical risk tweets are past-tense 60.2% of the time (33.0% present, 6.8% future), compared to 38.2% for transition risk tweets (49.8% present, 12.0% future).

In addition, in Figure 8 we identify the highest-volume top days for each risk type and report relevant events occurring on those days. On the physical risk side, we highlight the April 2015 Nepal earthquake and the September 2017 Hurricane Irma landfall. On the transition risk side, we annotate the pre-2020 period with COP21's opening in November 2015, the U.S. withdrawal from the Paris Agreement in June 2017, and the IPCC Special Report on 1.5°C in October 2018; and the post-2020 period

The full prompt is available at [C.4](#).

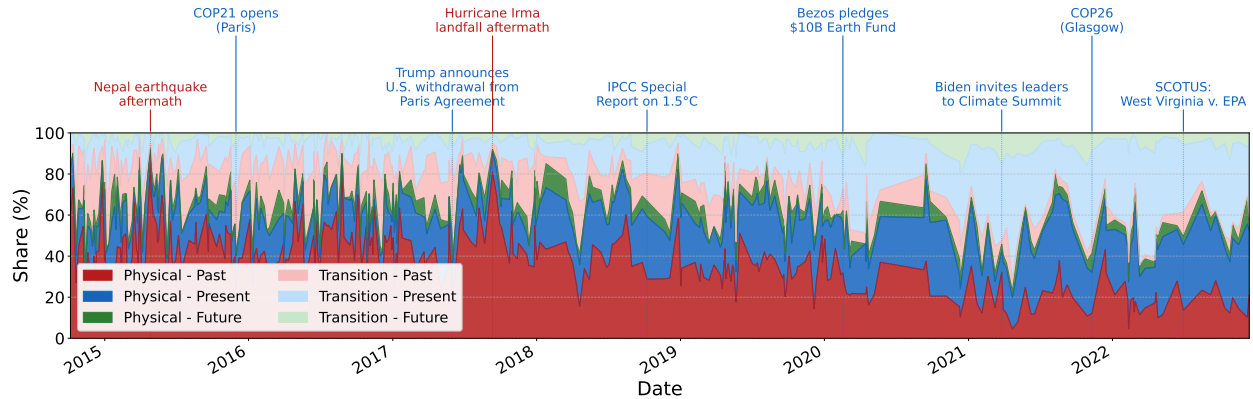


FIG. 8. TEMPORAL CLASSIFICATION ON TOP CAI DAYS (GPT CLASSIFIER)

Notes: This figure reports the share of climate-related tweets by risk type and temporal orientation on top CAI days, using GPT’s temporal classification detailed in Subsection 3. Each stacked area shows the fraction of tweets classified as physical risk past (dark red), physical risk present (dark blue), physical risk future (dark green), transition risk past (light pink), transition risk present (light blue), and transition risk future (light green). Annotated dates mark the highest-volume top days for each risk type and the corresponding real world events; event labels and connecting lines are colored by risk type (red = physical, blue = transition).

with Jeff Bezos’s \$10 billion Earth Fund pledge in February 2020, President Biden’s invitation to world leaders to a climate summit in March 2021, ahead of COP26 in November 2021, and the U.S. Supreme Court’s *West Virginia v. EPA* ruling in June 2022.

To summarize, our physical risk examples are retrospective accounts of damage already materialized, while the transition risk examples are regulatory, judicial, or corporate actions unfolding in real time or anticipated in the near term.

4 Climate News Shocks and Asset Prices

In this section, we explore the role of climate news shocks for currencies and equity returns.

4.1 Exchange Rates

In what follows, let $\Delta e_{(H,F)} > 0$ refer to an appreciation of the currency of country H versus country F. Proposition 1 states that countries with higher sensitivity to climate news should experience a currency appreciation if adverse news materialize.

In order to test this condition, we obtain bilateral exchange rate data from Bloomberg and sort our country pairs as follows. Let H and F be two countries in a country-pair. For every non-repeated pairwise combination of countries, $i \neq j$, we look at the exposure of the country-specific CAI to the global CAI as estimated in equation (13). If $\beta_i - \beta_j > 0$ ($\beta_i - \beta_j < 0$) we denote country i as H and j as F (j as H and i as F) and focus on the exchange rate $e_{(H,F)}$.

We test this assumption at a high-frequency level, particularly on days when climate-related news is exceptionally significant. Figure 9 illustrates the average cumulative exchange rate variation following these specific days,

$$\Delta e_{t,t+k} := \sum_{s=0}^{k-1} \Delta e_{t+s,t+s+1} \frac{100 \cdot N^{top}}{k}.$$

Cumulative returns are averaged over the time horizon k , they are in percentage, and they are multiplied by N^{top} , i.e., the average number of top-days in a year. In our benchmark analysis $N^{top} = 38$.

To ensure that our observations are not influenced by idiosyncratic patterns in our dataset, these averages are computed relative to the average cumulative variation observed in the remaining 85% of business days in our sample. Specifically, we estimate the relative cumulative returns with the following regression:

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + b_k \cdot \mathbb{1}_t^{Top} + \epsilon_{t,k}^{i,j},$$

where $\mathbb{1}_t^{Top}$ is a dummy variable that equals 1 if the day falls in the set of extreme climate news days. b_k captures the relative cumulative returns over a horizon of k days. We only include country pairs where β_i is significantly larger than β_j at 10% level (see figure C.1, Appendix C). Standard errors are clustered at both the date and country-pair level.

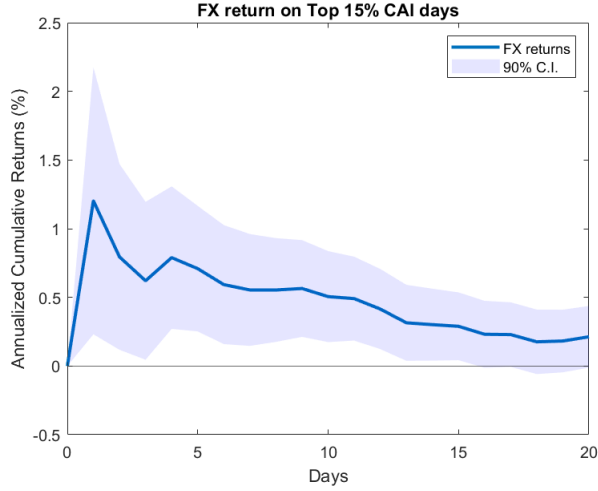


FIG. 9. FX AND INNOVATIONS TO WORLD CLIMATE ATTENTION INDEX (CAI)

Notes: This figure shows the average cumulative return in exchange rates around days with a positive spike (top-15% days in our sample) in the global CAI. This cumulative variation is net of the common cumulative variation that we obtain when considering all other days in our sample. Exchange rates are computed at Greenwich Mean Time in each day across 138 country pairs with exposures to the global CAI that are different from each other at the 10% confidence level. Exchange rates are quoted such that for each country pair an appreciation refers to the country with the higher exposure to global climate news shocks. Our Global Climate Attention Index is based on the methods described in section 3. Countries are equally weighted. The shaded areas present the 90% confidence interval which is constructed using standard errors clustered at both the date and country-pair level.

If climate-related news has no bearing on exchange rates, we would anticipate null cumulative effects. Conversely, our findings reveal that adverse climate news shocks are linked to the contemporaneous appreciation of countries that are comparatively more exposed (i.e., possessing higher β values). Furthermore, this appreciation exhibits persistence and it extends over multiple days, signifying a lasting impact on currencies. In a typical year, the currencies of high-exposure countries appreciate by an average of approximately 1.25% on the day following a large climate attention shock—an economically significant response. Appendix C.2 reports the corresponding pre-event trends.

EMs versus DMs. In Figure 10, we illustrate this effect among pairs of countries consisting solely of advanced economies (left panel), developing economies (right pan-

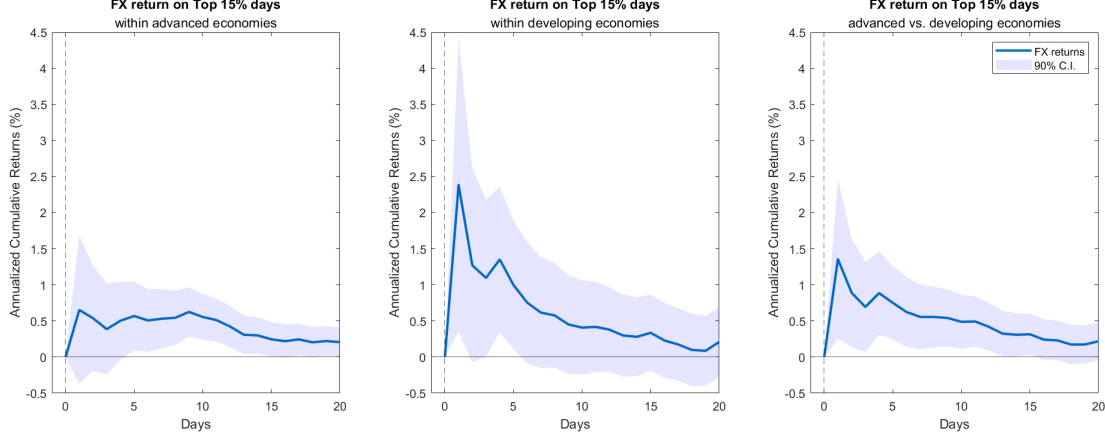


FIG. 10. FX AND INNOVATIONS TO WORLD CAI ACROSS COUNTRY PAIRS

Notes: This figure shows the average cumulative return in exchange rates after days with a top-15% positive spike in the global CAI. For more details, see the notes under figure 9 and section 3. We focus on country pairs in which (i) both countries are advanced economies (left panel), (ii) only one country is defined as an advanced economy (mid panel), and (iii) both countries are developing economies (right panel).

els), or a combination of advanced and developing countries (middle panel).¹² The appreciation pattern is observed across all country pairs. Furthermore, it tends to persist longer when at least one emerging country is involved in the analysis.

Type of news. We take advantage of our text analysis results and estimate the following regression:

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + (b_k + c_k \cdot S_t) \cdot \mathbb{1}_t^{Top} + \epsilon_{t,k}^{i,j}, \quad (14)$$

where S_t denotes the share of tweets focused on transition risks at day t , normalized between 0 and 1. This share is computed by using both positive and negative tweets classified as related to transition risk (see figure 7, dark red and dark green shaded areas). In figure 11, we depict the composite coefficient $b_k + c_k \cdot S_t$ for different values of S_t . This specification enables us to identify whether our results are primarily driven by transition or physical risk. We find that the results depicted in figure 9 are consistent with those obtained by estimating equation (14) and setting S_t to its

¹²We define advanced economies according to the IMF, where the advanced economies in our sample are: AU, CA, CN (HK), FR, DE, IT, JP, KR, NZ, NO, PT, ES, SE, CH, UK, US. The developing economies are: AR, BR, CL, CN, CO, IN, MX, SA, ZA.

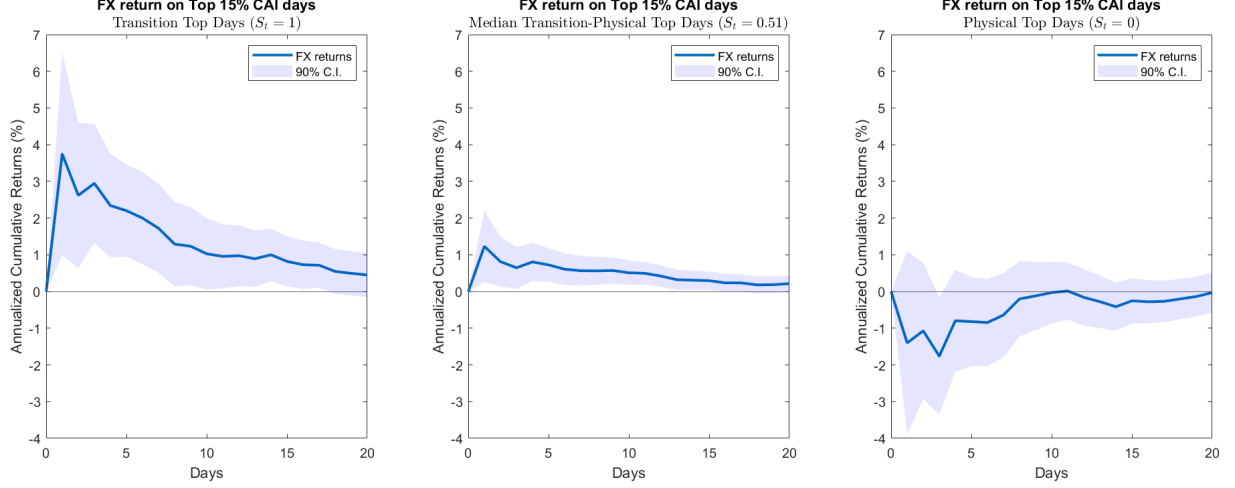


FIG. 11. FX AND WORLD CAI: TRANSITION VS. PHYSICAL DAYS.

Notes: This figures show the average cumulative return in exchange rates after days with a positive spike in the global CAI. The left (right) panel shows the FX responses conditional on the days that the tweets mostly talk about transition risk (equally talk about both physical and transition risk). The specification is

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + (b_k + c_k \cdot S_t) \cdot \mathbb{1}_t^{Top} + \epsilon_{t,k}^{i,j},$$

where S_t denotes the share of tweets talking about transition risks at day t , normalized between 0 and 1. The responses are constructed using the composite coefficient, $b_k + c_k \cdot S_t$, and by choosing different values of S_t .

median value. Most importantly, we find that the results are stronger when $S_t = 1$, i.e., in top days when attention is entirely about transition risk.

We repeated this exercise using the share of tweets related to transition risk with a negative sentiment (figure 7, dark red area) and obtained virtually identical results. Even though in the model we do not distinguish between physical and transition climate news, in the data we find that the currency-appreciation response is strongest on days when attention is concentrated on transition risk.

The role of sentiment. We examines whether the FX response to top CAI days is stronger when the underlying climate news is more negative in sentiment. Specifically, we estimate the following regression with an interaction term that allows the

FX response to vary with the daily share of negative climate-related tweets:

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + (b_k + c_k \cdot S_t^{neg}) \cdot \mathbf{1}_t^{Top} + \varepsilon_{t,k}^{i,j}, \quad (15)$$

where S_t^{neg} is the share of negative climate-related tweets among non-neutral climate-related tweets on day t . Since neutral tweets are excluded, this measure is closely related to a bad-minus-good climate sentiment measure.

Figure C.2 in the Appendix plots the implied FX response evaluated at the maximum, median, and minimum observed values of S_t^{neg} across top CAI days. The response is strongest when the negative-sentiment share is high, close to the benchmark response at the median, and statistically weak when the negative-sentiment share is low.

FX second moment. We assess whether top climate-attention days are associated with higher second moments of FX returns. Specifically, we estimate equation (14) after placing event-window realized volatility at the left-hand side. Our volatility measure is constructed from squared daily bilateral FX returns. For each currency pair (i, j) and horizon k , we define

$$RV_{t,t+k}^{i,j} = 100 \times \sqrt{\frac{250}{k} \sum_{h=1}^k (r_{t+h}^{i,j})^2}, \quad (16)$$

where $r_{t+h}^{i,j}$ is the daily log return on the bilateral exchange rate between countries i and j .

The point estimates are positive throughout the event window, suggesting that top CAI days are associated with elevated FX realized volatility (see figure C.3 in the Appendix).

4.2 Robustness

Weighting scheme and top-days threshold. In Appendix C, we show that our results are unchanged when we focus on either a GDP-weighted global index (figure C.4(a)) or a Twitter volume-weighted global index (figure C.4(b)). We also replicate

the same exercise by focusing on GDP- and Volume-weighted CAI indeces (figures C.4(a)-C.4(b)). Our results are qualitatively unchanged. In addition, if we change the threshold of top days from 15% to either 10% or 20%, our results remain unchanged (figures C.5(a)-C.5(b)).

Heterogeneity in sensitivity. Figures C.6(a)-C.6(b) confirm that our results are not sensitive to the choice of the significance level that we use in order to select country-pairs with different exposure coefficients. Our results apply regardless of whether we include all country pairs (including those with modest heterogeneity in β) or just those with a more pronounced degree of heterogeneity (5% confidence level).

We further divide the 138 benchmark currency pairs into three equally sized groups according to their climate-beta difference. Figure C.7 shows larger appreciation responses in the middle- and high-difference groups than in the low-difference group. At horizon five, the annualized average responses are 0.37%, 0.79%, and 0.97%, respectively. These estimates complement and support our benchmark results.

The role of the EU. Figure C.8(a) shows that our results are confirmed also when we aggregate the five Euro Area countries in our dataset (DE, FR, IT, PT, ES) into a single entity with a common currency.

Vulnerability. Most importantly, figure C.8(b) confirms that our results are not driven by vulnerability, but rather by the unique information content of our estimated exposure coefficients. Specifically, we orthogonalize our exposure coefficients with respect to vulnerability by considering the residuals of the cross sectional regression depicted in figure 3. The results in figure C.8(b) are based on the residual exposure and are consistent with our benchmark results. Equivalently, using heterogeneity in vulnerability across countries produces insignificant responses. The same conclusions apply to absolute latitude.

Data frequency. Figure C.9(a) confirms our main findings when using exposures estimated at the monthly frequency. Figure C.9(b) is constructed as figure 9 in main text, but it uses weekly observations and it includes weekends. The resulting weekly

CAI index covers a total of 430 weeks. During this process, climate-related tweets are aggregated on a weekly basis, with each week of the time series commencing on a Saturday. This weekly aggregation provides a comprehensive perspective on climate-related discussions in social media, encompassing an extended period that includes weekends.¹³ Similar to our daily frequency analysis, we identify weeks in which our index records top values and evaluate their cumulative impact on exchange rates.

Local versus global news. In figure C.10, we examine local top-attention days, as opposed to global top-attention days. We find that local shocks have no significant impact on FX markets. More broadly, local shocks also appear less relevant in the equity analysis presented next. For brevity, we exclude local shocks from the subsequent analysis.

Macroeconomic announcements. High climate-attention days may overlap with macro announcement days. In order to isolate the role of “pure” climate days, we either exclude or control for days on which the countries in a currency pair have scheduled macro announcements.

For each one of our 25 countries, Bloomberg provides announcement-level data, which we consolidate into four broad categories: Prices & Inflation, Labor Market, Monetary, and GDP Growth & Productivity.¹⁴ To avoid double counting intraday releases within the same category, we collapse the data to one observation per country-date-category triplet, so that the unit of observation is a unique announcement day for a given country and broad category. In addition, macroeconomic announcements for the EU count for each EU country in our sample.

We consider two specifications. In the first specification, we keep the top climate-

¹³To identify the top weeks in terms of climate attention, we calculate the growth rate of the weekly global CAI, then the top 10% of observations of this time series are considered as indicative of heightened climate attention. We only keep top weeks that are at least 5 weeks apart to avoid the confounding effects.

¹⁴Table C.1 summarizes the mapping of Bloomberg event categories into one of the four broad categories described above. The full event-level mapping (495 Bloomberg event types across our 25 countries) is available in the paper’s replication package.

attention day indicator as specified before and add an announcement-day control:

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + b_k \cdot \mathbf{1}_t^{Top} + c_k \cdot \mathbf{1}_{(i,j),t}^{ANN} + \varepsilon_{t,k}^{i,j},$$

where $\mathbf{1}_{(i,j),t}^{ANN}$ equals one if at least one of the two countries in the pair experiences a macroeconomic announcement on day t , and zero otherwise. We also consider a less restrictive definition under which the indicator equals one if both countries have macroeconomic announcements, and we report results under both definitions.

In the second specification, we redefine top climate-attention days by excluding days on which the announcement indicator is switched on,

$$\widetilde{\mathbf{1}}_t^{Top} = \mathbf{1}_t^{Top} \cdot (1 - \mathbf{1}_{(i,j),t}^{ANN}),$$

and run the following regression:

$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + b_k \cdot \widetilde{\mathbf{1}}_t^{Top} + \varepsilon_{t,k}^{i,j}.$$

This specification removes top climate-attention days that coincide with relevant macro announcements, and yet it retains 60% (93%) of the original selected days when we account for at least one country (both country) to have an announcement. Figure C.11 reports the results. The top two panels refer to the definition of $\mathbf{1}_{(i,j),t}^{ANN}$ requiring an announcement in at least one country in the pair, while the bottom two panels use the alternative definition requiring announcements in both countries. Across all possible cases, our main results continue to hold.

4.3 Equity returns

Proposition 2 suggests that brown stocks should be more exposed to climate news shocks than green stocks. In addition, this exposure differential should be more pronounced in countries in which sensitivity to climate news is higher (higher b in the model).

Similar to our currency analysis, we examine the behavior of average cumulative returns following days marked by a positive spike in our global index (CAI). Consistent with the model, we account for both firm-level “brown-ness”—measured by

emission intensity—and country-level sensitivity—captured by our estimated β_i coefficients in Table 2. Emission intensity (defined as tons of Scope 1 CO₂ equivalent emissions per unit of sales) is sourced from the Trucost dataset. Our sample includes 16,774 firms, totaling 23,305,563 firm-day observations. Their distribution across countries is stable and it is depicted in figure C.12 in the Appendix. Working with such a granular cross section, rather than aggregate country-level equity data, enhances our inference.

Specifically, we estimate the following regression:

$$r_{t,t+k}^{i,c} = \alpha_k^i + \zeta_{t,k}^{i,c} \cdot \mathbb{1}_t^{Top} + \epsilon_{t,k}^{i,c}, \quad (17)$$

$$\zeta_{t,k}^{i,c} = \gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k} \cdot \beta_c) \cdot E_{i,t_a}, \quad (18)$$

where c denotes the country in which the headquarter of the firm is located, i refers to one of our firms, β_c represents our equally weighted sensitivity, E_{i,t_a} is the emission intensity of firm i assessed the year prior to day t , and $\mathbb{1}_t^{Top}$ is a dummy variable that equals 1 if the day falls in the set of extreme climate news days. In equation (17), the left-hand side variable is defined as follows:

$$r_{t,t+k}^{i,c} = \sum_{s=0}^{k-1} r_{t+s,t+s+1}^{i,c} \frac{100 \cdot N^{top}}{k},$$

that is, it is averaged over the time horizon k , it is in percentage, and it is multiplied by the average number of top-days in a year.

Since emission intensity is very persistent at the annual frequency, the coefficient $\zeta_{t,k}^{i,c}$ is almost time-invariant in our dataset. Consequently, this composite coefficient primarily reflects heterogeneity across different countries and stocks. The literature has proxied the exposure of a firm to climate risk either by examining its emissions or considering geographical factors such as distance from the ocean or latitude.

As in the model, we hypothesize that a firm's equity returns' sensitivity to climate news reflects the interaction of its brownness and the degree of investor sensitivity to climate news within a specific country. This analysis is agnostic on whether the valuation adjustment is driven by uncertainty about physical damages or transition risk. Given the low correlation between our β_i s and vulnerability, we interpret these valuation adjustments as driven mostly by heterogeneous sensitivity to global climate

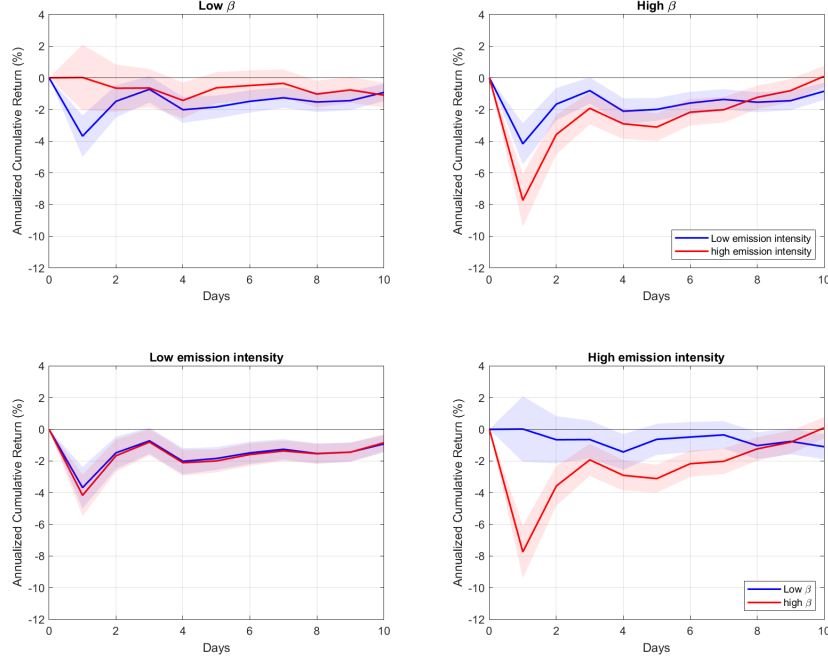


FIG. 12. USD EQUITY RETURNS AND INNOVATIONS TO WORLD CAI

Notes: This figure shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI. For more details, see the notes under figure 9. We depict the estimates of $\zeta_{t,k}^{i,c}$ as defined in equations (17)-(18) across different daily horizons, k . Standard errors are clustered at the industry $\times$ date and firm level. We follow the Fama-French 49 classification. The sample comprises a total of about 23 million firm-day observations. Daily returns are annualized using the average number of top climate days in a year. In the four panels, high (low) β refers to the top-decile (bottom-decile) of betas across countries in our sample, and high (low) emission intensity is using the top- and bottom-5% of emission-sorted firms in our sample. The shaded areas present the 90% confidence interval.

attention (discount rate channel).

According to our model, global climate news should play an important role mostly among firms with high emissions in highly sensitive countries ($\gamma_{2,k} < 0$). Our estimates confirm this prediction and are reported in table B.5. In figure 12, we depict the estimates of our composite parameter $\hat{\zeta}_{t,k}^{i,c}$ over different daily horizons, k .

In the top portion of the figure, we first sort countries according to their exposure β . The top-decile of the distribution of β s is 2.27 (Saudi Arabia). The bottom-decile is 0.197 (South Africa). We then select the values of carbon emission intensity for a representative green (brown) firm using the 5th (95th) percentile of the emission

intensity distribution, which corresponds to the median value within the top-decile (bottom-decile).¹⁵ Given these numbers, we can depict the implied estimated composite coefficient $\zeta_{t,k}^{i,c}$ and its associated standard error across different daily horizons, k .

The bottom two panels convey the same information as the top panels but from a different perspective. Specifically, the bottom left (right) panel presents the composite coefficients for a representative green (brown) firm in a low- β country and in a high- β country.

In the model, our brown assets are perfectly distinct from green technologies. In reality, however, firms typically embody a mix of both brown and green technologies, making it difficult to map the model’s predictions directly to firm-level data. As custom in the asset pricing literature, we address this concern by sorting firms according to their emissions. In the data, we focus on the *return spread* across firms that are brown technology-intensive (top-5% in the emissions distribution) and firms that are green technology-intensive (bottom-5% in the emissions distribution).

Our analysis yields several significant findings. Firstly, firms situated in countries with moderate sensitivity to climate attention experience a more moderate loss of value. This effect holds for both high- and low-emission firms. Conversely, firms in countries with high β values suffer a more severe loss of value, an effect that is particularly pronounced for high-emission firms. Across all these scenarios, the negative impact on cumulative returns is very persistent.

Changing the order of the sorting variables does not alter our conclusions. Specifically, the bottom two panels of figure 12 confirm that firms located in ‘naturally hedged’ countries (that is, countries with low β) are generally less affected by adverse climate attention shocks. However, when examining stocks with high emission intensity, the negative impact on valuation is more severe for firms in countries with a higher climate news sensitivity, β . The economic magnitude is comparable to related tweet-based asset-pricing evidence. For example, Bianchi et al. (2024) show that congressional tweets generate economically meaningful stock-price reactions: firms receiving highly critical tweets earn abnormal returns of about -12% annually. ... By comparison, the annualized response of high-emission firms in high- β countries in

¹⁵The 5th (95th) percentile of the log emission intensity in our sample is 0.43 (6.88). The thresholds remain consistent both across and within countries because the country-level distribution of emission intensities is stable. See figure C.12(a) in the Appendix for more details.

figure 12 is about -8% .¹⁶

Appendix C.2 reports the behavior of cumulative returns over the five trading days preceding a top climate-attention day. Returns are positive and approximately flat for both high- and low-emission firms, as the greenium would suggest, and display no negative pre-event drift.

We also examine responses under a coarser sorting, reported in Appendix Figure C.16. We split countries into above- and below-median climate betas and, within each group, we sort firms every year according to the median of their lagged emission intensity. The country-level ordering is unchanged: firms located in high- β countries experience the larger loss of value. The differential across emission groups attenuates, consistent with our industry-level evidence, which attributes most of the variation to differences across industries rather than within them.

Industry-level results. We decompose emission intensity into cross-industry and within-industry components. First, we compute each firm’s average emission intensity over time:

$$\bar{E}_i = \frac{1}{T_i} \sum_t E_{i,t}.$$

We then obtain the within-industry component of emissions, $\tilde{E}_i$, as the residual of a cross-sectional regression with industry fixed effects, $E_{s(i)}$:

$$\bar{E}_i = E_{s(i)} + \tilde{E}_i.$$

In this decomposition, $s(i)$ denotes the industry of firm i (Fama-French 49 industry classifications), $E_{s(i)}$ captures cross-industry variation in emission intensity, while $\tilde{E}_i$ captures firm-specific emission variation within industries. Figure C.13 in the appendix shows substantial variation in the distribution of emission intensity across industries.

Next, we replace E_{i,t_a} in equation (18) with either $E_{s(i)}$ or $\tilde{E}_i$ and visualize the results in figure C.14 in the Appendix. Our main results are driven by industry variation. The return response to climate news therefore reflects the repricing of sectors

¹⁶Our results about both the exchange rate and equity reactions continue to hold when we control for one-day lagged realizations.

exposed to transition policy, rather than a discrimination between cleaner and dirtier firms operating within the same sector. This is consistent with the transition-risk channel documented above: regulation, carbon pricing, and policy commitments bite at the sector level, and investors appear to price them accordingly.¹⁷

Type of news. Following our analysis of currency markets, we examine whether these results are primarily driven by top-attention days in which news focuses on either physical or transition risk. Specifically, we estimate the following specification:

$$\begin{aligned} r_{t,t+k}^{i,c} &= \alpha_k^i + \zeta_{t,k}^{i,c} \cdot 1_t^{Top} + \epsilon_{t,k}^{i,c}, \\ \zeta_{t,k}^{i,c} &= (\gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k} \cdot \beta_c) \cdot E_{i,ta}) + (\kappa_{0,k} + (\kappa_{1,k} + \kappa_{2,k} \cdot \beta_c) \cdot E_{i,ta}) \cdot S_t, \end{aligned}$$

where S_t represents the share of tweets discussing transition risks on day t , normalized between 0 and 1. Figure C.15 in the Appendix presents our results. The responses are constructed using the composite coefficient $\zeta_{t,k}^{i,c}$, with $S_t = 0.50$ in the left panel and $S_t = 1$ in the right panel. For a median top-attention day, this specification reproduces the results reported in Figure 12. When news is primarily about transition risk, we observe more pronounced depreciations across all panels. As with currency markets, equity depreciations are most pronounced on days when the news is concentrated on transition risk specifically. When we focus exclusively on the share of transition-related tweets with negative sentiment, the results are amplified by a factor of two.

Additional validation. We provide additional validation of our findings by documenting that brown firms explicitly discuss transition-related risks after spikes in climate news. We use firm-quarter climate-risk exposure measures from earnings calls constructed by Sautner et al. (2023b). These measures capture several dimensions of firms' climate-related discussions, including climate change, opportunity, regulatory, and physical-risk topics.

Because these nonnegative measures comprise many zeros, we estimate the regressions using Poisson pseudo-maximum likelihood (PPML) to accommodate zero

¹⁷Among the Fama–French 49 industries, Coal, Utilities, Agriculture, and Petroleum and Natural Gas have the highest average emission intensity in our sample and therefore constitute the high-emission industry component driving the cross-industry result.

outcomes of the dependent variable. Specifically, we estimate the following equation:

$$E[ClimateDiscussion_{i,q}|X_{i,q}] = \exp(\alpha_i + \beta_1 TopShare_q + \beta_2 Emission_{i,q-4} + \beta_3 TopShare_q \times Emission_{i,q-4}),$$

where $ClimateDiscussion_{i,q}$ is a firm-quarter measure of climate-related discussion from earnings calls, $TopShare_q$ is the share of global top climate-days in quarter q , defined as the number of top climate-attention days divided by the number of days in the quarter, and $Emission_{i,q-4}$ is measured by one-year-lagged log emission intensity. All specifications include firm fixed effects, and standard errors are two-way clustered by firm and calendar quarter.¹⁸ The coefficient of interest is β_3 . A positive estimate indicates that firms with higher emission intensity discuss climate risk more in quarters containing more global top climate days. All specifications include firm fixed effects, and standard errors are two-way clustered by firm and calendar quarter.

Table 3 reports the results for three dependent variables most directly related to transition concerns: (i) climate change risk, (ii) regulatory exposure, and (iii) regulatory risk. The interaction between $TopShare_q$ and lagged log emission intensity is positive and statistically significant in all three columns. Firms with higher emission intensity therefore discuss climate and regulatory risks more extensively in quarters containing more global climate-news shocks.

Robustness. In Appendix C, we show that our results are robust across various methodological changes. Panel (a) in Figure C.17 uses GDP-weighted CAI, while Panel (b) employs Twitter Volume-weighted CAI. Both panels confirm the robustness of our findings as documented in Figure 12. Furthermore, Figure C.18 compares the response of equity returns using thresholds of 10% and 20% to identify spikes in CAI, respectively. This figure documents that our results remain qualitatively unchanged regardless of the chosen threshold.

Figure C.19 provides supplementary analyses to validate our findings. In Panel (a) we cluster standard errors at the country×industry×date and firm levels, while in Panel (b) we use residual β s obtained by regressing our exposure coefficients on the

¹⁸We have firms with zero outcome in every quarter. The observations associated with these firms are not used in our regressions.

TABLE 3. FIRMS' CLIMATE-RISK DISCUSSION & CLIMATE TOP DAYS

	(1) Climate Change Risk	(2) Regulatory Exposure	(3) Regulatory Risk
$TopShare_q$	-3.443*** (-4.73)	-6.761*** (-2.88)	-8.480*** (-2.74)
$Emission_{i,q-4}$	-0.0550 (-1.46)	-0.0847 (-1.49)	-0.0252 (-0.16)
$TopShare_q \times Emission_{i,q-4}$	0.252** (2.43)	0.447** (2.02)	1.160** (2.35)
Constant	0.309** (2.22)	0.903*** (3.49)	0.569 (0.75)
Observations	54,525	53,752	8,571

Notes: This table reports PPML estimates of firms' climate-risk discussion during earnings calls. The dependent variables are firm-quarter measures from [Sautner et al. \(2023b\)](#): climate change risk, regulatory exposure, and regulatory risk. $TopShare_q$ is the share of global climate top days in quarter q , defined as the number of top climate-attention days divided by the number of days in the quarter. $Emission_{i,q-4}$ is log emission intensity measured one year before the earnings-call quarter. The coefficient on $TopShare_q \times Emission_{i,q-4}$ tests whether brown firms increase climate-risk discussion more after quarters with more global climate top days. All specifications include firm fixed effects, and standard errors are two-way clustered by firm and calendar quarter. t -statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

University of Notre Dame's vulnerability index. These results corroborate our main conclusions.

Most importantly, our results are not solely driven by the exchange rate adjustment. As shown in figure [C.20](#), our results hold also when we focus on returns expressed in local currency.

Similarly to what we have done for currencies, we isolate the role of pure climate days from other macroeconomic announcement days. As done with currencies, we consider two specifications. In the first specification, we keep the top climate-attention day indicator as in our benchmark specification and add a country-specific announcement-day control, $1_{c,t}^{ANN}$, which is equal to 1 when t is an announcement day in country c :

$$r_{t,t+k}^{i,c} = \alpha_k^i + \zeta_{t,k}^{i,c} \cdot \mathbf{1}_t^{Top} + d_k \cdot \mathbf{1}_{c,t}^{ANN} + \varepsilon_{t,k}^{i,c}.$$

As in the main equity specification, the return response to climate news loads on

firm-level emissions and country-level climate-news exposure:

$$\zeta_{t,k}^{i,c} = \gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k}\beta_c) \cdot E_{i,t_a},$$

where β_c is country c 's exposure to the global CAI, and E_{i,t_a} is firm i 's emission intensity measured in the year prior to day t .

In the second specification, we redefine top climate-attention days by excluding global announcement days. Since our top climate-attention indicator is itself constructed from the global CAI, we exclude on a global rather than a country-specific basis. For each trading day t , we compute the share of sample countries with at least one scheduled macroeconomic release,

$$\text{Share}_t = \frac{1}{N} \sum_{c=1}^N \mathbf{1}_{c,t}^{ANN},$$

where $N = 25$. We define a *global* macro announcement day as any trading day on which Share_t falls in the top decile of its empirical distribution. The threshold is computed over the trading days between October 2014 and December 2022 with at least one scheduled release, and is then applied to every trading day in the sample, including those with no scheduled release. The dummy variable $\mathbf{1}_{glob,t}^{ANN}$ is equal to 1 on global announcement days, and 0 otherwise.

We remove top climate days that coincide with global macro announcement days by creating the following redefined top climate-day indicator,

$$\tilde{\mathbf{1}}_t^{Top} = \mathbf{1}_t^{Top} \cdot (1 - \mathbf{1}_{glob,t}^{ANN}),$$

and run the standard regression

$$r_{t,t+k}^{i,c} = \alpha_k^i + \tilde{\zeta}_{t,k}^{i,c} \cdot \tilde{\mathbf{1}}_t^{Top} + \varepsilon_{t,k}^{i,c},$$

where the exposure coefficient is defined analogously as

$$\tilde{\zeta}_{t,k}^{i,c} = \gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k}\beta_c) \cdot E_{i,t_a}.$$

Figure C.21 confirms that our results also hold after accounting for other macroeco-

nomic announcements.

5 International Flows

In this section, we test the prediction of our Proposition 3, that is, the current account of countries with higher climate sensitivity should decline when adverse sentiment news materialize. In the data, we are interested in estimating the following system of equations:

$$\Delta CAI_{i,t} = \beta_i \cdot \overline{\Delta CAI}_t + u_{i,t} \quad (19)$$

$$\begin{aligned} \Delta \left(\frac{NX_{i,t}}{GDP_{i,t}} \right) - \Delta \left(\frac{NX_{j,t}}{GDP_{j,t}} \right) = & \Gamma \cdot (\beta_i - \beta_j) \overline{\Delta CAI}_t + \dots \\ & + \theta' \cdot (control_{i,t} - control_{j,t}) + \epsilon_{ij,t}, \end{aligned} \quad (20)$$

where $\frac{NX_{i,t}}{GDP_{i,t}}$ is the net export over GDP of country i at time t . We add the first difference to make sure that all variables are stationary. $\Delta CAI_{i,t}$ denotes the growth rate of the climate attention index for country i and quarter t , $\overline{\Delta CAI}_t$ is the cross-sectional average of ΔCAI_t at date t , i.e., the global CAI index.¹⁹ The term *control* is a vector of control variables including the change in the industrial production index (ΔIPI), to control for country-specific productivity shocks, and the share of Twitter volume on days when the CAI is at the bottom 5%, to control for other events that affect Twitter activities.

We provide results based on the cases in which $\overline{\Delta CAI}_t$ is constructed in one of three ways: (i) by equally weighting all countries, (ii) by weighting each country proportionally to its share of global GDP, and (iii) by weighting each country proportionally to its share of Twitter volumes. Our estimation procedure is detailed in the Appendix (see section C.3).

Given our interest on the role of global climate news shocks on the dynamics of international trade, we focus on the estimated parameter Γ in equation (20). Accord-

¹⁹In order to eliminate the effect of the Covid-19 and the beginning of the war in Ukraine, we have regressed $\Delta CAI_{i,t}$ on two dummies of 2020Q2 and 2022Q1 for each country, and obtain the residual for our analysis.

ing to our model, this coefficient should be negative. We estimate the parameters of interest using an expanding window to show whether the estimated coefficient is stable over different sample periods. Specifically, we fix our starting period at 2015:Q1 (where our sample begins), and expand the end of sample from 2016:Q4 to 2022:Q4. The estimated Γ coefficient along with its 90% confidence interval are presented in figure 13.

Our estimates support the prediction of our model, as they suggest that an adverse global climate shock produces an outflow of resources away from countries with a lower exposure. This result is confirmed across different ways to aggregate our country-level CAI in order to form a global index. In addition, our result is driven mainly by country pairs in which there is at least one developed economy. This is broadly consistent with the spirit of our model since we assume that risk sharing is implemented with frictionless trade. We show that these effects on the current account can persist for six months (see Figure C.24 in the Appendix).

Robustness. We examine whether our results on the net-export reactions are robust to additional controls. Because the net-export analysis is conducted at the quarterly frequency, the estimates may be more sensitive to common macroeconomic conditions than the high-frequency exchange-rate and equity-return results. We therefore re-estimate the expanding-window GMM specification for the full sample under two additional sets of controls.

First, we add year-quarter fixed effects to absorb common global shocks that may affect all country pairs in the same quarter, such as global trade-cycle fluctuations, pandemic-related disruptions, or geopolitical shocks. Second, we augment the baseline specification with additional macroeconomic controls, namely quarterly real GDP growth and CPI inflation. In both exercises, we keep the baseline controls for industrial production growth and non-climate Twitter activity. We report the results separately for the equally weighted, GDP-weighted, and Twitter-volume-weighted global CAI measures.

We plot the expanding-window estimates of the coefficient on the global climate-news component in Figure C.25 in the Appendix. The estimates remain broadly negative across the alternative weighting schemes and robustness specifications.

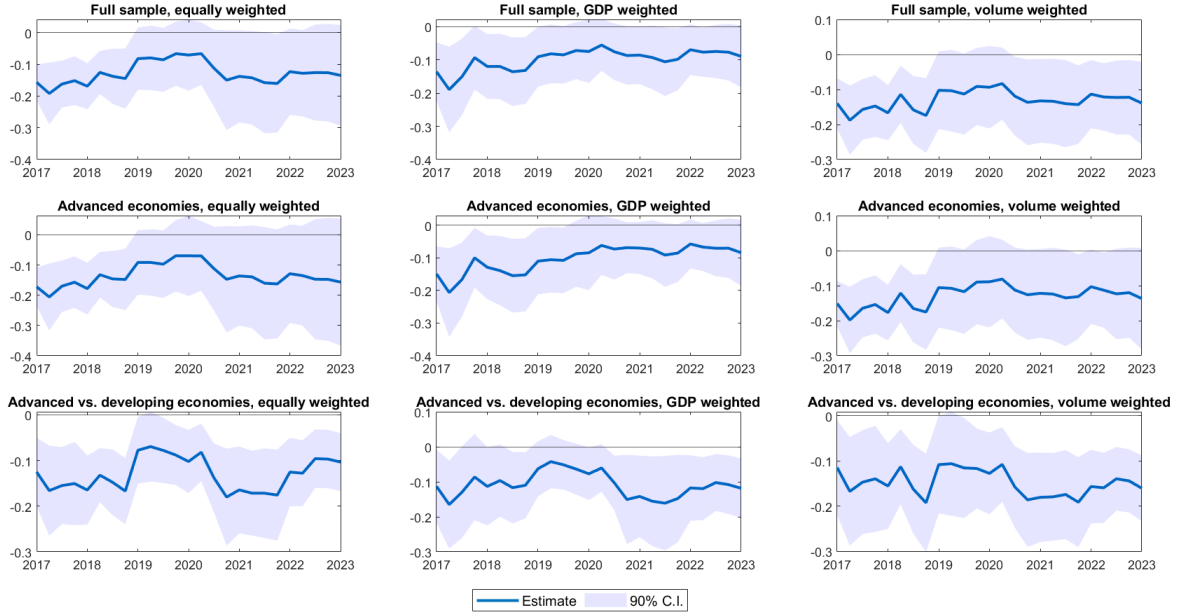


FIG. 13. GMM ESTIMATION OF Γ .

Notes: The figure shows estimates of Γ (defined in equation (20)) using expanding time windows. Each window starts in 2015:Q1 and ends on the date reported on the x-axis. The parameter Γ is estimated using the GMM described in section 2. Panels in different columns refer to results obtained from different ways to aggregate country-level CAI indices in order to form a global component. Panels in different rows refer to results obtained from different groups of country-pairs. Advanced economies are defined according to the IMF. Standard errors are HAC-adjusted. The shaded areas present the 90% confidence interval.

6 Concluding Remarks

Our research significantly contributes to our understanding of the impact of global climate attention on economic and financial outcomes. The availability of high frequency data is particularly valuable as it enables a more direct analysis of how climate news shocks reverberate throughout global financial markets and influence currencies on a global scale. The establishment of a clear connection between climate attention shocks and bilateral trade among nations represents a noteworthy advancement for both the macro-finance literature and policymakers.

Moreover, the economic model we introduce, along with the climate attention index we propose, open up various avenues for future research. For instance, our economic theory implies that heightened global news attention to climate issues may influence decisions regarding investments in green technology. Furthermore, the climate

attention index we have developed can be leveraged to explore the relationship between climate-related news and returns on a wide range of assets above and beyond stocks, both within individual countries and across international borders. These potential research directions hold promise for gaining deeper insights into the complex interplay between climate awareness and economic and financial dynamics.

Establishing a connection between attention to climate news and currency movements is of paramount importance in the current global landscape. As climate change becomes an increasingly pressing concern, understanding how shifts in climate attention impact financial markets and trade can inform both economic policies and investment strategies. It provides a valuable tool for policymakers to anticipate and respond to economic implications of climate-related events and offers investors insights into the potential risks and opportunities associated with climate awareness. By shedding light on these intricate connections, our research serves not only the academic community but also contributes to the broader efforts to address the multifaceted challenges posed by climate change on a global scale.

Importantly, while our paper focuses on the response of current accounts to climate news shocks, future research should also explore the implications for international capital and financial accounts. In the spirit of the literature on global imbalances and the international transmission of risk (see [Jiang et al. 2024b](#), [Dahlquist et al. 2025](#)), an important next step is to examine how climate-related shocks affect cross-border financial positions and to identify the key financial centers that intermediate global climate risk. Such work would further our understanding of how climate risk is hedged and redistributed through international financial markets, complementing our findings on trade and currency responses.

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Internet Appendix

Appendix A. Model

In this section, we present both the full description and key derivations of our model. For a more detailed analysis of international models of risk-sharing with recursive preferences and news shocks, see [Colacito et al. \(2018\)](#).

A.1 Setup

We generalize the two-period setting presented in [Colacito et al. \(2018\)](#) to include: (i) green investment, (ii) disutility from climate shocks, and (iii) climate news shocks. We assume that the intertemporal elasticity of substitution (henceforth IES) is equal to 1 and the preferences are defined as

$$u_t = (1 - \beta) \log \left(\tilde{C}_t \cdot \mathcal{G}_t \right) + \frac{\beta}{1 - \gamma} \log E_t [\exp \{ (1 - \gamma) u_{t+1} \}], \quad t = \{0, 1, 2\},$$

where $\mathcal{G}_t$ captures climate news shocks, and $\tilde{C}_t$ includes both a standard consumption bundle C_t and an endogenous disruption term D_t : $\tilde{C}_t = C_t \cdot D_t^{-1}$. The sentiment process evolves as $\mathcal{G}_t = e^{-b \cdot g_t}$, where g_t is a zero-mean normally distributed global climate news shock. In what follows, we derive all equilibrium conditions using an equivalent formulation of the model:

$$u_t = (1 - \beta) \log \left(C_t \tilde{D}_t^{-a} \right) + \frac{\beta}{1 - \gamma} \log E_t [\exp \{ u_{t+1} (1 - \gamma) \}],$$

where $\tilde{D}_t = (D_t / \mathcal{G}_t)^{1/a} = D_t^{1/a} e^{bg_t/a}$. This definition is a same-date algebraic reparameterization: it combines D_t and $\mathcal{G}_t$ without changing the timing of either object. At time $t = 1$, two news shocks, $\theta \sim N(0, \sigma_\theta^2)$ and $\tilde{g} \sim N(0, \sigma_g^2)$, occur. The shock θ determines the productivity of capital at $t = 2$, whereas $\tilde{g} > 0$ is an adverse climate news shock. At time $t = 1$, uncertainty is resolved and utilities of the home and foreign countries are

$$u_1 = (1 - \beta) \log \left(C_1 \tilde{D}_1^{-a} \right) + \tilde{\beta} \log \left(C_2 \tilde{D}_2^{-a} \right), \quad u_1^* = (1 - \beta) \log \left(C_1^* \tilde{D}_1^{*-a} \right) + \tilde{\beta} \log \left(C_2^* \tilde{D}_2^{*-a} \right), \quad (A.1)$$

where $\tilde{\beta} = \beta(1 - \beta)$ and

$$C_t = X_t^\lambda Y_t^{(1-\lambda)}, \quad C_t^* = X_t^{*(1-\lambda)} Y_t^{*\lambda}, \quad \forall t = \{1, 2\}$$

denote the consumption bundles, the parameter $a \in (0, 1)$ captures the disutility from climate-related damage, and $\tilde{D}_t$ is the amount of damage times the climate news shock. The damage

depends on both aggregate output and green investment. At time $t = 1$, $\tilde{D}_1$ and $\tilde{D}_1^*$ are given, while $\tilde{D}_2$ and $\tilde{D}_2^*$ denote the composite climate-related damages at time $t = 2$ in the home and foreign countries, respectively. Specifically:

$$D_2 = \left[\left(e^\theta B / I_{g,1} \right)^{\lambda_g} \left(e^{-\theta} B^* / I_{g,1}^* \right)^{1-\lambda_g} \right]^a, \quad D_2^* = \left[\left(e^{-\theta} B^* / I_{g,1}^* \right)^{\lambda_g} \left(e^\theta B / I_{g,1} \right)^{1-\lambda_g} \right]^a,$$

$$\tilde{D}_2 = e^{b\tilde{g}} D_2^{1/a} = e^{b\tilde{g}} \left(e^\theta B / I_{g,1} \right)^{\lambda_g} \left(e^{-\theta} B^* / I_{g,1}^* \right)^{1-\lambda_g}, \quad \tilde{D}_2^* = e^{\tilde{g}} (D_2^*)^{1/a} = e^{\tilde{g}} \left(e^{-\theta} B^* / I_{g,1}^* \right)^{\lambda_g} \left(e^\theta B / I_{g,1} \right)^{1-\lambda_g},$$

where b captures the sensitivity of the home country to global climate news (the exposure of the foreign country is normalized to unity), λ_g captures the home bias in climate-related damage; B and B^* are aggregates of time $t = 1$ investment for the home and foreign countries and are defined as

$$B = I_{x,1}^{\lambda_I} I_{x,1}^{*1-\lambda_I}, \quad B^* = I_{y,1}^{1-\lambda_I} I_{y,1}^{*\lambda_I}. \quad (\text{A.2})$$

At time $t = 2$, the Appendix parameterizes the transformed disruption as $\tilde{D}_2 = D_2^{1/a} e^{b\tilde{g}}$. This continuation-utility representation differs from the same-date reparameterization above only in the normalization and timing of the climate news shock. In the Appendix, the climate-news contribution to u_1 is $-\tilde{\beta}ab\tilde{g} = -\beta(1-\beta)ab\tilde{g}$, whereas in the main text the time-1 shock contributes $-(1-\beta)bg$. The two representations are therefore identical when $g = \beta a\tilde{g}$, which implies $g \sim N(0, (\beta a\sigma_g)^2)$. The factor a comes from the disruption reparameterization, while the factor β comes from moving the shock from time-2 continuation utility to time-1 utility. Time $t = 2$ output in the two countries is determined as $e^\theta B$ and $e^{-\theta} B^*$. The economy-wide resource constraints are

$$\begin{aligned} 1 &= X_1 + X_1^* + I_{x,1} + I_{y,1} + I_{g,1}, & 1 &= Y_1 + Y_1^* + I_{y,1}^* + I_{x,1}^* + I_{g,1}^* \\ e^\theta B &= X_2 + X_2^*, & e^{-\theta} B^* &= Y_2 + Y_2^*. \end{aligned}$$

Throughout this Appendix, we retain the following assumption on the domain of the parameters.

Assumption A1. Assume that there is home bias in consumption, investment, and climate-related damage, that is, $0.5 < \lambda, \lambda_I, \lambda_g < 1$. In addition, assume $\frac{1-a}{2} < \hat{\lambda} = \lambda - a\lambda_g < 1 - a$ (home bias in the consumption-damage composite).

A.2 Social planner problem and pseudo-Pareto weights.

The social planner chooses allocations of consumption goods $\{X_t, X_t^*, Y_t, Y_t^*\}_{t=1,2}$, and allocations of investments $I_{x,1}, I_{y,1}, I_{g,1}, I_{y,1}^*, I_{x,1}^*, I_{g,1}^*$ to maximize

$$W = \mu_0 E_0[u_1] + (1 - \mu_0) E_0[u_1^*],$$

where μ_0 and $(1 - \mu_0)$ denote the date $t = 0$ Pareto weights attached to the two agents. Define $\tilde{S}_0 := \frac{\mu_0}{1 - \mu_0}$ and let $\tilde{S}_1$ be the ratio of pseudo-Pareto weights after the news shocks are realized. Colacito et al. (2018) show that:

$$\tilde{S}_1 = \tilde{S}_0 \exp \{(u_1 - u_1^*)(1 - \gamma)\}.$$

If we initialize the economy under the assumption that the distribution of wealth at date $t = 0$ is identical across the two countries, then

$$\tilde{s}_1 := \log \tilde{S}_1 = (u_1 - u_1^*)(1 - \gamma). \quad (\text{A.3})$$

In what follows, we document that the solutions for u_1 and u_1^* are linear in the time $t = 1$ shocks, θ and $\tilde{g}$. It follows that the solution for $\tilde{s}_1$ is also linear:

$$\tilde{s}_1 = \lambda_s^\theta \cdot \theta + \lambda_s^g \cdot \tilde{g}, \quad (\text{A.4})$$

where the coefficients λ_s^θ and λ_s^g are determined in equilibrium. Finally, since all uncertainty is resolved at time $t = 1$, we have $\tilde{S}_2 = \tilde{S}_1$.

First-order conditions. Let W_0 and W_0^* be the time-0 utility functions of the home and foreign agents. Let μ_0 be the Pareto weight assigned to the home country. The associated Lagrangian problem can be stated as follows:

$$\begin{aligned} \mathcal{W} = & \mu_0 W_0 + (1 - \mu_0) W_0^* + \dots \\ & + \lambda_{1t} \left[F(I_{x,t-1}, I_{x,t-1}^*) - X_t - X_t^* - I_{x,t} - I_{y,t} - I_{g,t} \right] \\ & + \lambda_{2t} \left[F^*(I_{y,t-1}, I_{y,t-1}^*) - Y_t - Y_t^* - I_{y,t}^* - I_{x,t}^* - I_{g,t}^* \right] \\ & + \lambda_{3t} \left[O(I_{g,t-1}, I_{g,t-1}^*) - O_t \right] + \lambda_{4t} \left[O^*(I_{g,t-1}, I_{g,t-1}^*) - O_t^* \right] \\ & + \lambda_{5t} \left[f(F_t, F_t^*) - E_t \right] + \lambda_{6t} \left[f^*(F_t, F_t^*) - E_t^* \right] \dots \end{aligned} \quad (\text{A.5})$$

where $t = 1, 2$, and all variables indexed by $t = 0$ are pre-determined. In addition, at time $t = 2$ the economy ends and hence at time $t = 2$ all investment-related variables are optimally set to zero. The first-order condition with respect to $I_{x,1}$ is

$$\lambda_{12} \frac{\partial F}{\partial I_{x,1}} - \lambda_{11} + \lambda_{52} \frac{\partial f}{\partial F} \frac{\partial F}{\partial I_{x,1}} + \lambda_{62} \frac{\partial f^*}{\partial F} \frac{\partial F}{\partial I_{x,1}} = 0,$$

and by rearranging it, we get:

$$\frac{\lambda_{12}}{\lambda_{11}} \frac{\partial F}{\partial I_{x,1}} \left(1 + \frac{\lambda_{52}}{\lambda_{12}} \frac{\partial f}{\partial F} + \frac{\lambda_{62}}{\lambda_{12}} \frac{\partial f^*}{\partial F} \right) = 1.$$

Using the stochastic discount factor between time 1 and 2,

$$M_{1|2} := \frac{\lambda_{12}}{\lambda_{11}} = \frac{1}{R_{1|2}^f},$$

we obtain the familiar no-arbitrage equation

$$1 = M_{1|2} \cdot \frac{\partial F}{\partial I_{x,1}} \left(1 - a \frac{X_2}{F_2} \frac{\lambda_g}{\lambda} - a \frac{X_2^*}{F_2} \frac{1 - \lambda_g}{1 - \lambda} \right).$$

The FOCs with respect to E_2 and X_2 are:

$$-\lambda_{52} + \mu_0 \frac{\partial W_0}{\partial W_2} \frac{\partial W_2}{\partial E_2} = 0,$$

$$-\lambda_{12} + \mu_0 \frac{\partial W_0}{\partial W_2} \frac{\partial W_2}{\partial c_2} \frac{\partial c_2}{\partial X_2} = 0,$$

respectively. By taking the appropriate ratios:

$$\frac{\lambda_{52}}{\lambda_{12}} = \frac{\partial W_2 / \partial E_2}{\partial W_2 / \partial c_2 \cdot \partial c_2 / \partial X_2} = -\frac{1}{\lambda} \frac{X_2}{E_2},$$

This is the shadow value of emissions expressed in units of good X . Similarly, for foreign emissions we have:

$$-\lambda_{62} + (1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial E_2^*} = 0,$$

and

$$\frac{\lambda_{62}}{\lambda_{12}} = \frac{(1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial E_2^*}}{\mu_0 \frac{\partial W_0}{\partial W_2} \frac{\partial W_2}{\partial c_2} \frac{\partial c_2}{\partial x_2}} = -\frac{1}{\lambda} \frac{X_2}{E_2^*} \frac{1}{\widetilde{S}_2}.$$

This is the shadow value of foreign emissions in units of good X . Considering the optimality with respect to the allocation of good Y ,

$$-\lambda_{22} + (1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial c_2^*} \frac{\partial c_2^*}{\partial Y_2^*} = 0,$$

we get

$$\frac{\lambda_{62}}{\lambda_{22}} = \frac{\partial W_2^* / \partial E_2^*}{\partial W_2^* / \partial c_2^* \cdot \partial c_2^* / \partial Y_2^*} = -\frac{1}{\lambda} \frac{Y_2^*}{E_2^*}.$$

This is the shadow value of foreign emissions in units of good Y , and this expression is symmetric to $\frac{\lambda_{52}}{\lambda_{12}}$. Since the relative price of the two goods must satisfy both

$$\frac{\lambda_{2t}}{\lambda_{1t}} = \frac{(1 - \mu_0) \frac{\partial W_0^*}{\partial W_t^*} \frac{\partial W_t^*}{\partial c_t^*} \frac{\partial c_t^*}{\partial Y_t^*}}{\mu_0 \frac{\partial W_0^*}{\partial W_t^*} \frac{\partial W_t^*}{\partial c_t^*} \frac{\partial c_t^*}{\partial X_t^*}} = \frac{X_t}{Y_t^*} \frac{1}{S_t}, \quad t = 1, 2,$$

and

$$\frac{\lambda_{2t}}{\lambda_{1t}} = \frac{(1 - \lambda) \frac{1}{Y_t}}{\lambda \frac{1}{X_t}} = \frac{1 - \lambda}{\lambda} \frac{X_t}{Y_t} = \frac{\lambda}{1 - \lambda} \frac{X_t^*}{Y_t^*}, \quad t = 1, 2,$$

the cost of foreign emissions in units of good X can equivalently be expressed as:

$$\frac{\lambda_{62}}{\lambda_{22}} \cdot \frac{\lambda_{22}}{\lambda_{12}} = -\frac{1}{\lambda} \frac{Y_2^*}{E_2^*} \cdot \frac{X_2^*}{1 - \lambda} \cdot \frac{\lambda}{Y_2^*} = -\frac{1}{1 - \lambda} \frac{X_2^*}{E_2^*}.$$

$$\frac{\partial f^*}{\partial F_2} = a(1 - \lambda_g) \frac{f^*}{F_2} = a(1 - \lambda_g) \frac{E_2^*}{F_2}.$$

Turning our attention to the green technology, the first-order condition with respect to green investment, $I_{g,1}$, is

$$\lambda_{32} \frac{\partial O_2}{\partial I_{g,1}} + \lambda_{42} \frac{\partial O_2^*}{\partial I_{g,1}} - \lambda_{11} = 0.$$

Rearranging,

$$\frac{\lambda_{12}}{\lambda_{11}} \left(\frac{\lambda_{32}}{\lambda_{12}} \frac{\partial O_2}{\partial I_{g,1}} + \frac{\lambda_{42}}{\lambda_{12}} \frac{\partial O_2^*}{\partial I_{g,1}} \right) = 1.$$

Note that

$$-\lambda_{32} + \mu_0 \frac{\partial W_0}{\partial W_2} \frac{\partial W_2}{\partial O_2} = 0 \quad \Rightarrow \quad \boxed{\lambda_{32} = \mu_0 \frac{\partial W_0}{\partial W_2} \frac{1}{O_2}},$$

and

$$-\lambda_{42} + (1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial O_2^*} = 0 \quad \Rightarrow \quad \boxed{\lambda_{42} = (1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{1}{O_2^*}}.$$

Given these results, we are able to compute the relative price of domestic offsets expressed in units of good X ,

$$\frac{\lambda_{32}}{\lambda_{12}} = \frac{\mu_0 \frac{\partial W_0}{\partial W_2} \frac{1}{O_2}}{\mu_0 \frac{\partial W_0}{\partial W_2} \frac{\partial W_2}{\partial c_2} \frac{\partial c_2}{\partial X_2}} = \frac{1}{\lambda} \frac{X_2}{O_2},$$

the relative prices of foreign offsets expressed in units of good X ,

$$\frac{\lambda_{42}}{\lambda_{12}} = \frac{\lambda_{42}}{\lambda_{22}} \frac{\lambda_{22}}{\lambda_{12}} = \frac{1}{O_2^*} \frac{Y_2^*}{\lambda} \cdot \frac{\lambda}{1 - \lambda} \frac{X_2^*}{Y_2^*} = \frac{1}{1 - \lambda} \frac{X_2^*}{O_2^*}.$$

and in units of good Y ,

$$\frac{\lambda_{42}}{\lambda_{22}} = \frac{(1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial O_2^*}}{(1 - \mu_0) \frac{\partial W_0^*}{\partial W_2^*} \frac{\partial W_2^*}{\partial c_2^*} \frac{\partial c_2^*}{\partial Y_2^*}} = \frac{1}{\lambda} \frac{Y_2^*}{O_2^*}.$$

Using the stochastic discount factor definition and the previous results, we get:

$$M_{1|2} \left(a \frac{\lambda_g}{\lambda} \frac{X_2}{I_{g,1}} + a \frac{1 - \lambda_g}{1 - \lambda} \frac{X_2^*}{I_{g,1}} \right) = 1.$$

Recall that

$$\begin{aligned} \mu_0 \frac{\partial W_0}{\partial W_t} \frac{\partial W_t}{\partial c_t} \frac{\partial c_t}{\partial X_t} &= \lambda_{1t}, \quad t = 1, 2 \\ (1 - \mu_0) \frac{\partial W_0^*}{\partial W_t^*} \frac{\partial W_t^*}{\partial c_t^*} \frac{\partial c_t^*}{\partial X_t^*} &= \lambda_{1t}. \end{aligned}$$

and hence

$$\tilde{S}_t \lambda \frac{1}{X_t} = (1 - \lambda) \frac{1}{X_t^*} \implies \frac{X_t}{\tilde{S}_t \lambda} = \frac{X_t^*}{1 - \lambda}, \quad t = 1, 2$$

$$\boxed{X_t = X_t^* \frac{\tilde{S}_t \lambda}{1 - \lambda}} \quad \text{and} \quad \boxed{X_t^* = X_t \frac{1 - \lambda}{\tilde{S}_t \lambda}}, \quad t = 1, 2.$$

Combining terms, the first order condition with respect to green investment becomes:

$$M_{1|2} \left(a \frac{X_2}{\lambda I_{g,1}} \left[\lambda_g + \frac{1 - \lambda_g}{\tilde{S}_2} \right] \right) = 1,$$

or, equivalently, in terms of X_2^*

$$M_{1|2} \left(a \left[\frac{\lambda_g \tilde{S}_2}{1 - \lambda} + \frac{1 - \lambda_g}{1 - \lambda} \right] \frac{X_2^*}{I_{g,1}} \right) = 1,$$

which simplifies to

$$M_{1|2} a \left(\frac{\lambda_g \tilde{S}_2 + (1 - \lambda_g)}{1 - \lambda} \right) \frac{X_2^*}{I_{g,1}} = 1.$$

A.3 Solution of the Model: Backward Induction

Since the model features a finite horizon, we begin the analysis by solving the terminal period and proceed recursively backward.

Time 2. The optimization problem at time $t = 2$ is

$$\begin{aligned} \max_{X_2, X_2^*, Y_2, Y_2^*} \quad & \tilde{S}_2 u_2 + u_2^* \\ \text{s.t.} \quad & e^\theta B = X_2 + X_2^*, \quad e^{-\theta} B^* = Y_2 + Y_2^*. \end{aligned}$$

Note that the objective function can be written as

$$\tilde{S}_2 \left(\lambda \log X_2 + (1 - \lambda) \log Y_2 - a \log \tilde{D}_2 \right) + \left((1 - \lambda) \log X_2^* + \lambda \log Y_2^* - a \log \tilde{D}_2^* \right).$$

The optimal condition is

$$\frac{\tilde{S}_2 \lambda}{X_2} = \frac{1 - \lambda}{X_2^*}, \quad \frac{\tilde{S}_2 (1 - \lambda)}{Y_2} = \frac{\lambda}{Y_2^*}.$$

Using the condition that $\tilde{S}_2 = \tilde{S}_1$, the solution for the home country is

$$X_2 = \frac{\lambda \tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} e^\theta B, \quad Y_2 = \frac{(1 - \lambda) \tilde{S}_1}{\lambda + (1 - \lambda) \tilde{S}_1} e^{-\theta} B^*.$$

The solution for the foreign country follows by symmetry. We can log-linearize this solution around $\bar{s}_1 = 0$ (which means equal initial size of the two countries, i.e., zero net foreign asset position) to obtain

$$\log X_2 \approx \log \lambda + (1 - \lambda) \tilde{s}_1 + \theta + \log B, \quad \log Y_2 \approx \log(1 - \lambda) + \lambda \tilde{s}_1 - \theta + \log B^*.$$

It follows that:

$$\begin{aligned} \log C_2 \tilde{D}_2^{-a} &= \text{const} + (2\lambda - 1)\theta + 2\lambda(1 - \lambda)\tilde{s}_1 + \lambda \log B + (1 - \lambda) \log B^* - a \log \tilde{D}_2 \quad (\text{A.6}) \\ \log C_2^* \tilde{D}_2^{*-a} &= \text{const} - (2\lambda - 1)\theta - 2\lambda(1 - \lambda)\tilde{s}_1 + (1 - \lambda) \log B + \lambda \log B^* - a \log \tilde{D}_2^* \end{aligned}$$

where

$$\log \tilde{D}_2 = \lambda_g(\log B - \log I_{g,1} + \theta) + (1 - \lambda_g)(\log B^* - \log I_{g,1}^* - \theta) + b\tilde{g} \quad (\text{A.7})$$

$$\log \tilde{D}_2^* = \lambda_g(\log B^* - \log I_{g,1}^* - \theta) + (1 - \lambda_g)(\log B - \log I_{g,1} + \theta) + \tilde{g}, \quad (\text{A.8})$$

respectively.

Time 1. At time $t = 1$, the social planner chooses the allocations of consumption goods, X_1 , X_1^* , Y_1 , Y_1^* , of investment goods in the brown technology, $I_{x,1}$, $I_{y,1}$, $I_{y,1}^*$, $I_{x,1}^*$, and investment goods in the green technology, $I_{g,1}$, $I_{g,1}^*$ to maximize

$$\tilde{S}_1 \left(\log C_1 + \beta \log C_2 \tilde{D}_2^{-a} \right) + \left(\log C_1^* + \beta \log C_2^* \tilde{D}_2^{*-a} \right) \quad (\text{A.9})$$

subject to the two economy-wide feasibility constraints:

$$1 = X_1 + X_1^* + I_{x,1} + I_{y,1} + I_{g,1}, \quad 1 = Y_1 + Y_1^* + I_{y,1}^* + I_{x,1}^* + I_{g,1}^*, \quad (\text{A.10})$$

Since we set $\tilde{D}_1 = \tilde{D}_1^* = 1$ and $(1 - \beta)$ is a multiplicative coefficient of the objective function, we can omit them from this maximization. Using the time $t = 2$ equilibrium solutions for $\log(C_2 \tilde{D}_2^{-a})$ and $\log(C_2^* \tilde{D}_2^{*-a})$ in (A.6), the definitions of $\log(\tilde{D}_2)$ and $\log \tilde{D}_2^*$ in (A.7), and the definitions of B and B^* in (A.2), we can expand the objective function in (A.9) as

$$\begin{aligned} & \tilde{S}_1 \left\{ \lambda \log X_1 + (1 - \lambda) \log Y_1 + \beta \left[(2\lambda - 1)\theta + 2\lambda(1 - \lambda)\tilde{s}_1 + \hat{\lambda} (\lambda_I \log I_{x,1} + (1 - \lambda_I) \log I_{x,1}^*) \right. \right. \\ & \quad \left. \left. + (1 - a - \hat{\lambda}) (\lambda_I \log I_{y,1}^* + (1 - \lambda_I) \log I_{y,1}) - a [-\lambda_g \log I_{g,1} - (1 - \lambda_g) \log I_{g,1}^* + (2\lambda_g - 1)\theta + b\tilde{g}] \right] \right\} \\ & + \lambda \log Y_1^* + (1 - \lambda) \log X_1^* + \beta \left[- (2\lambda - 1)\theta - 2\lambda(1 - \lambda)\tilde{s}_1 + (1 - a - \hat{\lambda}) (\lambda_I \log I_{x,1} + (1 - \lambda_I) \log I_{x,1}^*) \right. \\ & \quad \left. + \hat{\lambda} (\lambda_I \log I_{y,1}^* + (1 - \lambda_I) \log I_{y,1}) - a [-\lambda_g \log I_{g,1}^* - (1 - \lambda_g) \log I_{g,1} - (2\lambda_g - 1)\theta + \tilde{g}] \right], \end{aligned} \quad (\text{A.11})$$

where $\hat{\lambda} \equiv \lambda - a\lambda_g$. Since the economy-wide feasibility constraints in (A.10) are linear in the choice variables, we can use them to express $I_{g,1}$ and $I_{g,1}^*$ as functions of the remaining choice variables. Substituting these expressions into the objective function (A.11) yields the following solution from the first-order necessary conditions:

1. Consumption goods:

$$X_1 = \frac{1}{K} \tilde{S}_1 \lambda, \quad X_1^* = \frac{1}{K} (1 - \lambda), \quad Y_1^* = \frac{1}{K^*} \lambda, \quad Y_1 = \frac{1}{K^*} \tilde{S}_1 (1 - \lambda)$$

2. Investments in the brown technology:

$$I_{x,1} = \frac{\beta\lambda_I}{K} \left(\tilde{S}_1\hat{\lambda} + 1 - a - \hat{\lambda} \right), \quad I_{y,1} = \frac{\beta(1-\lambda_I)}{K} \left(\tilde{S}_1(1-a-\hat{\lambda}) + \hat{\lambda} \right),$$

$$I_{y,1}^* = \frac{\beta\lambda_I}{K^*} \left(\hat{\lambda} + \tilde{S}_1(1-a-\hat{\lambda}) \right), \quad I_{x,1}^* = \frac{\beta(1-\lambda_I)}{K^*} \left((1-a-\hat{\lambda}) + \tilde{S}_1\hat{\lambda} \right)$$

3. Investments in the green technology:

$$I_{g,1} = \frac{\beta a}{K} \left(\tilde{S}_1\lambda_g + 1 - \lambda_g \right), \quad I_{g,1}^* = \frac{\beta a}{K^*} \left(\lambda_g + \tilde{S}_1(1-\lambda_g) \right),$$

where

$$K = 1 - \lambda + \beta\lambda_I(1-a-\hat{\lambda}) + \beta(1-\lambda_I)\hat{\lambda} + \beta a(1-\lambda_g) + \left[\lambda + \beta\lambda_I\hat{\lambda} + \beta(1-\lambda_I)(1-a-\hat{\lambda}) + \beta a\lambda_g \right] \tilde{S}_1,$$

$$K^* = \lambda + \beta\lambda_I\hat{\lambda} + \beta(1-\lambda_I)(1-a-\hat{\lambda}) + \beta a\lambda_g + \left[1 - \lambda + \beta\lambda_I(1-a-\hat{\lambda}) + \beta(1-\lambda_I)\hat{\lambda} + \beta a(1-\lambda_g) \right] \tilde{S}_1.$$

Note that if $\tilde{S}_1 = 1$, $K = K^* = (1 + \beta)$.

Log-linearizing the optimal allocations around $\bar{s}_1 = 0$, equivalently $\tilde{S}_1 = 1$, we obtain:

$$\begin{aligned} \log X_1 &\approx \log \frac{\lambda}{1+\beta} + \lambda_x^s \cdot \tilde{s}_1, & \log X_1^* &\approx \log \frac{1-\lambda}{1+\beta} + \lambda_{x^*}^s \cdot \tilde{s}_1, \\ \log Y_1 &\approx \log \frac{1-\lambda}{1+\beta} - \lambda_{x^*}^s \tilde{s}_1, & \log Y_1^* &\approx \log \frac{\lambda}{1+\beta} - \lambda_x^s \tilde{s}_1, \\ \log I_{x,1} &\approx \log \frac{\beta\lambda_I(1-a)}{1+\beta} + \lambda_{I_x}^s \cdot \tilde{s}_1, & \log I_{y,1} &\approx \log \frac{\beta(1-\lambda_I)(1-a)}{1+\beta} + \lambda_{I_y}^s \cdot \tilde{s}_1, \\ \log I_{x,1}^* &\approx \log \frac{\beta(1-\lambda_I)(1-a)}{1+\beta} - \lambda_{I_y}^s \tilde{s}_1, & \log I_{y,1}^* &\approx \log \frac{\beta\lambda_I(1-a)}{1+\beta} - \lambda_{I_x}^s \tilde{s}_1, \\ \log I_{g,1} &\approx \log \frac{\beta a}{1+\beta} + \lambda_{I_g}^s \cdot \tilde{s}_1, & \log I_{g,1}^* &\approx \log \frac{\beta a}{1+\beta} - \lambda_{I_g}^s \tilde{s}_1, \end{aligned} \tag{A.12}$$

where

$$\lambda_x^s = \frac{1 - \lambda + \beta\lambda_I(1-a-\hat{\lambda}) + \beta(1-\lambda_I)\hat{\lambda} + \beta a(1-\lambda_g)}{1+\beta}, \quad \lambda_{x^*}^s = \lambda_x^s - 1 \tag{A.13}$$

$$\lambda_{I_x}^s = \frac{a(\lambda - \lambda_g)}{1-a} + \frac{\beta(1-\lambda_I)(2\hat{\lambda} - (1-a))}{1+\beta}, \quad \lambda_{I_y}^s = \lambda_{I_x}^s - \frac{2\hat{\lambda} - 1 + a}{1-a}, \quad \lambda_{I_g}^s = \lambda_{I_x}^s + \frac{\lambda_g - \lambda}{1-a}.$$

Signs of the slopes of the log-linearized solution. We can characterize the slopes of the log-linearized solutions in (A.13). In what follows, we retain the parametric restrictions of Assumption A1.

TABLE A.1. SIGNS OF THE SLOPES OF THE LOG-LINEARIZED SOLUTION

	λ_x^s	$\lambda_{x^*}^s$	$\lambda_{I_x}^s$	$\lambda_{I_y}^s$	$\lambda_{I_g}^s$
$\lambda > \lambda_g$	+	−	+	−	Undetermined
$\lambda < \lambda_g$	+	−	Undetermined	−	+

Notes: The table reports the signs of the slopes of the log-linearized solution of the model reported in (A.13). These sign conclusions hold based on the domain restrictions assumed in Assumption A1.

1. Given that $\lambda, \lambda_g < 1$ and $\hat{\lambda} < 1 - a$, we can easily verify that $\lambda_x^s > 0$.

2. Consider the inequality:

$$1 - \lambda + \beta\lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} + \beta a(1 - \lambda_g) < 1 + \beta\lambda_I(1 - a) + \beta(1 - \lambda_I)(1 - a) + \beta a = 1 + \beta.$$

From this, we conclude that $\lambda_x^s < 1$, and thus $\lambda_{x^*}^s = \lambda_x^s - 1 < 0$.

3. Since

$$\lambda_{I_x}^s = \frac{a(\lambda - \lambda_g)}{1 - a} + \frac{\beta(1 - \lambda_I)(2\hat{\lambda} - (1 - a))}{1 + \beta},$$

it follows that if $\lambda > \lambda_g$ and $\hat{\lambda} > \frac{1-a}{2}$, then $\lambda_{I_x}^s > 0$. Otherwise, $\lambda_{I_x}^s$ can be negative.

4. Note that

$$\begin{aligned} \lambda_{I_y}^s &= \frac{(1 + \beta)a(\lambda - \lambda_g) + (1 + \beta a)(1 - a - 2\hat{\lambda}) + \beta\lambda_I(1 - a)(1 - a - 2\hat{\lambda})}{(1 + \beta)(1 - a)} \\ &= \frac{a(\lambda - \lambda_g) + 1 - a - 2\hat{\lambda} + \beta\lambda_I(1 - a)(1 - a - 2\hat{\lambda}) + \beta a(1 - a - 2\hat{\lambda} + \lambda - \lambda_g)}{(1 + \beta)(1 - a)} \\ &= \frac{(1 - \lambda)(1 - a) - \hat{\lambda} + \beta\lambda_I(1 - a)(1 - a - 2\hat{\lambda}) + \beta a((1 - \lambda_g)(1 - a) - \hat{\lambda})}{(1 + \beta)(1 - a)}. \end{aligned}$$

Since $(1 - \lambda)(1 - a) < \frac{1-a}{2}$, $(1 - \lambda_g)(1 - a) < \frac{1-a}{2}$, and $\hat{\lambda} > \frac{1-a}{2}$, the numerator is negative. Therefore, $\lambda_{I_y}^s < 0$.

5. Finally, since

$$\lambda_{I_g}^s = \lambda_g - \lambda + \frac{\beta(1 - \lambda_I)(2\hat{\lambda} - (1 - a))}{1 + \beta},$$

and $\hat{\lambda} > \frac{1-a}{2}$, we have $\lambda_{I_g}^s > 0$ when $\lambda < \lambda_g$. Otherwise, $\lambda_{I_g}^s$ may be negative.

Table A.1 summarizes the signs of the slopes of the log-linearized allocations in (A.13).

Equilibrium utilities. Starting from the time $t = 1$ utilities in (A.1), we substitute the log-linearized allocations from (A.12), along with the definitions of $\log \tilde{D}_2$ and $\log \tilde{D}_2^*$ in (A.7), and the definitions of B and B^* in (A.2), to obtain:

$$\begin{aligned} u_1 &= \text{const} + \lambda_u^\theta \theta + b \lambda_u^g \tilde{g} + \lambda_u^s \tilde{s}_1 = \text{const} + \left(\lambda_u^\theta + \lambda_u^s \lambda_s^\theta \right) \theta + (b \lambda_u^g + \lambda_u^s \lambda_s^g) \tilde{g} \\ u_1^* &= \text{const} - \lambda_u^\theta \theta + \lambda_u^g \tilde{g} - \lambda_u^s \tilde{s}_1 = \text{const} - \left(\lambda_u^\theta + \lambda_u^s \lambda_s^\theta \right) \theta - (-\lambda_u^g + \lambda_u^s \lambda_s^g) \tilde{g} \end{aligned} \quad (\text{A.14})$$

where the second equality follows from the linearity of $\tilde{s}_1$ in the two shocks (see A.4) and where

$$\begin{aligned} \lambda_u^\theta &= \tilde{\beta} \left(2\hat{\lambda} - (1 - a) \right), \quad \lambda_u^g = -\beta(1 - \beta)a, \\ \lambda_u^s &= (1 - \beta) \left(\lambda \lambda_x^s - (1 - \lambda) \lambda_{x^*}^s \right) + \tilde{\beta} \left(2\lambda(1 - \lambda) + (2\hat{\lambda} - 1 + a) \left(\lambda_I \lambda_{I_x}^s - (1 - \lambda_I) \lambda_{I_y}^s \right) + a(2\lambda_g - 1) \lambda_{I_g}^s \right). \end{aligned} \quad (\text{A.15})$$

and $\tilde{\beta} = \beta(1 - \beta)$.

Sensitivity of the Pareto weights to the shocks. Recall from equation (A.3) that the log-ratio of the Pareto weights at date $t = 1$ is equal to $\tilde{s}_1 = (u_1 - u_1^*)(1 - \gamma)$. Since utilities are linear functions of the shocks θ and $\tilde{g}$ (see (A.14)), it follows that $\tilde{s}_1$ is also linear in the shocks:

$$\tilde{s}_1 = \bar{s}_1 + \lambda_s^\theta \cdot \theta + \lambda_s^g \cdot \tilde{g}, \quad (\text{A.16})$$

where

$$\lambda_s^\theta = 2(1 - \gamma) \left(\lambda_u^\theta + \lambda_u^s \lambda_s^\theta \right), \quad \lambda_s^g = (1 - \gamma) \left((b - 1) \lambda_u^g + 2 \lambda_u^s \lambda_s^g \right),$$

where λ_u^g and λ_u^s are defined in (A.15), while λ_s^g is determined below. Solving these equations yields:

$$\lambda_s^\theta = \frac{2(1 - \gamma) \lambda_u^\theta}{1 + 2(\gamma - 1) \lambda_u^s}, \quad \lambda_s^g = \frac{(1 - \gamma)(b - 1) \lambda_u^g}{1 + 2(\gamma - 1) \lambda_u^s}. \quad (\text{A.17})$$

The coefficient λ_s^g in (A.17) multiplies the Appendix shock $\tilde{g}$. When the same relationship is expressed in terms of the main-text shock $g = \beta a \tilde{g}$, the coefficient on g is

$$\frac{\lambda_s^g}{\beta a} = \frac{(1 - \beta)(\gamma - 1)(b - 1)}{1 + 2(\gamma - 1) \lambda_u^s}.$$

Next we examine how the ratio of the pseudo Pareto weights responds to different shocks.

Lemma 1. *Under Assumption A1, $\lambda_u^s > 0$. In addition, when $\gamma > 1$ we have: (i) $\lambda_s^\theta < 0$, and*

(ii) $\lambda_s^g > 0$ (< 0) if $b > 1$ (< 1).

Proof. First, we prove that the coefficient λ_u^s defined in (A.15) is always strictly larger than zero. We notice the following:

1. When $\lambda > \lambda_g$, we conclude that $\lambda\lambda_x^s - (1 - \lambda)\lambda_{x^*}^s > 0$ and $\lambda_I\lambda_{I_x}^s - (1 - \lambda_I)\lambda_{I_y}^s > 0$, based on the characterization of the signs of the slopes of the log-linear solution for the allocations in Table A.1. We now need to verify that the negative contribution associated with $\lambda_{I_g}^s$ does not outweigh the positive terms. From the expression $(1 - \beta)(\lambda\lambda_x^s - (1 - \lambda)\lambda_{x^*}^s)$, we can extract a strictly positive term $(1 - \beta)(1 - \lambda) > 0$ (since $\lambda_{x^*}^s = \lambda_x^s - 1$) and retain the remaining positive part. The negative component coming from $\lambda_{I_g}^s$ is given by $(1 - \beta)\beta a(2\lambda_g - 1)(\lambda_g - \lambda)$. Thus, we need to verify that

$$(1 - \beta)(1 - \lambda) + (1 - \beta)\beta a(2\lambda_g - 1)(\lambda_g - \lambda) > 0,$$

or, equivalently,

$$1 - \lambda + \beta a(2\lambda_g - 1)(\lambda_g - \lambda) > 0$$

under the parameter restrictions in Assumption A1. Note that $\hat{\lambda} < 1 - a$ also implies:

$$\lambda < 1 - a + a\lambda_g.$$

Then,

$$\begin{aligned} 1 - \lambda + \beta a(2\lambda_g - 1)(\lambda_g - \lambda) &> 1 - (1 - a + a\lambda_g) + \beta a(2\lambda_g - 1)(\lambda_g - (1 - a + a\lambda_g)) \\ &= a(1 - \lambda_g) + \beta a(2\lambda_g - 1)(1 - a)(\lambda_g - 1) \\ &= a(1 - \lambda_g)(1 - \beta(2\lambda_g - 1)(1 - a)) \\ &> 0, \end{aligned}$$

where the final inequality holds because $\beta < 1$, $2\lambda_g - 1 < 1$, and $1 - a < 1$.

2. When $\lambda < \lambda_g$, the only negative component in λ_u^s is associated with $\lambda_{I_x}^s$, based on the sign characterization in Table A.1. The negative term is given by

$$(1 - \beta)\beta(2\hat{\lambda} - 1 + a)\lambda_I \frac{a}{1 - a}(\lambda - \lambda_g).$$

As before, given that $(1 - \beta)(1 - \lambda) > 0$, we need to verify that

$$1 - \lambda > \beta(2\hat{\lambda} - 1 + a)\lambda_I \frac{a}{1 - a}(\lambda_g - \lambda).$$

Note that

$$\frac{2\hat{\lambda} - 1 + a}{1 - a} = \frac{2\hat{\lambda}}{1 - a} - 1 < 1,$$

since $\hat{\lambda} < 1 - a$. It follows that

$$\beta(2\hat{\lambda} - 1 + a)\lambda_I \frac{a}{1 - a}(\lambda_g - \lambda) < \beta a \lambda_I(\lambda_g - \lambda) < \lambda_g - \lambda < 1 - \lambda.$$

The above derivations prove that $\lambda_u^s > 0$. The characterization of the signs of λ_u^θ and λ_u^g in (A.15) is straightforward:

$$\lambda_u^\theta = \beta(1 - \beta) \left(2\hat{\lambda} - (1 - a) \right) > 0, \quad \lambda_u^g = -\beta(1 - \beta)a < 0.$$

We conclude that $\lambda_s^\theta < 0$ when $\gamma > 1$. In addition, when $b > 1$ (< 1), $\lambda_s^g > 0$ (< 0). $\square$

A.4 Exchange rates

Since markets are complete, the growth rate of the exchange rate equals the ratio of the stochastic discount factors (SDFs). We can characterize this ratio using the equilibrium condition for the ratio of Pareto weights:

$$\tilde{S}_1 = \tilde{S}_0 \frac{M_1}{M_1^*} \frac{e^{\Delta c_1}}{e^{\Delta c_1^*}},$$

where M_1 and M_1^* denote the SDFs for the home and foreign countries, respectively. Taking logarithms yields:

$$\tilde{s}_1 = \tilde{s}_0 + m_1 - m_1^* + \Delta c_1 - \Delta c_1^*.$$

Note that $\Delta e_1 = m_1 - m_1^*$, where Δe_1 is the log change in the real exchange rate (measured as the amount of foreign currency per unit of domestic currency). Given that $\tilde{S}_0$, c_0 , and c_0^* are constants that do not depend on the shocks, we have:

$$\begin{aligned} \Delta e_1 &= \text{const} + \tilde{s}_1 - c_1 + c_1^* \\ &= \text{const} + \tilde{s}_1 - (\lambda \log X_1 + (1 - \lambda) \log Y_1) + (\lambda \log Y_1^* + (1 - \lambda) \log X_1^*) \\ &= \text{const} + \tilde{s}_1 - (\lambda \lambda_x^s - (1 - \lambda) \lambda_{x^*}^s) \tilde{s}_1 + (-\lambda \lambda_x^s + (1 - \lambda) \lambda_{x^*}^s) \tilde{s}_1 \\ &= \text{const} + \lambda_e^s \cdot \tilde{s}_1, \end{aligned}$$

where

$$\lambda_e^s = 1 - 2(\lambda\lambda_x^s - (1 - \lambda)\lambda_{x^*}^s). \quad (\text{A.18})$$

Thus, there is a one-to-one mapping between the pseudo-Pareto weight $\tilde{S}_1$ and changes in the real exchange rate. The nature of this relationship depends on the sign of λ_e^s . The following proposition establishes that $\lambda_e^s > 0$ under Assumption A1.

Proposition 1. *Let $\gamma > 1$. If $b > 1$ ($b < 1$), a global climate news shock $\tilde{g} > 0$ causes an appreciation of the home (foreign) real exchange rate.*

Proof. We need to characterize the sign of the composite coefficient λ_e^s in equation (A.18). Using $\lambda_{x^*}^s = \lambda_x^s - 1$ (see (A.13)) in the definition of λ_e^s , we need to show that

$$\lambda_e^s = 1 - 2(\lambda\lambda_x^s - (1 - \lambda)(\lambda_x^s - 1)) = 1 - 2((2\lambda - 1)\lambda_x^s + (1 - \lambda)) > 0$$

which is equivalent to $(2\lambda - 1)\lambda_x^s < \lambda - \frac{1}{2}$ or $\lambda_x^s < \frac{1}{2}$. By using the expression for λ_x^s in (A.13), we obtain:

$$1 - \lambda + \beta\lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} + \beta a(1 - \lambda_g) < \frac{1 + \beta}{2}.$$

Note that

$$\beta\lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} = \beta(\lambda_I(1 - a - 2\hat{\lambda}) + \hat{\lambda}).$$

Given $1 - a - 2\hat{\lambda} < 0$ (Assumption A1) and $\lambda_I > 0$, then $\beta\lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} < \beta\hat{\lambda}$. Then it is sufficient to prove that:

$$1 - \lambda + \beta\hat{\lambda} + \beta a(1 - \lambda_g) < \frac{1 + \beta}{2}.$$

Since $\hat{\lambda} = \lambda - a\lambda_g$, the left hand side of the above expression becomes

$$1 - \lambda + \beta\lambda + \beta a(1 - 2\lambda_g) < 1 - \lambda + \beta\lambda.$$

Since $1 - 2\lambda_g < 0$, a sufficient condition is then $1 - \lambda + \beta\lambda < \frac{1 + \beta}{2}$. After rearranging, we need $\beta(\lambda - \frac{1}{2}) < \lambda - \frac{1}{2}$. This condition holds because $\lambda - \frac{1}{2} > 0$ and $\beta < 1$. $\square$

A.5 Household problem: setup and environment

In this two-period model, the households have the same decision problem. We describe the setting from the home country point of view.

- **At $t = 0$** , the household is endowed with wealth W_0 that is invested across two equity indices, which are claims on aggregate dividends (defined in Section A.6) in the home or foreign country. The investment in home (foreign) equity is Z_0^H (Z_0^F).
- **At $t = 1$** , shocks are realized and uncertainty is resolved. The household chooses the consumption bundle (X_1, Y_1) and invests the rest in a global storage asset that pays a risk-free rate $R_{1|2}^f$ (denominated in units of good X).
- **At $t = 2$** , there is no continuation value; the household chooses the consumption bundle (X_2, Y_2) .

A.5.1 Consumption Allocation

Time 2 Household optimization problem:

Home country

$$\max_{X_2, Y_2} u_2 \equiv \lambda \log X_2 + (1 - \lambda) \log Y_2 - a \log \tilde{D}_2$$

$$s.t. \quad X_2 + p_2 Y_2 = W_2$$

Foreign country

$$\max_{X_2^*, Y_2^*} u_2^* \equiv \lambda \log Y_2^* + (1 - \lambda) \log X_2^* - a \log \tilde{D}_2^*$$

$$s.t. \quad X_2^* + p_2 Y_2^* = W_2^*$$

where p_2 is the relative price of the foreign good at time 2, with the price of the home good normalized to 1. W_2 (W_2^*) is the initial wealth of the home (foreign) country at time 2, both denominated in units of good X .

The solutions for the consumption decisions of the two countries are:

$$X_2 = \lambda W_2, \quad Y_2 = (1 - \lambda) \frac{W_2}{p_2}.$$

$$X_2^* = (1 - \lambda) W_2^*, \quad Y_2^* = \lambda \frac{W_2^*}{p_2}.$$

Time 1 Household optimization problem:

Home country

$$\max_{X_1, Y_1, Z_1} u_1 \equiv \log C_1 + \beta \log C_2 \tilde{D}_2^{-a}$$

$$\begin{aligned} s.t. \quad & X_1 + p_1 Y_1 + Z_1 = W_1 \\ & Z_1 R_{1|2}^f = W_2 \end{aligned}$$

Foreign country

$$\max_{X_1^*, Y_1^*, Z_1^*} u_1^* \equiv \log C_1^* + \beta \log C_2^* \tilde{D}_2^{*-a}$$

$$\begin{aligned} s.t. \quad & X_1^* + p_1 Y_1^* + Z_1^* = W_1^* \\ & Z_1^* R_{1|2}^f = W_2^* \end{aligned}$$

Given the solution at $t = 2$, the time-1 utility function of the home-country household can be written as

$$u_1 = \lambda \log X_1 + (1 - \lambda) \log Y_1 + \beta \log W_2 + \text{const.}$$

Therefore, the solution at $t = 1$ for the home country is

$$X_1 = \frac{\lambda}{1 + \beta} W_1, \quad Y_1 = \frac{1 - \lambda}{1 + \beta} \frac{W_1}{p_1}, \quad Z_1 = \frac{\beta}{1 + \beta} W_1.$$

For the foreign country, the solution is

$$X_1^* = \frac{1 - \lambda}{1 + \beta} W_1^*, \quad Y_1^* = \frac{\lambda}{1 + \beta} \frac{W_1^*}{p_1}, \quad Z_1^* = \frac{\beta}{1 + \beta} W_1^*.$$

The risk-free rate from time 1 to 2 is

$$R_{1|2}^f = \frac{W_2}{Z_1} = \frac{1 + \beta}{\beta} \frac{W_2}{W_1}.$$

A.5.2 Replicating the First Best

We show the conditions under which the competitive equilibrium replicates the social planner's first-best allocation.

Time 2 For the home country household:

Social Planner's Solution

$$\begin{cases} X_2 = \frac{\lambda \tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} e^{\theta B} \\ Y_2 = \frac{(1 - \lambda) \tilde{S}_1}{\lambda + (1 - \lambda) \tilde{S}_1} e^{-\theta B^*} \end{cases}$$

Decentralized Solution

$$\begin{cases} X_2 = \lambda W_2 \\ Y_2 = (1 - \lambda) \frac{W_2}{p_2} \end{cases}$$

where $\tilde{S}_1$ is the pseudo Pareto weight of the home country at $t = 1$ in the social planner's solution. To make the decentralized solution replicate the social planner's solution:

$$W_2 = \frac{\tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} e^{\theta} B. \quad (\text{A.19})$$

$$p_2 = \frac{1 - \lambda}{\lambda} \frac{X_2}{Y_2} = \frac{\lambda + (1 - \lambda) \tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} e^{2\theta} B/B^*. \quad (\text{A.20})$$

Solving for the foreign country,

$$W_2^* = \frac{1}{1 - \lambda + \lambda \tilde{S}_1} e^{\theta} B. \quad (\text{A.21})$$

Time 1

Social Planner's Solution

Decentralized Solution

$$\begin{cases} X_1 = \frac{1}{K} \lambda \tilde{S}_1 \\ Y_1 = \frac{1}{K^*} (1 - \lambda) \tilde{S}_1 \end{cases} \quad \begin{cases} X_1 = \frac{\lambda}{1 + \beta} W_1 \\ Y_1 = \frac{1 - \lambda}{1 + \beta} \frac{W_1}{p_1} \end{cases}$$

where

$$K(\tilde{S}_1) = \underbrace{1 - \lambda + \beta \lambda_I (1 - a - \hat{\lambda}) + \beta (1 - \lambda_I) \hat{\lambda} + \beta a (1 - \lambda_g)}_{K_0} + \underbrace{\left[\lambda + \beta \lambda_I \hat{\lambda} + \beta (1 - \lambda_I) (1 - a - \hat{\lambda}) + \beta a \lambda_g \right]}_{K_1} \tilde{S}_1$$

$$K^*(\tilde{S}_1) = K_1 + K_0 \tilde{S}_1$$

As a result, domestic and foreign wealth must evolve as follows:

$$W_1 = \frac{1 + \beta}{K(\tilde{S}_1)} \tilde{S}_1, \quad (\text{A.22})$$

$$W_1^* = \frac{1 + \beta}{K^*(\tilde{S}_1)}. \quad (\text{A.23})$$

The terms of trade are simply

$$p_1 = \frac{K^*(\tilde{S}_1)}{K(\tilde{S}_1)}, \quad (\text{A.24})$$

and the risk-free rate that replicates the first-best allocation is

$$R_{1|2}^f = \frac{1 + \beta}{\beta} \frac{W_2}{W_1} = \frac{1}{\beta} \frac{K(\tilde{S}_1)}{1 - \lambda + \lambda \tilde{S}_1} e^{\theta} B. \quad (\text{A.25})$$

A.6 Firm Problem

A.6.1 Regulation about emissions and offsets.

In what follows, we adopt the following terminology. E_2 and E_2^* denote the emissions of the home and foreign brown firm at time 2, respectively. Emissions are defined as follows:

$$E_2 = \left[GDP_2^{\lambda_g} GDP_2^{*1-\lambda_g} \right]^a \quad E_2^* = \left[GDP_2^{1-\lambda_g} GDP_2^{*\lambda_g} \right]^a.$$

The green firms produce carbon sequestration and offsets, which are denoted by O_2 and O_2^* in the home and foreign countries, respectively. Offsets are computed as follows:

$$O_2 = \left[I_{g,1}^{\lambda_g} I_{g,1}^{*1-\lambda_g} \right]^a, \quad \text{and} \quad O_2^* = \left[I_{g,1}^{*\lambda_g} I_{g,1}^{1-\lambda_g} \right]^a.$$

As a result of this, the disruption terms that affect the consumption bundle can be rewritten as:

$$D_2 = E_2/O_2, \quad D_2^* = E_2^*/O_2^*,$$

Thus, when a country is net zero, so that offsets and sequestration equal emissions, $E_2 = O_2$ ($E_2^* = O_2^*$), there is no disruption: $D_2 = 1$ ($D_2^* = 1$).

We postulate that in each country the government taxes climate-related damage, as measured by $D_2 - 1$ in the home country ($D_2^* - 1$ in the foreign country). Each government sets the local tax on emissions (τ in the home country, and τ^* in the foreign one), or, equivalently, it sets the supply of emission allowances in the local ETS in order to target a specific price per ton of CO_{2e} .

Importantly, each government recognizes the relevance of offsets and sequestration and it enables local offsets to count for the assessment of the damage. Very much as in the EU ETS, we assume that international offsets are not allowed to count for the reduction of net emissions. Equivalently, taxation is applied to $D_2 - 1$. As in the EU, each country adopts a Carbon Border Adjustment Mechanism (CBAM), and it taxes damages realized abroad. As a result, the home country collects a total amount of taxes defined as

$$T := \tau (D_2 - 1) + p_2 \tau^* (D_2^* - 1),$$

where the last term is expressed in units of good X . We assume that all proceeds are given to the green firm as a lump-sum subsidy.

In addition, the international spillovers of green investment are rewarded by a system of

international proportional transfers. The home green firm receives $p_2 p_o^{FH} I_{g,1}$ from abroad, and it rebates $p_o^{HF} I_{g,1}^*$ to the foreign green firm. We think of these transfers as consistent with the spirit of Article 6.4 of the Paris Agreement. More precisely, these transfers are set among private firms, not via government-to-government transactions (Article 6.2 of the Paris Agreement). We derive p_o^{FH} and p_o^{HF} in what follows, as we describe the firms' problem in the home country. The derivations for the foreign firms follow by symmetry.

A.6.2 The brown firm

The brown firm maximizes the present value of its dividends:

$$V_1^B = \max_{I_{x,1}, I_{x,1}^*, O_2} \underbrace{(1 - I_{x,1} - p_1 I_{x,1}^*)}_{Div_1^B} + \frac{1}{R_{1|2}^f} \underbrace{[GDP_2 - p_o O_2 - T]}_{Div_2^B}, \quad (\text{A.26})$$

where p_o denotes the price of offset purchased from the green firm at time 2. The demand for offsets is:

$$p_o = \tau \frac{E_2}{O_2},$$

implying that the value of offsets is anchored to the marginal reduction in the carbon tax. Optimal investment prescribes:

$$\lambda_I \frac{GDP_2}{I_{x,1}} \cdot \left(1 - a \lambda_g \frac{\tau E_2}{GDP_2 O_2} - a(1 - \lambda_g) \frac{\tau^* p_2 E_2^*}{GDP_2 O_2^*} \right) = R_{1|2}^f,$$

that is, the marginal product of investment net of taxation must equal the risk-free rate. A similar expression can be obtained for $I_{x,1}^*$.

Connection to first-best. Under the planner's problem, the marginal damage of domestic investment due to domestic disruption is:

$$a \frac{\lambda_g X_2}{\lambda GDP_2},$$

hence replicating this margin requires

$$\tau = \frac{X_2 O_2}{\lambda E_2},$$

and the marginal value of offsets becomes

$$p_o = \frac{X_2}{\lambda O_2},$$

like in the planner's problem.

By symmetry, suppose that

$$\tau^* = \frac{Y_2^* O_2^*}{\lambda E_2^*}, \quad \text{and} \quad p_o^* = \tau^* \frac{E_2^*}{O_2^{*2}} = \frac{Y_2^*}{\lambda O_2^*}.$$

In addition, recall that

$$p_2 = \frac{\lambda}{1 - \lambda} \frac{X_2^*}{Y_2^*},$$

and hence

$$p_2 p_o^* = \frac{1}{1 - \lambda} \frac{X_2^*}{O_2^*}$$

As a result, the marginal damage of domestic investment due to foreign disruption is:

$$a \frac{1 - \lambda_g}{1 - \lambda} \frac{X_2^*}{GDP_2},$$

as in the planner's problem.

The green firm. The green firm maximizes the present value of its dividends:

$$V_1^G = \max_{I_{g,1}, I_{g,1}^*} \underbrace{-I_{g,1}}_{Div_1^G} + \frac{1}{R_{1|2}^f} \underbrace{[p_o O_2 - p_o^{HF} I_{g,1}^* + p_2 p_o^{FH} I_{g,1} + T]}_{Div_2^G}, \quad (\text{A.27})$$

where $p_o O_2$ denotes the total sales of carbon credits in the home country; $p_2 p_o^{FH} I_{g,1} - p_o^{HF} I_{g,1}^*$ is the net international transfer that accounts for synergies across domestic and foreign green investments.

The optimal condition for domestic green investment is:

$$1 = \frac{1}{R_{1|2}^f} \left(p_o \lambda_g a \frac{O_2}{I_{g,1}} + p_2 p_o^{FH} \right),$$

accounts for marginal benefits originating from both the domestic sale of offsets and the international compensation. The transfer from the home country to the foreign one is obtained

from the optimal demand of foreign green investment:

$$p_o^{HF} = p_o(1 - \lambda_g)a \frac{O_2}{I_{g,1}^*},$$

and, by symmetry,

$$p_o^{FH} = p_o^*(1 - \lambda_g)a \frac{O_2^*}{I_{g,1}}.$$

Connection to first-best. According to the planner, optimal domestic green investment should be:

$$1 = \frac{1}{R_{1|2}^f} \left(a \frac{\lambda_g}{\lambda} \frac{X_2}{I_{g,1}} + a \frac{1 - \lambda_g}{1 - \lambda} \frac{X_2^*}{I_{g,1}} \right).$$

Given our previous result on $p p_o$, the decentralized economy has the same investment behavior as the planner's one.

A.7 Household's Time-0 Portfolio Weights

Time-0 wealth is exogenous and normalized to 1 for both the home and foreign countries. The household invests a fraction Z_0^H in the local stock market, and $Z_0^F = 1 - Z_0^H$ in the foreign one. As a result, the home-country wealth at time 1 is:

$$W_1 = Z_0^H (\mathcal{V}_1^A + Div_1^A) + Z_0^F (\mathcal{V}_1^{A*} + Div_1^{A*})$$

where $\mathcal{V}_1^A$ ($\mathcal{V}_1^{A*}$) is the time-1 ex-dividend aggregate stock value in the home (foreign) country; Div_1^A (Div_1^{A*}) is the time-1 aggregate dividend in the home (foreign) country. All values are denominated in units of good X .

We derive $\mathcal{V}_1^A$ and Div_1^A in the Sections [A.7.1](#) and [A.7.2](#), respectively. In the main text, $V_1^A = \mathcal{V}_1^A + Div_1^A$.

A.7.1 Time-1 Ex-Dividend Values

At time 2, the tax paid by the brown firm is transferred to the green firm. As a result, aggregate dividends are simply

$$Div_2^A = Div_2^g + Div_2^b = GDP_2 - p_o^{HF} I_{g,1}^* + p_2 p_o^{FH} I_{g,1}.$$

After substituting the optimal prices and transfers, we obtain

$$GDP_2 - p_o(1 - \lambda_g)a \frac{O_2}{I_{g,1}^*} I_{g,1}^* + p_2 p_o^*(1 - \lambda_g)a \frac{O_2^*}{I_{g,1}} I_{g,1},$$

which simplifies to

$$Div_2^A = GDP_2 - a(1 - \lambda_g)(p_o O_2 - p_2 p_o^* O_2^*) = GDP_2 - a(1 - \lambda_g) \left(\frac{X_2}{\lambda} - \frac{X_2^*}{1 - \lambda} \right).$$

Since

$$\frac{X_2}{\lambda} - \frac{X_2^*}{1 - \lambda} = \frac{X_2}{\lambda} \left(1 - \frac{1}{\tilde{S}_1} \right),$$

we obtain

$$Div_2^A = GDP_2 - a \frac{1 - \lambda_g}{\lambda} X_2 \left(1 - \frac{1}{\tilde{S}_1} \right).$$

This expression can be further simplified, since under the first-best allocation

$$X_2 = \frac{\lambda \tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} GDP_2,$$

and hence

$$\begin{aligned} Div_2^A &= GDP_2 - a \frac{1 - \lambda_g}{\lambda} \frac{\lambda \tilde{S}_1}{1 - \lambda + \lambda \tilde{S}_1} GDP_2 \left(1 - \frac{1}{\tilde{S}_1} \right) \\ &= GDP_2 \left[1 - a \frac{1 - \lambda_g}{\lambda} \frac{\lambda(\tilde{S}_1 - 1)}{1 - \lambda + \lambda \tilde{S}_1} \right] \\ &= GDP_2 \frac{1 + (\lambda - a(1 - \lambda_g))(\tilde{S}_1 - 1)}{1 + \lambda(\tilde{S}_1 - 1)}. \end{aligned}$$

When $a = 0$, $Div_2^A = GDP_2$ for all $\tilde{S}_1$, and there is no international transfer. When $a > 0$,

$$\log Div_2^A = \log GDP_2 + \log \left(1 + (\lambda - a(1 - \lambda_g))(\tilde{S}_1 - 1) \right) - \log \left(1 + \lambda(\tilde{S}_1 - 1) \right)$$

After log-linearizing, we obtain

$$\log Div_2^A = \log GDP_2 + (\lambda - a(1 - \lambda_g) - \lambda) \tilde{s}_1 = \log GDP_2 - a(1 - \lambda_g) \tilde{s}_1.$$

According to the planner's solution,

$$R_{1|2}^f = \frac{1}{\beta} \frac{K(\tilde{S}_1)}{1 + \lambda(\tilde{S}_1 - 1)} GDP_2,$$

and hence the ex-dividend value of the aggregate stock market at $t = 1$ is

$$\mathcal{V}_1^A = \frac{1}{R_{1|2}^f} Div_2^A = \beta \frac{1 + [\lambda - a(1 - \lambda_g)](\tilde{S}_1 - 1)}{K(\tilde{S}_1)}.$$

A.7.2 Time-1 Aggregate Dividend

Under the first-best, the investment flows are

$$\begin{aligned} I_{x,1} &= \frac{\beta \lambda_I}{K} \left(\tilde{S}_1 \hat{\lambda} + 1 - a - \hat{\lambda} \right), & I_{y,1} &= \frac{\beta(1-\lambda_I)}{K} \left(\tilde{S}_1(1 - a - \hat{\lambda}) + \hat{\lambda} \right), \\ I_{y,1}^* &= \frac{\beta \lambda_I}{K^*} \left(\hat{\lambda} + \tilde{S}_1(1 - a - \hat{\lambda}) \right), & I_{x,1}^* &= \frac{\beta(1-\lambda_I)}{K^*} \left((1 - a - \hat{\lambda}) + \tilde{S}_1 \hat{\lambda} \right) \\ I_{g,1} &= \frac{\beta a}{K} \left(\tilde{S}_1 \lambda_g + 1 - \lambda_g \right), & I_{g,1}^* &= \frac{\beta a}{K^*} \left(\lambda_g + \tilde{S}_1 (1 - \lambda_g) \right). \end{aligned}$$

As a result, aggregate dividends evolve as follows:

$$\begin{aligned} Div_1^A &= 1 - I_{g,1} - I_{x,1} - p_1 I_{x,1}^* = 1 - \frac{\beta}{K} \left(\tilde{S}_1 \lambda + 1 - \lambda \right), \\ Div_1^{A*} &= p_1 \left(1 - I_{g,1}^* - I_{y,1}^* - \frac{1}{p_1} I_{y,1} \right) = p_1 - \frac{\beta}{K} \left(\tilde{S}_1(1 - \lambda) + \lambda \right). \end{aligned}$$

A.7.3 Time-1 Portfolio Weights

The cum-dividend value at $t = 1$ is

$$\begin{aligned} \mathcal{V}_1^A + Div_1^A &= \beta \frac{1 + [\lambda - a(1 - \lambda_g)](\tilde{S}_1 - 1)}{K(\tilde{S}_1)} + 1 - \beta \frac{1 + \lambda(\tilde{S}_1 - 1)}{K(\tilde{S}_1)} \\ &= 1 - \frac{\beta}{K(\tilde{S}_1)} a(1 - \lambda_g)(\tilde{S}_1 - 1). \end{aligned}$$

Similarly, the foreign stock cum-dividend value at $t = 1$ is

$$\mathcal{V}_1^{A*} + Div_1^{A*} = p_1 + \frac{\beta}{K(\tilde{S}_1)} a(1 - \lambda_g)(\tilde{S}_1 - 1).$$

The equation that pins down the share Z_0^H is

$$Z_0^H(\mathcal{V}_1^A + Div_1^A) + Z_0^F(\mathcal{V}_1^{A*} + Div_1^{A*}) = W_1 = \frac{1 + \beta}{K(\tilde{S}_1)} \tilde{S}_1$$

Given the cum-dividend values and the price $p_1 = \frac{K^*(\tilde{S}_1)}{K(\tilde{S}_1)}$, the above equation becomes

$$Z_0^H \left(1 - \frac{\beta}{K(\tilde{S}_1)} a(1 - \lambda_g)(\tilde{S}_1 - 1) \right) + Z_0^F \left(\frac{K^*(\tilde{S}_1)}{K(\tilde{S}_1)} + \frac{\beta}{K(\tilde{S}_1)} a(1 - \lambda_g)(\tilde{S}_1 - 1) \right) = \frac{1 + \beta}{K(\tilde{S}_1)} \tilde{S}_1. \quad (\text{A.28})$$

or, equivalently,

$$Z_0^H(K(\tilde{S}_1) - \beta a(1 - \lambda_g)(\tilde{S}_1 - 1)) + Z_0^F(K^*(\tilde{S}_1) + \beta a(1 - \lambda_g)(\tilde{S}_1 - 1)) = (1 + \beta)\tilde{S}_1$$

Given that $K(\tilde{S}_1) = K_0 + K_1\tilde{S}_1$ and $K^*(\tilde{S}_1) = K_1 + K_0\tilde{S}_1$, we obtain two equations by matching both the constant and the slope coefficient across the two sides of the above expression:

$$\begin{aligned} Z_0^H(K_0 + \beta a(1 - \lambda_g)) + Z_0^F(K_1 - \beta a(1 - \lambda_g)) &= 0 \\ Z_0^H(K_1 - \beta a(1 - \lambda_g)) + Z_0^F(K_0 + \beta a(1 - \lambda_g)) &= 1 + \beta \end{aligned}$$

Given that $K_0 + K_1 = 1 + \beta$, we have

$$\begin{cases} Z_0^H = \frac{K_1'}{K_1' - K_0'} \\ Z_0^F = -\frac{K_0'}{K_1' - K_0'} \end{cases}$$

where

$$K_0' = K_0 + \beta a(1 - \lambda_g) = 1 - \lambda + \beta \lambda_I(1 - a - \hat{\lambda}) + \beta(1 - \lambda_I)\hat{\lambda} + 2\beta a(1 - \lambda_g),$$

$$K_1' = K_1 - \beta a(1 - \lambda_g) = \lambda + \beta \lambda_I \hat{\lambda} + \beta(1 - \lambda_I)(1 - a - \hat{\lambda}) + \beta a(2\lambda_g - 1).$$

Since K_0' and K_1' are both positive, in general we could have either $Z_0^H > 1$ (the home country goes long in domestic equity and shorts foreign equity) or $Z_0^H < 0$ (the home country shorts domestic equity in order to invest in foreign equity). Under a reasonable calibration, we have $Z_0^H > 1$. In what follows, we analyse the portfolio composition step by step.

Figure A.1 shows the weights Z_0^H and Z_0^F as a function of a when (i) $\beta = 0.95$, and (ii)

$\lambda = \lambda_I = \lambda_g$ ranges from 0.55 to 0.95.

- When λ s are small, there exists a^* such that $K_0 + \beta a(1 - \lambda_g) = K_1 - \beta a(1 - \lambda_g)$. That is, the equation does not pin down a unique solution for Z_0^H and Z_0^F .
 - When $a < a^*$, the home-country household is long home equity and short foreign equity.
 - When $a > a^*$, the home-country household is short home equity and long foreign equity.
- When λ s are large enough, the home-country household is always long home equity and short foreign equity. In addition, as a increases, the home country increases its long position in home equity.

Figure A.2 shows the weights Z_0^H and Z_0^F as a function of λ_g when (i) $\beta = 0.95$, (ii) $a = 0.1$, and (iii) $\lambda = \lambda_I$ ranges from 0.55 to 0.95.

The home-country household is always long home equity and short foreign equity. In addition, as λ_g increases, the home country decreases its long position in home equity.

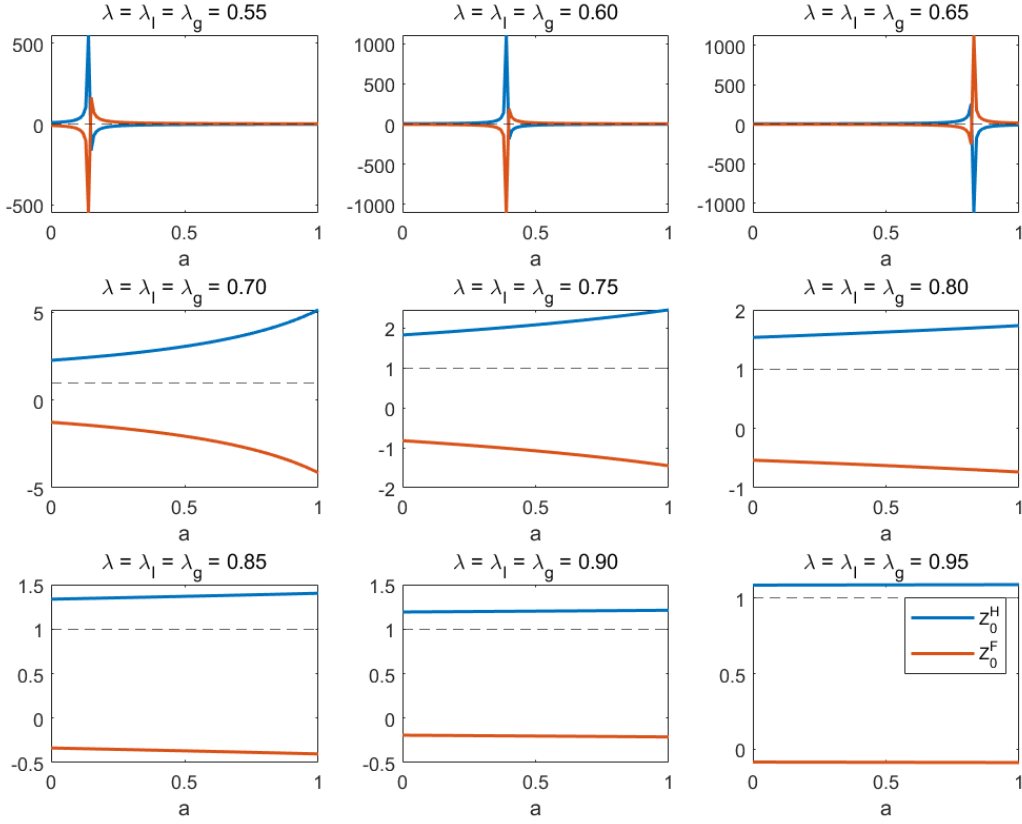


FIG. A.1. PORTFOLIO WEIGHTS AS A FUNCTION OF a

Notes: This figure shows Z_0^H and Z_0^F as a function of a when (i) $\beta = 0.95$, and (ii) $\lambda = \lambda_I = \lambda_g$ ranges from 0.55 to 0.95.

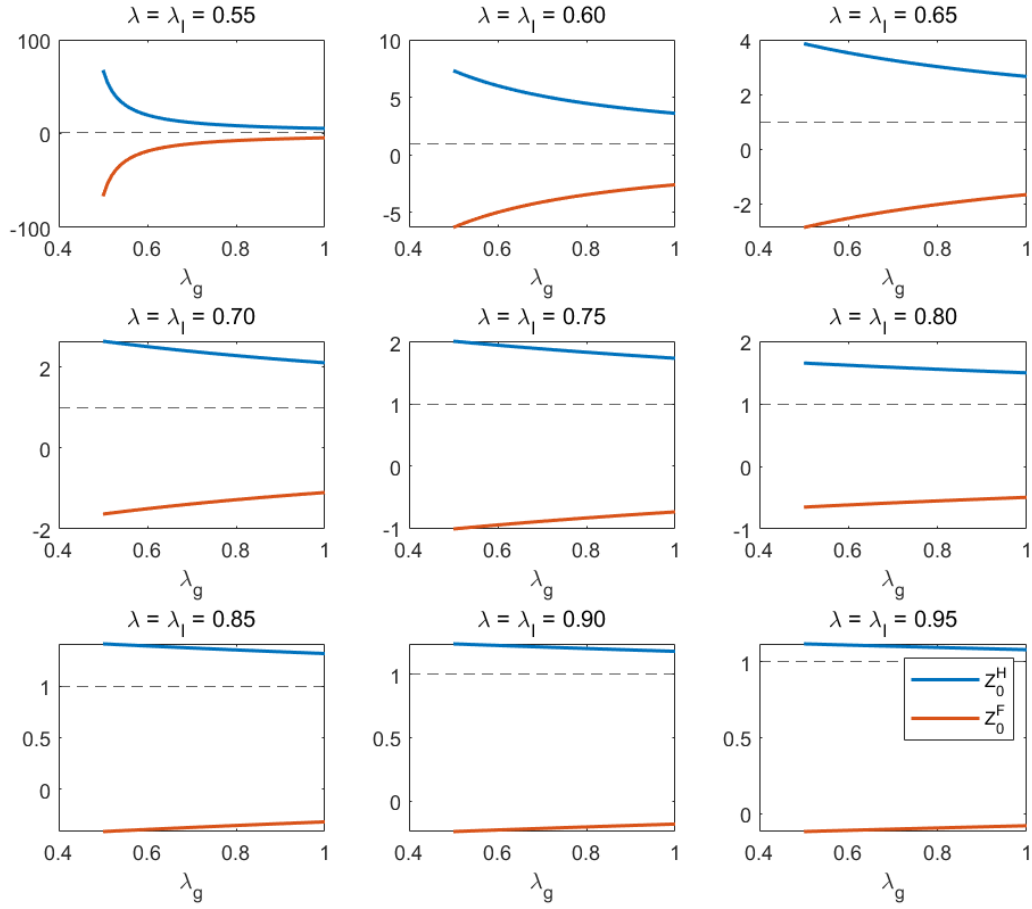


FIG. A.2. PORTFOLIO WEIGHTS AS A FUNCTION OF λ_g

Notes: This figure shows Z_0^H and Z_0^F as a function of λ_g when (i) $\beta = 0.95$, (ii) $a = 0.1$, and (iii) $\lambda = \lambda_I$ ranges from 0.55 to 0.95.

A.8 Brown-minus-Green Returns

Constructing green and brown returns. Define gross returns of the green and brown firms as follows:

$$R_{0|1}^i = \frac{V_1^i}{V_0^i - Div_0^i}, \quad \forall i \in \{G, B\}. \quad (\text{A.29})$$

Since there are no shocks and no investment or consumption decisions made at $t = 0$, we assume $Div_0^i = 0$ and $V_0^i = E_0(V_1^i)$. We can get

$$r_{0|1}^{BMG} = \log \left(\frac{R_{0|1}^B}{R_{0|1}^G} \right) = \log(V_1^B) - \log(V_1^G) + \text{const}$$

In the following part, we will linearize the part $\log(V_1^B) - \log(V_1^G)$; we ignore the constant part because we want to focus on how the exogenous shocks affect $r_{0|1}^{bmG}$ instead of the identifying premium.

Note that, in our two-period model, for green firms:

$$V_1^G = Div_1^G + \frac{1}{R_{1|2}^f} Div_2^G \quad (\text{A.30})$$

where

$$Div_1^G = -I_{g,1},$$

$$Div_2^G = p_o O_2 - p_o^{HF} I_{g,1}^* + p_2 p_o^{FH} I_{g,1} + \tau \left(\frac{E_2}{O_2} - 1 \right) + \tau^* p_2 \left(\frac{E_2^*}{O_2^*} - 1 \right)$$

For brown firms:

$$V_1^B = Div_1^B + \frac{1}{R_{1|2}^f} Div_2^B \quad (\text{A.31})$$

where

$$Div_1^B = 1 - I_{x,1} - p_1 I_{x,1}^*,$$

$$Div_2^B = GDP_2 - p_o O_2 - \tau \left(\frac{E_2}{O_2} - 1 \right) - \tau^* p_2 \left(\frac{E_2^*}{O_2^*} - 1 \right)$$

At the first-best,

$$Div_2^G = \frac{X_2}{\lambda} - a(1 - \lambda_g) \left(\frac{X_2}{\lambda} - \frac{X_2^*}{1 - \lambda} \right) + T$$

$$Div_2^B = GDP_2 - \frac{X_2}{\lambda} - T$$

where T is the lump-sum transfer from brown firms to green firms,

$$T = \tau \left(\frac{E_2}{O_2} - 1 \right) + \tau^* p_2 \left(\frac{E_2^*}{O_2^*} - 1 \right) = \frac{X_2}{\lambda} \left(1 - \frac{O_2}{E_2} \right) + \frac{X_2^*}{1 - \lambda} \left(1 - \frac{O_2^*}{E_2^*} \right)$$

Proposition 2. *By taking the first-order Taylor expansion of $\log(V_1^B) - \log(V_1^G)$ around $\tilde{s}_1 = 0$, we get the linearized form of $r_{0|1}^{BMG}$ as a function of the log pseudo-Pareto weight $\tilde{s}_1$:*

$$r_{0|1}^{BMG} = \overline{r_{0|1}^{BMG}} + \lambda_{BMG}^s \cdot \tilde{s}_1 \quad (\text{A.32})$$

where

$$\lambda_{BMG}^s = \frac{1}{2} \left(\frac{a - 1}{1 - \Phi(a)} + \frac{2\lambda - 1 + 2\beta(2\lambda\lambda_I - \lambda - \lambda_I) + a\beta(2\lambda_I - 3)(2\lambda_g - 1)}{1 + \beta\Phi(a)} \right)$$

with

$$\Phi(a) = 2 \left(\frac{a}{(1 - a)\lambda_I^{\lambda_I}(1 - \lambda_I)^{1 - \lambda_I}} \right)^a + a - 2.$$

When a is close to zero, the sensitivity λ_{BMG}^s is negative for $\lambda, \lambda_I < 1$.

Proof. Taking the limit as $a \rightarrow 0^+$, we obtain

$$\lim_{a \rightarrow 0^+} \lambda_{BMG}^s = \lambda - 1 + \beta(2\lambda\lambda_I - \lambda_I - \lambda).$$

When $\lambda, \lambda_I < 1$, we have $\lambda - 1 < 0$ and $2\lambda\lambda_I - \lambda_I - \lambda = \lambda(\lambda_I - 1) + \lambda_I(\lambda - 1) < 0$. Therefore, the limiting value of λ_{BMG}^s is strictly negative.

By continuity, there exists $\epsilon > 0$ such that $\lambda_{BMG}^s < 0$ for every $a \in (0, \epsilon)$. $\square$

Figure A.3 plots λ_{BMG}^s against a under different values of λ, λ_I , and λ_g . After a positive global climate news shock, $\tilde{g} > 0$, the log pseudo-Pareto weight $\tilde{s}_1$ increases and the brown firm's return drops more than that of the green firm. This effect is stronger when the home country has higher exposure to the global climate news shock.

Calibration and numerical exercise. The previous proposition requires a to be small enough. Here we show that, through the lens of our model, this condition is not restrictive. The previous proposition fails only for levels of a that imply excessive damage-to-GDP values.

First, we set the parameters $\lambda = \lambda_I = 0.9$ and $\beta = 0.97$. Second, to see what is a reasonable value of a , we calculate the model-implied damage as a fraction of the GDP, which is given by

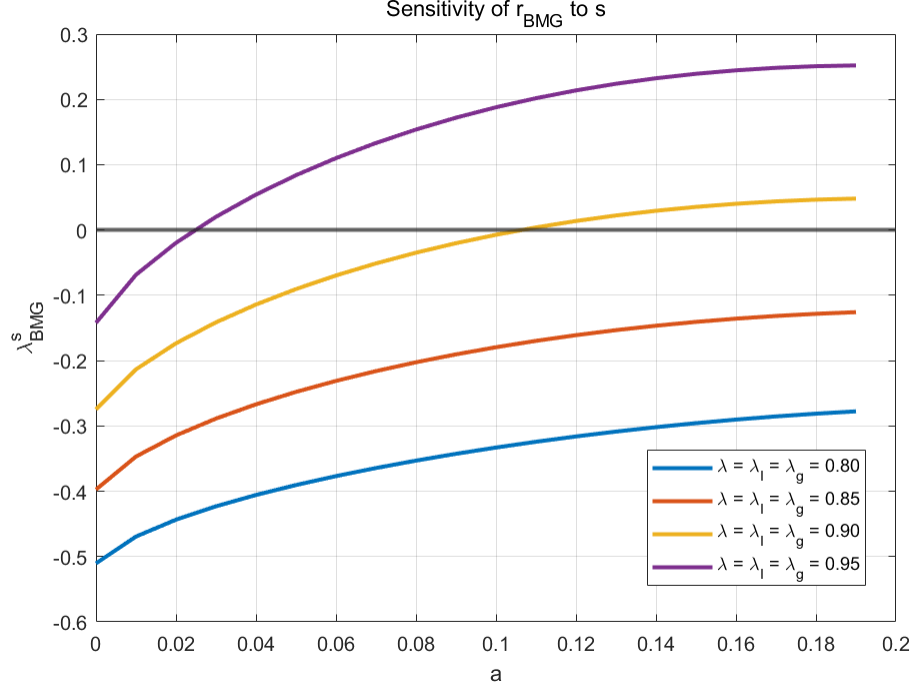


FIG. A.3 - Sensitivity of the brown-minus-green return to the pseudo Pareto weight

$D_2 - 1 \equiv \frac{E_2}{O_2} - 1$. Figure A.4 shows the equilibrium climate damage as a function of a under different parameter values of $\lambda = \lambda_I = \lambda_g$. We plot the damage in the steady state where shocks are muted.

To get a sense of the climate damage in real world, Nordhaus (2014) uses an estimate of 1.12% when the temperature anomaly increases to 2-degree Celsius. This is indicated by the horizontal dashed line in Figure A.4. Therefore, a reasonable upper bound of the parameter a in our model is around 0.002. Under this small value of a , the sensitivity λ_{BMG}^s is well below zero, as shown in Figure A.3.

A.9 Net exports

Using the definition of net exports of consumption goods, we obtain:

$$\begin{aligned}
 \frac{NX_1^C}{X_1} &= \frac{X_1^*}{X_1} - \left(\frac{1-\lambda}{\lambda} \frac{X_1}{Y_1} \right) \frac{Y_1}{X_1} = \frac{1-\lambda}{\lambda} \left(\frac{1}{\tilde{S}_1} - 1 \right) \\
 &= - \left(\frac{1-\lambda}{\lambda} \right) \left[\frac{(\exp\{\tilde{s}_1\} - 1)}{1 + (\exp\{\tilde{s}_1\} - 1)} \right].
 \end{aligned} \tag{A.33}$$

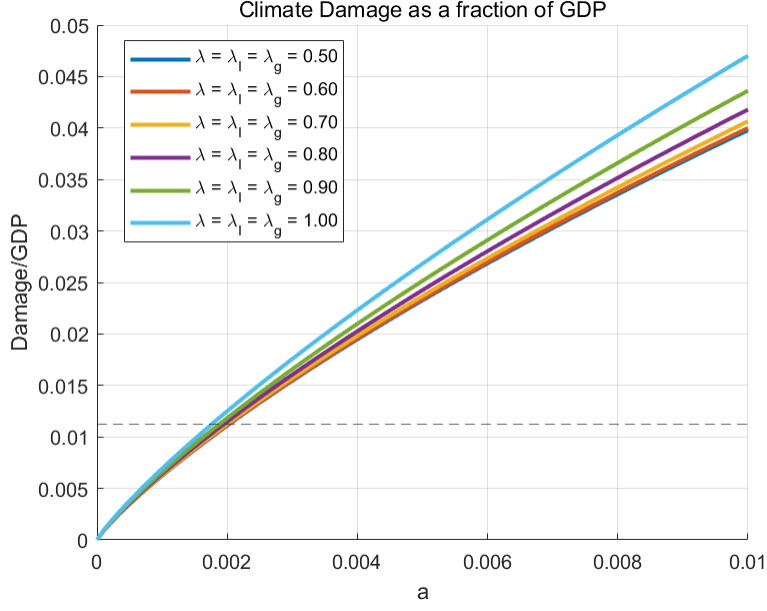


FIG. A.4. MODEL-IMPLIED CLIMATE DAMAGE AS A FRACTION OF GDP

Similarly, using the definition of net exports of investment goods, we get:

$$\begin{aligned} \frac{NX_1^I}{I_{x,1}} &= \frac{I_{y,1}}{I_{x,1}} - \left(\frac{1 - \lambda_I}{\lambda_I} \frac{I_{x,1}}{I_{x,1}^*} \right) \frac{I_{x,1}^*}{I_{x,1}} = \frac{1 - \lambda_I}{\lambda_I} \left(\frac{\tilde{S}_1 + \hat{\kappa}}{\hat{\kappa}\tilde{S}_1 + 1} - 1 \right) \\ &= - \left(\frac{1 - \lambda_I}{\lambda_I} \right) \left[\frac{(\hat{\kappa} - 1) \cdot (\exp \{\tilde{s}_1\} - 1)}{1 + \hat{\kappa} \cdot (\exp \{\tilde{s}_1\} - 1)} \right], \end{aligned} \quad (\text{A.34})$$

where $\hat{\kappa} = \frac{\hat{\lambda}}{1 - a - \hat{\lambda}} = \frac{\lambda - a\lambda_g}{1 - \lambda - a(1 - \lambda_g)} > 1$. We can characterize the response of net exports to the shocks in the following proposition.

Proposition 3. *Let $\gamma > 1$. If $b > 1$, a global climate news shock $\tilde{g} > 0$ decreases total net exports in the home country, implying that there is a reallocation of resources toward the country that hedges the most ex ante.*

Proof. Both net export functions in (A.33) and (A.34) are decreasing in $(\exp \{\tilde{s}_1\} - 1)$. Combining this observation with Lemma 1 establishes the result. $\square$

These derivations set the stage for our empirical analysis. For the home country, net ex-

ports of consumption goods are given by

$$\frac{NX_1^C}{X_1} = \frac{1-\lambda}{\lambda} \left(\frac{1}{\tilde{S}_1} - 1 \right) = \mathbf{const} - \frac{1-\lambda}{\lambda} e^{-\bar{s}_1} \tilde{S}_1.$$

By symmetry, net exports of the foreign country are

$$\frac{NX_1^{C*}}{Y_1^*} = \frac{1-\lambda}{\lambda} \left(\tilde{S}_1 - 1 \right) = \mathbf{const} + \frac{1-\lambda}{\lambda} e^{\bar{s}_1} \tilde{S}_1.$$

Thus,

$$\frac{NX_1^C}{X_1} - \frac{NX_1^{C*}}{Y_1^*} = \mathbf{const} - \frac{1-\lambda}{\lambda} (e^{\bar{s}_1} + e^{-\bar{s}_1}) \tilde{S}_1.$$

Given that

$$\tilde{s}_1 = \bar{s}_1 + \lambda_s^\theta \theta + \lambda_s^g \tilde{g},$$

it follows that

$$\frac{NX_1^C}{X_1} - \frac{NX_1^{C*}}{Y_1^*} = \mathbf{const} + \beta_\theta^{NX} \cdot 2\theta + \beta_g^{NX} (b-1) \tilde{g},$$

where, according to Lemma 1,

$$\begin{aligned} \beta_\theta^{NX} &= -\frac{1}{2} \frac{1-\lambda}{\lambda} (e^{\bar{s}_1} + e^{-\bar{s}_1}) \lambda_s^\theta > 0, \\ \beta_g^{NX} &= -\frac{1-\lambda}{\lambda} (e^{\bar{s}_1} + e^{-\bar{s}_1}) \frac{\lambda_s^g}{b-1} < 0. \end{aligned}$$

Similarly, the same relationship holds for net exports of investment goods. As a result, total net export differentials are positively related to a positive productivity shock ($\theta > 0$) and negatively related to a global climate news shock ($\tilde{g} > 0$), provided that $b > 1$.

A.10 Additional Model Results

Stochastic discount factor. We next examine how the stochastic discount factor responds to different shocks. Note that for the home country, the SDF at time 1 is given by

$$M_1 = \frac{\partial u_0 / \partial C_1}{\partial u_0 / \partial C_0}$$

where u_0 is the utility at time 0,

$$u_0 = (1-\beta) \log C_0 \tilde{D}_0^{-a} + \frac{\beta}{1-\gamma} \log E_0 [\exp \{u_1(1-\gamma)\}]$$

As a result, we get:

$$M_1 = \beta \frac{C_0}{C_1} \frac{\exp(u_1(1-\gamma))}{E_0[\exp(u_1(1-\gamma))]},$$

and in log-units:

$$m_1 = \log(\beta) + \log(C_0) - \log(C_1) + (1-\gamma)u_1 - \log E_0[\exp(u_1(1-\gamma))].$$

Substituting the expressions for $\log C_1$ and u_1 ,

$$m_1 = \text{const} - (\lambda\lambda_x^s - (1-\lambda)\lambda_{x^*}^s)\tilde{s}_1 + (1-\gamma)(\lambda_\theta\theta + \lambda_g\tilde{g}),$$

where all time-0 terms end up in the constant. In addition, note that

$$(1-\gamma)\lambda_\theta = \frac{1}{2}\lambda_s^\theta,$$

$$(1-\gamma)\lambda_g = (1-\gamma)(b\lambda_u^g + \lambda_u^s\lambda_s^g) = (1-\gamma)\underbrace{\left(\frac{b-1}{2}\lambda_u^g + \lambda_u^s\lambda_s^g\right)}_{\frac{1}{2}\lambda_s^g} + (1-\gamma)\frac{b+1}{2}\lambda_u^g.$$

Thus

$$m_1 = \text{const} + \underbrace{\left(\frac{1}{2} - (\lambda\lambda_x^s - (1-\lambda)\lambda_{x^*}^s)\right)}_{\frac{1}{2}\lambda_e^s}\tilde{s}_1 + \frac{1}{2}(1-\gamma)(1+b)\lambda_u^g\tilde{g}.$$

Similarly, the SDF for the foreign country is

$$m_1^* = \text{const} - \frac{1}{2}\lambda_e^s\tilde{s}_1 + \frac{1}{2}(1-\gamma)(1+b)\lambda_u^g\tilde{g}$$

The expressions for the two SDFs also verify that

$$\Delta e_1 = m_1 - m_1^* = \text{const} + \lambda_e^s\tilde{s}_1$$

We have shown that $\lambda_e^s > 0$ in Proposition 1. In addition, $\lambda_u^g = -\beta(1-\beta)a < 0$. Hence if we hold the Pareto weight constant, a global climate news shock increases the SDFs for both countries.

Proposition 4. *Holding $\tilde{g}$ constant, there is a one-to-one mapping between the SDF and the pseudo-Pareto weight $\tilde{S}_1$. An increase in $\tilde{S}_1$ corresponds to an increase in m_1 and a decrease in m_1^* .*

Thus, the productivity shock θ has an unambiguous effect on the SDF of the two countries.

The effect of the global climate news shock $\tilde{g}$ is unambiguous conditional on the following parameter restrictions.

Proposition 5. *Let $\gamma > 1$. If $b > 1$ ($b < 1$), a positive global climate news shock increases (decreases) the SDF of the home country relative to the foreign country.*

Appendix B. Data collection

TABLE B.1: Newspaper Twitter Handles by Country

Country	Newspaper name	Twitter handle	Language
Argentina	Clarín Diario Crónica Infobae La Nación	clarincom cronica infobae LANACION	Spanish
Australia	The Australian The Daily Telegraph The Australian Financial Review The Age	australian dailytelegraph FinancialReview theage	English
Brazil	Estadão Folha de S.Paulo GauchaZH Jornal O Globo	Estadao folha GauchaZH JornalOGlobo	Portuguese
Canada	The Globe and Mail Montreal Gazette Ottawa Citizen Toronto Star The Vancouver Sun	globeandmail mtlgazette OttawaCitizen TorontoStar VancouverSun	English
Switzerland	20 Minuten 24 heures Le Temps Neue Zürcher Zeitung	20min 24heuresch LeTemps NZZ	German French
Chile	El Mercurio El Mostrador La Tercera	ElMercurio_cl elmostrador latercera	Spanish
China	China Daily People's Daily China China Xinhua News	ChinaDaily PDChina XHNews	English
Colombia	El Colombiano El Espectador El Heraldó Colombia	elcolombiano elespectador elheraldoco	Spanish
Germany	BILD Frankfurter Allgemeine Handelsblatt ZEIT ONLINE	BILD faznet handelsblatt zeitonline	German
Spain	ABC El País El Mundo La Vanguardia	abc_es el_pais elmundoes LaVanguardia	Spanish
France	Le Figaro Le Parisien Le Monde Libération	Le_Figaro le_Parisien lemondefr libe	French
China (Hong Kong)	Hong Kong Free Press RTHK News	HongKongFP rthk_enews	English

Country	Newspaper name	Twitter handle	Language
	SCMP News	SCMPNews	
India	Economic Times Hindustan Times The Hindu Times of India	EconomicTimes htTweets the_hindu timesofindia	English
Italy	Corriere della Sera Repubblica Il Sole 24 ORE	Corriere repubblica sole24ore	Italian
Japan	Asahi Shimbun AJW The Japan Times Japan Today The Japan News	AJWasahi japantimes JapanToday The_Japan_News	English
Korea	The Korea JoongAng Daily The Korea Times The Korea Herald Yonhap News	JoongAngDaily koreatimescokr TheKoreaHerald YonhapNews	English
Mexico	El Universal La Jornada Milenio Reforma	El_Universal_Mx lajornadaonline Milenio Reforma	Spanish
Norway	Aftenposten Dagbladet VG	Aftenposten dagbladet vgnett	Norwegian
New Zealand	Dominion Post ODT - Otago Daily Times The Press Newsroom	DomPost odtnews PressNewsroom	English
Portugal	Correio da Manhã Expresso Jornal de Notícias	cmjornal expresso JornalNoticias	Portuguese
Saudi Arabia	Al Jazirah Alwatan OKAZ	al_jazirah AlwatanSA OKAZ_online	Arabic
Sweden	Dagens Nyheter Goteborgs Posten Svenska Dagbladet Sydsvenskan	dagensnyheter GoteborgsPosten SvD sydsvenskan	Swedish
United Kingdom	BBC News Financial Times The Guardian The Times and The Sunday Times	BBCNews FinancialTimes guardiannews thetimes	English
United States	Boston Globe Chicago Tribune Dallas News Houston Chronicle LA Times Miami Herald NY Times SF Chronicle USA Today	BostonGlobe chicagotribune dallasnews HoustonChron latimes MiamiHerald nytimes sfchronicle USATODAY	English

TABLE B.2. AI-GENERATED SENTENCES

	Physical Risk	Transition Risk
1	Hurricanes are becoming more intense	New carbon taxes increase energy costs
2	Rising sea levels threaten coastal cities	Governments are enforcing stricter emission limits
3	Wildfires are spreading faster each year	Climate policies impact fossil fuel investments
4	Floods are more frequent in urban areas	Carbon pricing is affecting industry competitiveness
5	Droughts are lasting longer worldwide	Environmental regulations are tightening worldwide
6	Heatwaves are breaking records globally	Firms must comply with stricter sustainability laws
7	Cyclones are forming earlier than expected	Carbon offsets are becoming a mandatory requirement
8	Tornadoes are appearing in new regions	Subsidies for renewable energy are rising
9	Storm surges endanger coastal communities	Stricter reporting rules affect corporate disclosures
10	Melting glaciers contribute to rising seas	Regulatory risks impact financial markets
11	Arctic ice is melting at record speeds	Fossil fuel demand is declining
12	Permafrost is thawing, releasing methane	Renewable energy adoption is accelerating
13	Global temperatures are steadily increasing	Investors are divesting from coal industries
14	Extreme cold events are less frequent	Gas prices fluctuate due to energy transitions
15	Ocean temperatures are rising rapidly	Green bonds are reshaping capital markets
16	Coral reefs are dying due to warming seas	The electric vehicle market is expanding rapidly
17	Changing seasons affect crop production	Oil companies face increasing financial pressure
18	More frequent heatwaves strain electricity grids	Hydrogen energy is gaining policy support
19	Polar vortex disruptions cause unusual cold	Nuclear power investments are rising
20	Snowfall patterns are shifting globally	Battery storage is critical for renewable energy growth
21	Rivers are drying up due to prolonged droughts	Energy efficiency technologies are reshaping industries
22	Flash floods destroy homes and roads	Carbon capture projects are scaling up
23	Drinking water sources are shrinking	Solar power efficiency is improving rapidly
24	Groundwater levels are depleting	Wind farms are replacing coal plants
25	Heavy rainfall increases landslide risks	Smart grids are enabling decentralized energy systems
26	Freshwater lakes are shrinking worldwide	Climate-friendly materials are replacing traditional ones
27	Ice sheets are collapsing faster than expected	Green hydrogen is becoming more viable
28	Saltwater intrusion threatens freshwater supplies	Electrification of transportation is accelerating
29	Dams are failing under extreme weather	Sustainable agriculture technologies are emerging
30	Reservoirs are running dry in many regions	Circular economy practices are gaining traction
31	Desertification is expanding in dry regions	Lawsuits against polluting companies are increasing
32	Livestock suffers from water shortages	Investors are demanding climate risk disclosures
33	Pests spread as temperatures rise	Regulators require firms to report climate risks
34	Changing climates affect coffee production	Non-compliance with emissions laws leads to penalties
35	Soil erosion worsens with heavy rain	Litigation risks threaten fossil fuel companies
36	Crop yields are declining due to heat stress	Banks must assess climate risks in lending portfolios
37	Unpredictable rainfall disrupts farming	Corporate climate pledges face legal scrutiny
38	Fisheries collapse due to warming oceans	Greenwashing claims lead to reputational damage
39	Extreme weather damages food supply chains	Disclosure failures result in investor lawsuits
40	More frequent frost events harm fruit crops	Companies face pressure to adopt net-zero targets
41	Roads crack under extreme heat	Job losses in fossil fuel industries are rising
42	Floods overwhelm drainage systems	Green jobs are reshaping the labor market
43	Airports close due to extreme weather	Energy prices are volatile due to policy shifts
44	Power outages increase with heatwaves	Consumer preferences are shifting toward sustainability
45	Rising insurance costs strain homeowners	Supply chains are adapting to climate policies
46	Heat damages rail tracks and highways	Financial institutions are reassessing climate risks
47	Coastal erosion threatens real estate	Insurance premiums rise due to climate-related policies
48	Ports are closing due to severe storms	Carbon-intensive industries face declining revenues
49	Cold snaps increase energy demand	Developing economies struggle with climate transition costs
50	Business disruptions rise with climate events	Global trade patterns are shifting due to decarbonization

Notes: This table shows sentences randomly generated by ChatGPT about climate physical and transition risk.

TABLE B.3. TOP DAILY INNOVATIONS IN THE GLOBAL CLIMATE NEWS INDEX

Date	ΔCAI	Event	Description	Tweet Examples
2017-01-02	1.246	Trump transition and expected rollback of U.S. climate policy	Climate coverage centered on how U.S. climate action would change under the incoming Trump administration, including expected attacks on Obama-era policies such as the Clean Power Plan and broader uncertainty over U.S. climate leadership.	“Perhaps the biggest threat to the climate agenda is not anything that Trump might do, but rather what he will not do.” (@nytimes, US) “Can one of the most promising technologies for fighting global warming survive during the Trump administration?” (@nytimes, US)
2016-09-26	1.169	World Environmental Health Day	World Environmental Health Day drew attention to environmental risks to human health, coincided with major WHO coverage of air pollution as a global environmental-health and climate risk.	“Support for world leading action on climate change has skyrocketed.” (@theage, AU) “Serious levels of air pollution recorded across Hong Kong.” (@SCMPNews, HK) “Climate chief: UK must not use Brexit to water down environment laws.” (@guardiannews, UK)
2015-11-30	0.715	COP21 Paris climate conference opens	The Paris UN Climate Conference began, launching the negotiations that led to the Paris Agreement. The spike reflects global attention to climate diplomacy and the expected new international climate framework.	“Negotiators from 195 countries are meeting in Paris to hammer out a new climate pact.” (@WSJ, US) “World leaders call for swift action to reach climate deal at #ParisClimateConference.” (@XHNews, CN) “150 heads of state seek the first global pact against climate change.” (@el_pais, ES, orig. Spanish)
2018-10-08	0.667	IPCC Special Report on Global Warming of 1.5°C	The IPCC released its 1.5°C special report, highlighting the large difference between 1.5°C and 2°C warming and the need for rapid, far-reaching emissions reductions.	“Global warming is likely to reach 1.5°C between 2030 and 2052 if it continues to increase at the current rate.” (@the_hindu, IN) “Global CO ₂ emissions must almost halve within 12 years to avoid catastrophic loss of coral reefs and Arctic ice, says a UN report.” (@thetimes, UK)
2015-11-09	0.614	Record greenhouse-gas concentrations and climate-poverty warning	Climate news emphasized record atmospheric greenhouse-gas concentrations and a World Bank warning that climate change could push more than 100 million people into poverty by 2030, reinforcing the urgency before COP21.	“Concentration of greenhouse gases in the Earth’s atmosphere hit another record high in 2014.” (@USATO-DAY, US) “Climate change could push more than 100 million people into extreme poverty by 2030.” (@BostonGlobe, US) “The World Bank warns: climate change could push 100 million people into poverty.” (@faznet, DE, orig. German)

Note: The table reports the ten largest daily innovations in the global climate attention index. Event labels are based on manual inspection of major climate-related news on or around each spike date. The final column reports representative tweets posted on the corresponding day. Non-English tweets are reported in English translation; original languages noted in parentheses.

TABLE B.4. TOP DAILY INNOVATIONS IN THE GLOBAL CLIMATE NEWS INDEX (CONT.)

Date	ΔCAI	Event	Description	Tweet Examples
2021-08-09	0.496	IPCC AR6 Working Group I report	The IPCC released the physical-science basis of its Sixth Assessment Report, warning that climate change was widespread, rapid, and intensifying.	“Code red for humanity”: global warming was dangerously close to being out of control.” (@OttawaCitizen, CA) “Rising seas, melting ice caps are ‘unequivocally’ driven by human greenhouse gas emissions, a U.N. climate panel report said.” (@WSJ, US)
2015-08-24	0.484	Obama clean-energy actions after the Clean Power Plan	The White House announced additional renewable-energy and energy-efficiency initiatives, reinforcing the U.S. mitigation agenda shortly after the Clean Power Plan and ahead of COP21.	“White House announces plans to boost renewable energy use.” (@globeandmail, CA) “Renewable energy on PM’s US agenda.” (@time-sofindia, IN)
2017-06-05	0.468	World Environment Day and Paris withdrawal reaction	World Environment Day 2017 occurred immediately after the U.S. announcement that it would withdraw from the Paris Agreement, generating broad climate and environmental-policy coverage.	“We Are Still In’ say Google, Facebook, Amazon and Apple when it comes to Paris climate agreement.” (@chicagotribune, US) “Kerry, Gore denounce Trump for exiting Paris climate accord.” (@japantimes, JP) “US pullout of #ParisAgreement does not call into question the worldwide dynamic established in the fight against climate change.” (@XHNews, CN)
2016-07-04	0.455	Paris Agreement adequacy and fossil-fuel transition debate	Coverage questioned whether the Paris Agreement’s targets could be met without a rapid decline in fossil-fuel use and a large expansion of renewable energy.	“Island nation in the central Pacific planning its own demise in face of climate change.” (@TorontoStar, CA) “The Paris climate accord will likely enter into force much sooner than expected.” (@faznet, DE, orig. German)
2015-03-16	0.436	Cyclone Pam and climate-disaster risk in Vanuatu	After Cyclone Pam devastated Vanuatu, news coverage linked the disaster to climate vulnerability, sea-level rise, and the need for resilience in small island states.	“Vanuatu’s president blames climate change for extreme weather.” (@guardiannews, UK) “UN: 24 dead, 3,300 displaced in Vanuatu after Cyclone Pam devastates the islands.” (@washingtonpost, US) “Cyclone Pam: Vanuatu’s president blames climate change.” (@le_Parisien, FR, orig. French)

Note: The table reports the ten largest daily innovations in the global climate attention index. Event labels are based on manual inspection of major climate-related news on or around each spike date. The final column reports representative tweets posted on the corresponding day. Non-English tweets are reported in English translation; original languages noted in parentheses.

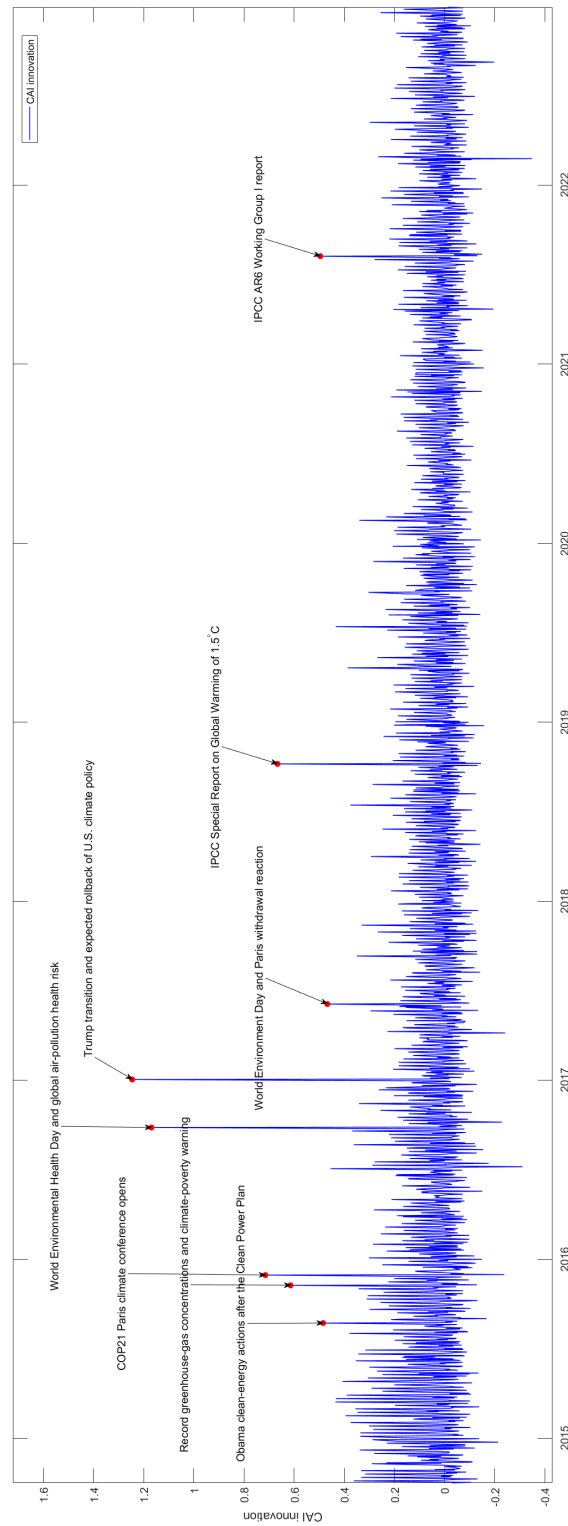


FIG. B.2. GLOBAL CLIMATE ATTENTION INDEX: LABELLED TOP SPIKES

Note: This figure shows innovations to our equal-weights Global Climate Attention Index. Red circles indicate the selected largest spikes, with text arrows identifying the associated climate-related news events. The labelled events include major climate-policy developments, IPCC reports, COP21-related news, environmental-health news, and climate-disaster coverage.

TABLE B.5. REGRESSION COEFFICIENTS OF THE EQUITY RETURNS ON GLOBAL TOP CLIMATE DAYS

	(1) $k = 1$	(2) $k = 2$	(3) $k = 3$	(4) $k = 4$	(5) $k = 5$	(6) $k = 6$	(7) $k = 7$	(8) $k = 8$	(9) $k = 9$	(10) $k = 10$
$\gamma_{0,k}$	-2.518*** (0.544)	-0.914** (0.416)	-0.362 (0.340)	-1.136*** (0.329)	-1.108*** (0.288)	-0.851*** (0.277)	-0.639** (0.252)	-0.847*** (0.247)	-0.819*** (0.235)	-0.533** (0.215)
$\gamma_{1,k}$	0.413** (0.164)	0.095 (0.118)	0.004 (0.096)	0.014 (0.089)	0.110 (0.078)	0.055 (0.075)	0.052 (0.069)	0.011 (0.067)	0.053 (0.064)	0.012 (0.060)
$\gamma_{2,k}$	-0.321*** (0.068)	-0.115** (0.046)	-0.043 (0.036)	-0.027 (0.033)	-0.083*** (0.029)	-0.039 (0.027)	-0.040 (0.025)	0.011 (0.023)	-0.001 (0.023)	0.019 (0.021)
Constant	-0.139 (0.106)	-0.371*** (0.082)	-0.464*** (0.071)	-0.368*** (0.062)	-0.366*** (0.057)	-0.382*** (0.052)	-0.387*** (0.049)	-0.361*** (0.046)	-0.364*** (0.043)	-0.382*** (0.041)
N	22612548	22356529	22106056	21866517	21638829	21441534	21246682	21057132	20874990	20696662
R^2	0.002	0.003	0.004	0.006	0.007	0.009	0.010	0.012	0.013	0.015

Notes: This table shows the coefficients of the following regression:

$$r_{t,t+k}^{i,c} = \alpha_k^i + \zeta_{t,k}^{i,c} \cdot \mathbb{1}_t^{Top} + \epsilon_{t,k}^{i,c}, \quad (\text{B.1})$$

$$\zeta_{t,k}^{i,c} = \gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k} \cdot \beta_c) \cdot E_{i,t_a}, \quad (\text{B.2})$$

with $k = 1, 2, \dots, 10$. Standard errors are clustered at the industry $\times$ date and firm level. We follow the Fama-French 49 classification.

Appendix C. Additional Empirical Results

In this Appendix, we share additional empirical results. We split the section in three subsections.

C.1. Robustness and sensitivity

Significance of climate attention β s. Figure C.1 depicts the values of the t-stat used to test the following null assumption $H_0 : \beta_i - \beta_j = 0, i \neq j$ for all possible country pairs in our dataset. The test is based on a joint estimation of equation (13) for each possible pair of country i and j . Dark colors refer to values of the t-stat with an associated confidence level smaller than 10%. We depict the country-pair test results only in the bottom portion of the figure because the upper portion is symmetric.

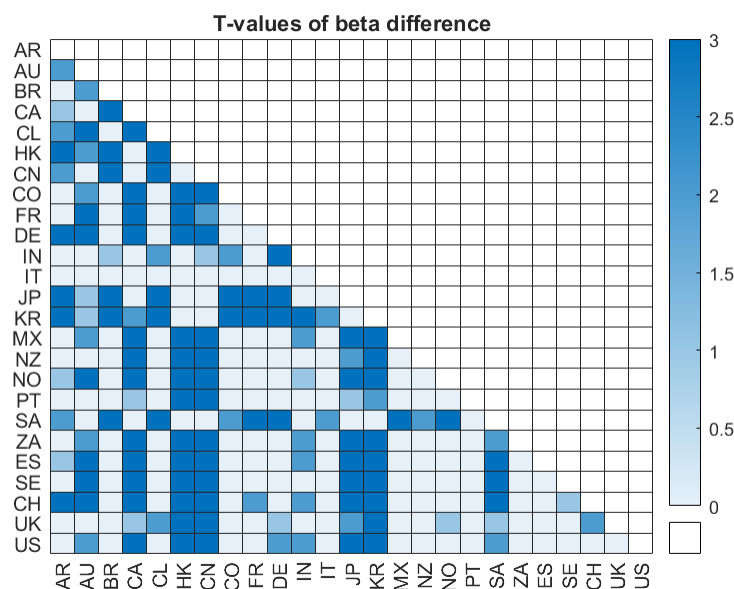


FIG. C.1. HETEROGENEOUS EXPOSURES ACROSS COUNTRY PAIRS

Notes: This figure shows the values of the t-stat used to test the following null assumption $H_0 : \beta_i - \beta_j = 0, i \neq j$ for all possible country pairs in our dataset. The test is based on a joint estimation of equation (13) for each possible pair of country i and j . Dark colors refer to values of the t-stat with an associated confidence level smaller than 10%. We depict the country-pair test results only in the bottom portion of the figure because the upper portion is symmetric.

The role of sentiment. Figure C.2 plots the implied FX response evaluated at the maximum,

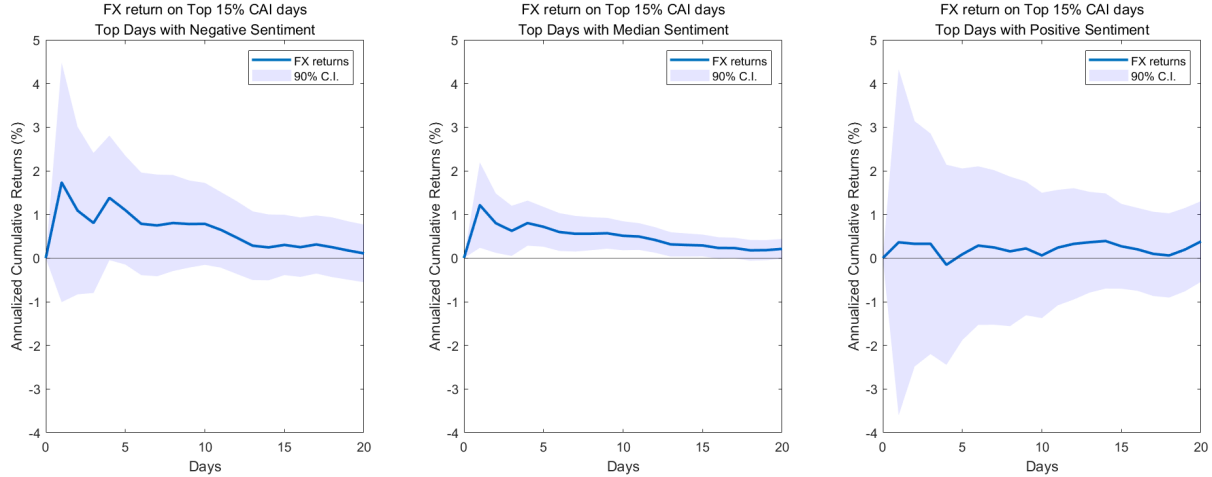


FIG. C.2. FX RESPONSES BY NEGATIVE CLIMATE SENTIMENT

Note: This figure reports average cumulative FX returns after top CAI days, allowing the response to vary with the share of negative climate-related tweets. The regression interacts the top-CAI-day indicator with S_t^{neg} , the share of negative climate-related tweets among non-neutral climate-related tweets on day t . The three panels plot the implied response evaluated at the maximum, median, and minimum observed values of S_t^{neg} across top CAI days. The low-negative-sentiment panel does not correspond to positive-sentiment days; it represents top CAI days with the lowest observed negative-sentiment share, which remains above one half. Shaded areas denote 90% confidence intervals.

median, and minimum observed values of S_t^{neg} across top CAI days. The response is strongest when the negative-sentiment share is high, close to the benchmark response at the median, and statistically weak when the negative-sentiment share is low.

FX second moment. We depict the reaction of volatility around top-days in Figure C.3. The point estimates are positive throughout the event window, suggesting that top CAI days are associated with elevated FX realized volatility.

Robustness of FX response. Figures C.4(a)-C.4(b) are constructed as figure 9 reported in main text, except that we compute our global CAI by weighting our countries by their GDP and tweets volume, respectively.

In figures C.5(a)-C.5(b) we consider top-10% and top-20% days, respectively. Our results are not sensitive to the percentile that we choose.

Figures C.6(a)-C.6(b) confirm that our results are not sensitive to the choice of the significance level that we use in order to select country-pairs with different exposure coefficients.

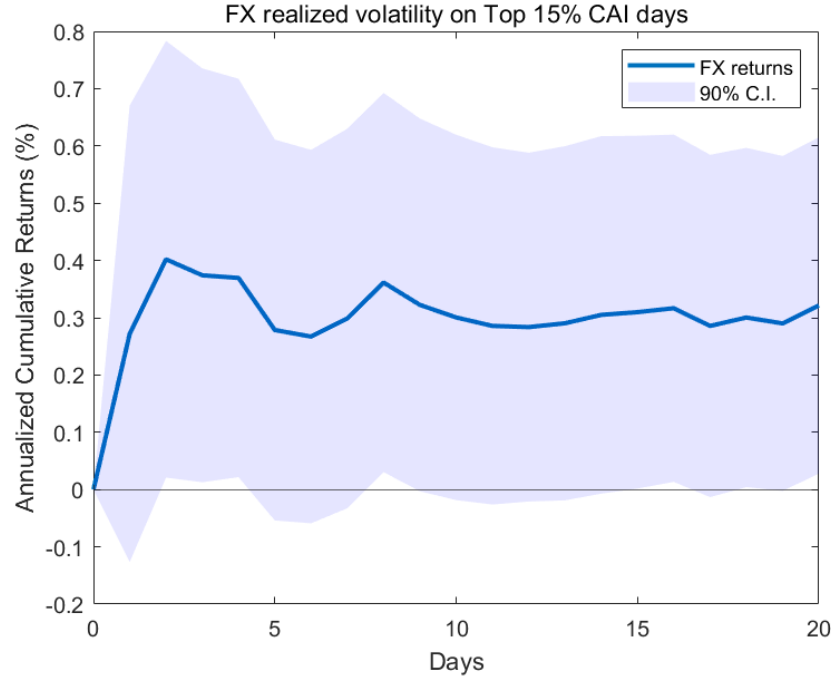


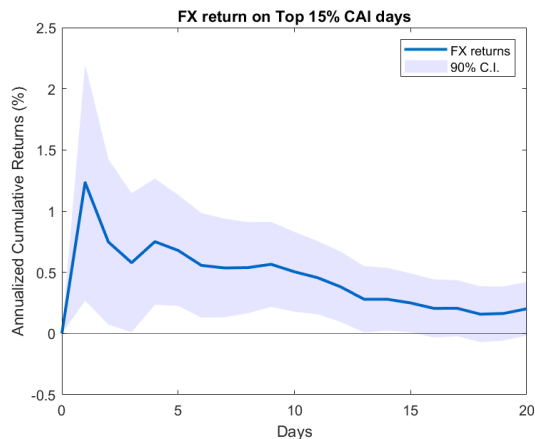
FIG. C.3. FX REALIZED VOLATILITY AROUND TOP CAI DAYS

Note: This figure reports the response of event-window realized FX volatility after top CAI days. Realized volatility is constructed from squared daily bilateral FX returns and annualized as $100 \times \sqrt{250/k \sum_{h=1}^k (r_{t+h}^{i,j})^2}$. The regression follows the benchmark FX specification, replacing cumulative FX returns with realized volatility as the dependent variable. The sample uses top 15% global CAI days. Shaded areas denote 90% confidence intervals.

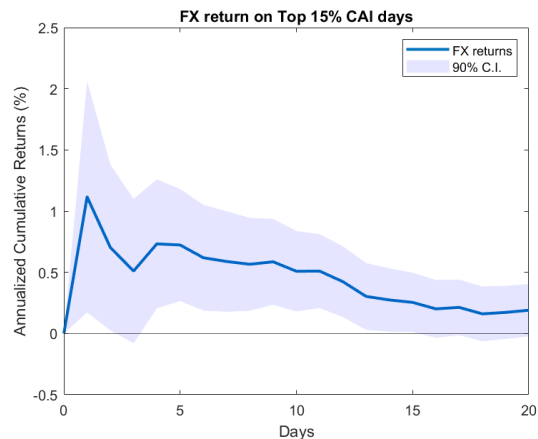
Our results apply regardless of whether we include all country pairs (including those with modest heterogeneity in beta) or just those with a more pronounced degree of heterogeneity (5% confidence level).

Figure C.8(a) shows that our results are confirmed also when we aggregate five Euro Area countries in our dataset (DE, FR, IT, PT, ES) in just one entity with one currency. Most importantly, figure C.8(b) confirms that our results are not driven by vulnerability, but rather by the unique information content of our estimated exposure coefficients.

Figure C.9(a) confirms our main findings when using exposures estimated at the monthly frequency. Figure C.9(b) is constructed as figure 9 in main text, but it uses weekly observations and it includes weekends. In figure C.10, we examine local top-attention days, as opposed to global top-attention days. We find that local shocks have no significant impact on FX markets.



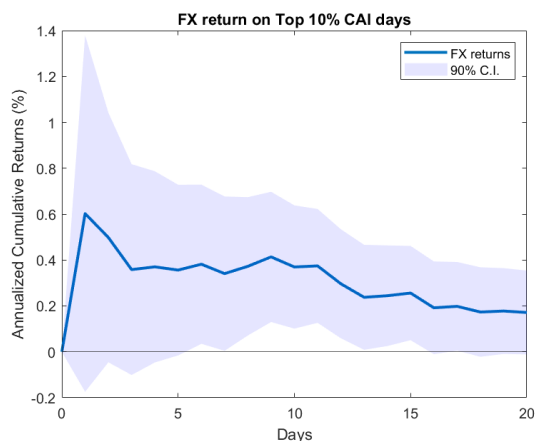
C.4(a) GDP-Weighted



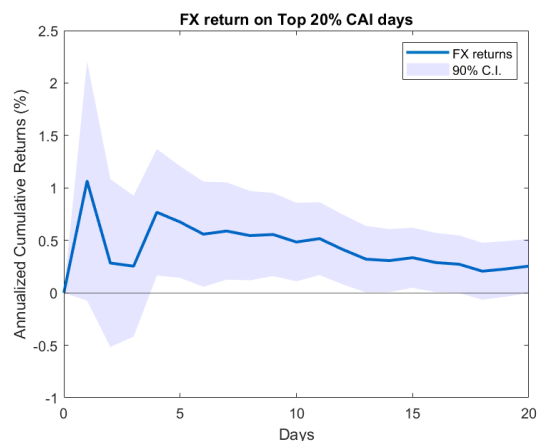
C.4(b) Volume-Weighted

FIG. C.4. FX AND WORLD CAI: THE ROLE OF THE WEIGHTING SCHEME.

Notes: average cumulative return in exchange rates after days with a positive spike in the global CAI. Panel A is constructed as figure 9 except that country exposures are obtained by regressing country-level CAI on the GDP-weighted global CAI. Panel B is constructed as figure 9 except that country exposures are obtained by regressing country-level CAI on the Twitter Volume-weighted global CAI.



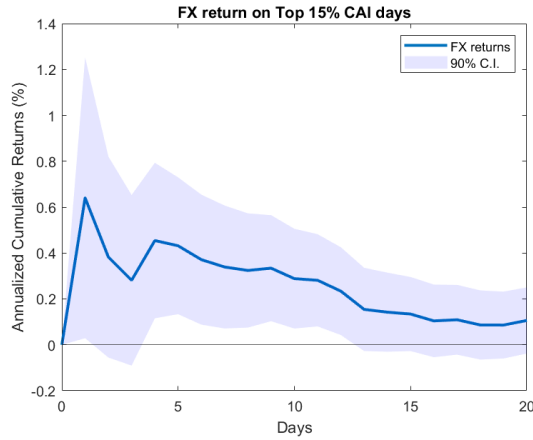
C.5(a) Top 10% Days



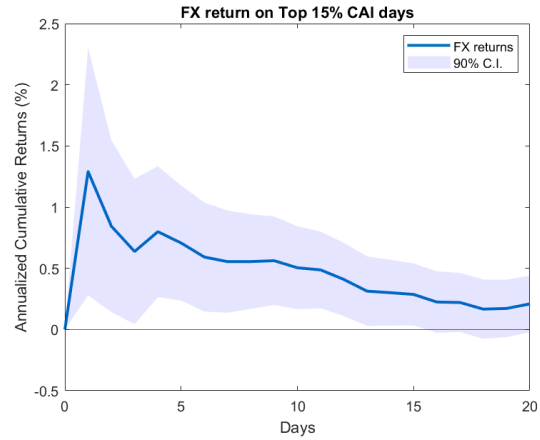
C.5(b) Top 20% Days

FIG. C.5. FX AND WORLD CAI: THE ROLE OF THE THRESHOLD.

Notes: average cumulative return in exchange rates around days with a positive spike in the global CAI. Panel A selects the top CAI days according to a top-10% threshold, while Panel B selects the top CAI days according to a top-20% threshold. For more details, see the notes under figure 9.



C.6(a) All Country Pairs



C.6(b) β Different at 5%

FIG. C.6. FX AND WORLD CAI: THE ROLE OF β .

Notes: each panel shows average cumulative return in FX rates after days with a positive spike in global CAI. Panel A uses all 300 country pairs. Panel B selects country pairs such that their betas are significantly different at least at the 5% confidence level. For more details, see the notes under figure 9.

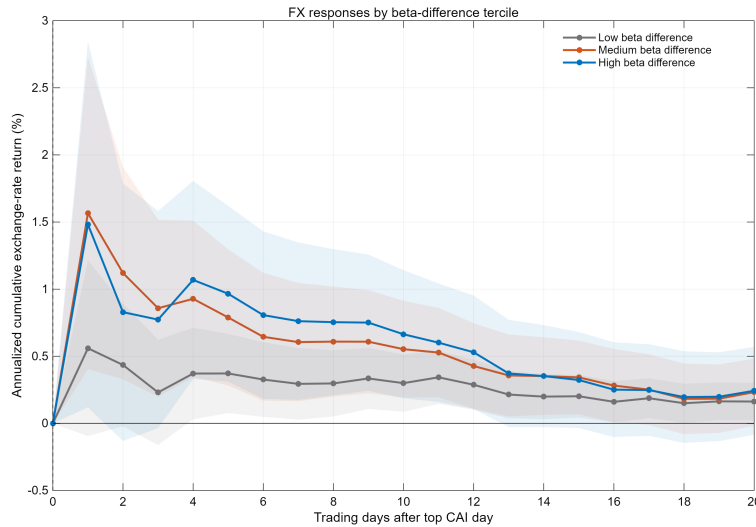
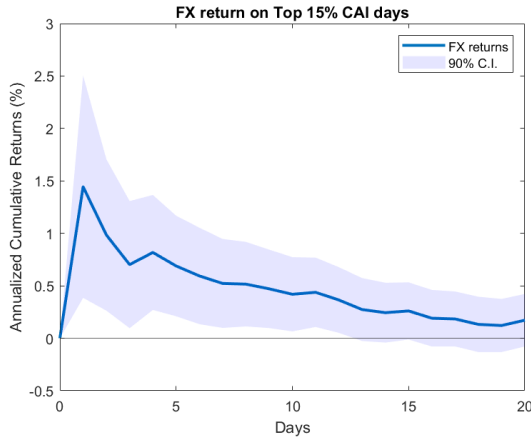
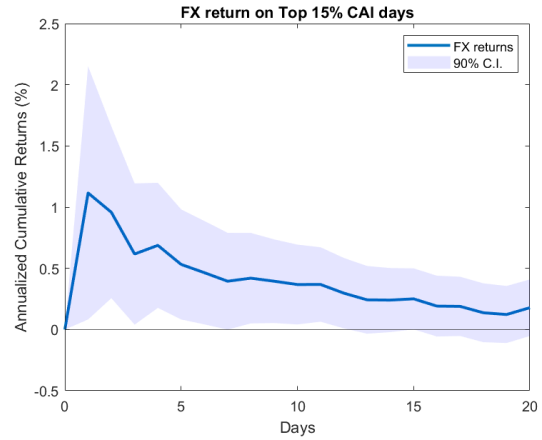


FIG. C.7. THREE GROUPS OF COUNTRY PAIRS

Notes: this panel shows average cumulative return in FX rates after days with a positive spike in global CAI. We split our 138 benchmark country pairs into equally sized terciles of the beta difference (46 pairs per tercile). The beta-difference ranges are [0.395, 1.090], (1.090, 1.506], and (1.506, 2.441]. For more details, see the notes under figure 9.



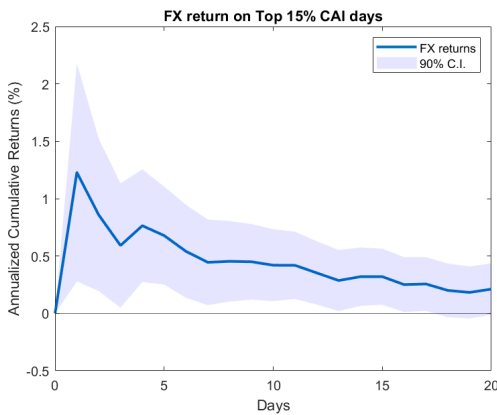
C.8(a) Combine EU



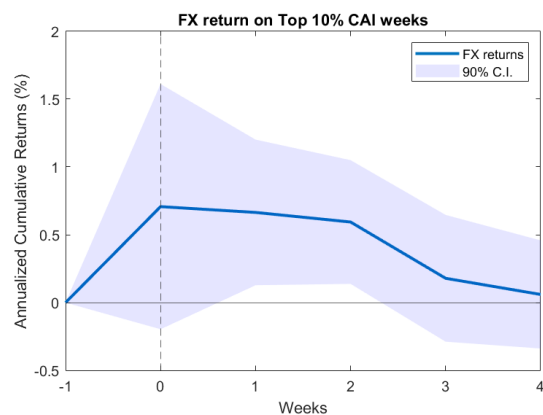
C.8(b) Residual Betas

FIG. C.8. FX AND WORLD CAI: ADDITIONAL ROBUSTNESS.

Notes: Panel (a) shows the average cumulative return in FX rates after days with a positive spike in the global CAI, combining the five EU countries (DE, FR, IT, PT, ES) by averaging their β exposures. Panel (b) uses the residual component of country exposures after regressing on the vulnerability index from the University of Notre Dame to repeat the analysis. See notes to figure 9 for details.



C.9(a) Monthly Data



C.9(b) Weekly Data

FIG. C.9. FX AND INNOVATIONS TO WORLD CAI: FREQUENCY ANALYSIS.

Notes: Panel (a) shows the average cumulative return in exchange rates after days with a positive spike in the global CAI, using monthly data to estimate country exposures instead of quarterly data. Panel (b) shows the average cumulative return after weeks with a positive spike in the global CAI, considering all 300 country pairs and selecting the top-10% weeks. To avoid confounding effects, weeks are chosen so that they do not overlap over a 5-week window. See notes to figure 9 for details.

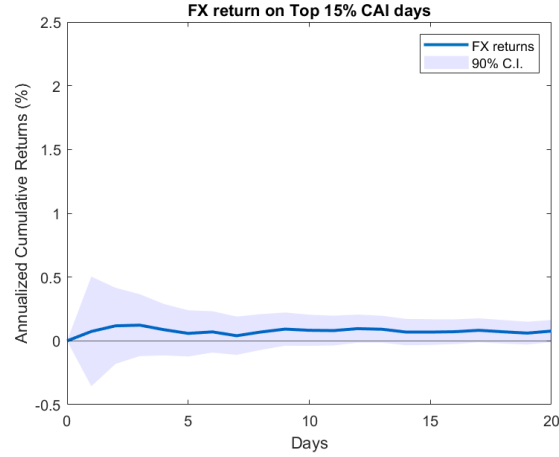


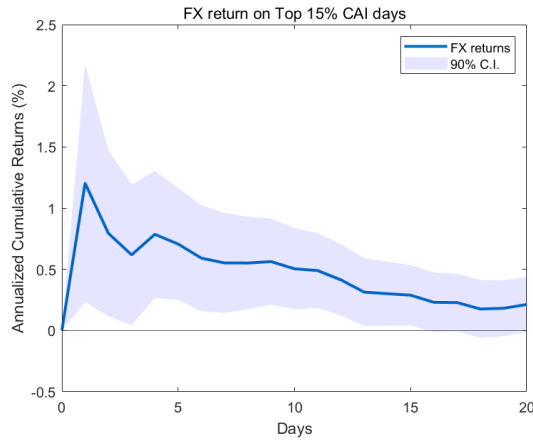
FIG. C.10. FX AND INNOVATIONS TO LOCAL CAI.

Notes: This figure shows the average cumulative return in exchange rates after days with a positive spike in the local CAI, defined as the residual from regressing a country's CAI on the global CAI. Specifically, we estimate

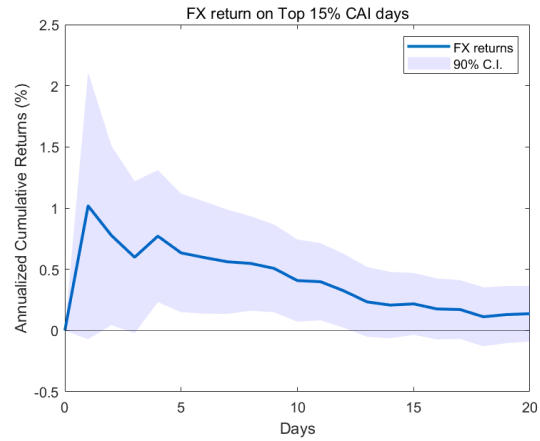
$$\Delta e_{t,t+k}^{i,j} = a_k^{i,j} + b_k \cdot \left(\mathbb{1}_{i,t}^{Top} - \mathbb{1}_{j,t}^{Top} \right) + \epsilon_{t,k}^{i,j},$$

For each country i , the local top-day dummy $\mathbb{1}_{i,t}^{Top}$ is based on the top-15% days of the country's local CAI. See notes to figure 9 for more details.

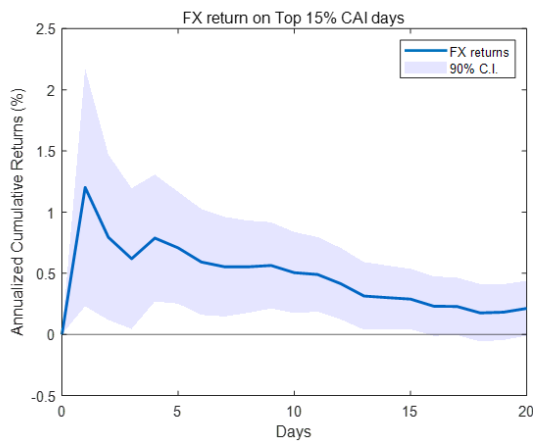
Figure C.11 confirms our main findings when we control for days with macroeconomic announcements, as described in section 4.2 in the main text.



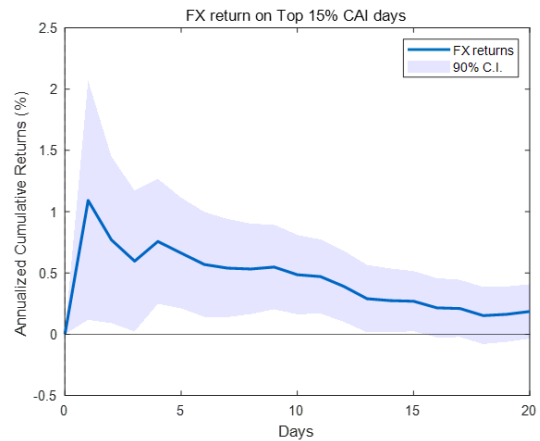
C.11(a) Controlling for Pair-Level Announcement Days



C.11(b) Excluding Pair-Level Announcement Days



C.11(c) Controlling for Pair-Level Announcement Days



C.11(d) Excluding Pair-Level Announcement Days

FIG. C.11. FX AND WORLD CAI - MACRO ANNOUNCEMENT DAYS

Notes: This figure reports average cumulative returns in exchange rates after days with a positive spike in the global CAI, controlling for country pair-level macro announcement days. Panel (a) and (c) use the specification that includes an announcement-day indicator as an additional control. Panel (b) and (d) use the specification that redefines top climate-attention days by excluding days that overlap with announcement days. In the top two panels, a day is classified as an announcement day if *at least one* country in the currency pair has a country-specific macroeconomic announcement; for France, Germany, Italy, Portugal, and Spain, EU-wide announcements are also included. In the bottom two panels, a day is classified as an announcement day only if *both* countries in the currency pair have macroeconomic announcements.

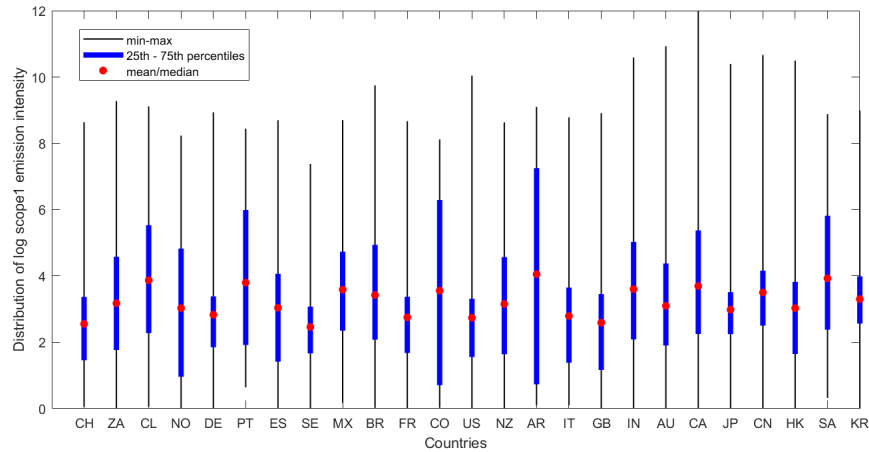
TABLE C.1. MAPPING OF BLOOMBERG ANNOUNCEMENT CATEGORIES TO BROAD CATEGORIES

Broad category	Bloomberg source category(ies)	Representative events
GDP Growth & Productivity	National Accounts	GDP, Economic Activity Index, Exports/Imports, Private Consumption, Government Spending, Net Exports Contribution to GDP, Inventory Contribution to GDP, Aggregate Supply and Demand
Labor Market	Labor Market	Unemployment Rate, Employment Change, Participation Rate, Average Hourly Earnings, Wages, Job Vacancies, Labour Costs, Average Weekly Hours, Challenger Job Cuts
Monetary	Monetary, Financial	ECB Main Refinancing Rate, FOMC Rate Decision, Bank of Canada Rate Decision, SNB Policy Rate, RBA Cash Rate Target, BOJ Policy Balance Rate, BOE Gilt Purchase Target, RBI Repurchase Rate, BOK Base Rate, Money Supply (M2/M3), Foreign Reserves
Prices & Inflation	Prices	CPI, CPI Core, CPI EU Harmonized, PPI, Wholesale Price Index, Import/-Export Price Index, GDP Deflator, Core PCE Price Index

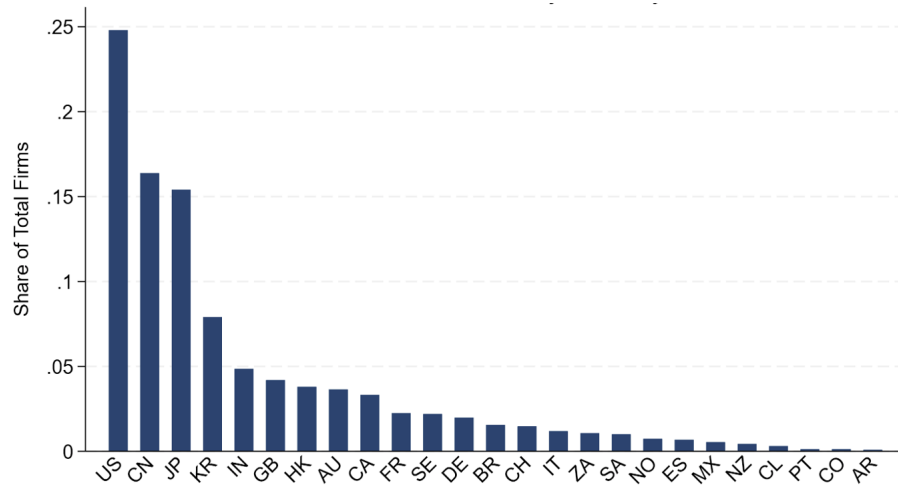
Notes: This table summarizes how each Bloomberg-classified macroeconomic announcement type is mapped into one of four broad categories used in Section 4.2. Eight events (of 495) are reassigned across their original Bloomberg source sheet based on economic content rather than the source label. The full event-level mapping (495 Bloomberg event types across 25 countries) is available in the paper's replication package.

Robustness of the equity analysis. Figure C.12(a) illustrates the distribution of emission intensity across firms, segmented by country. Emission intensity is calculated as the ratio of tons of CO2 emissions to sales, derived from the Trucost dataset. The dataset includes 16,774 firms, providing a comprehensive overview of country-level emission profiles.

Figure C.13 shows substantial variation in the distribution of emission intensity across industries. Figure C.14 confirms that our main results are driven by industry variation. Figure



C.12(a) Emission Intensity by Country



C.12(b) Firms Distribution across Countries

FIG. C.12. FIRMS DATA BY COUNTRY

Notes: The top (bottom) portion of this figure shows the distribution of emission intensity across firms (share of firm numbers) by country. Emission intensity is computed as tons of CO2 divided by sales and it is obtained from the Trucost dataset. We have a total of 16,774 firms.

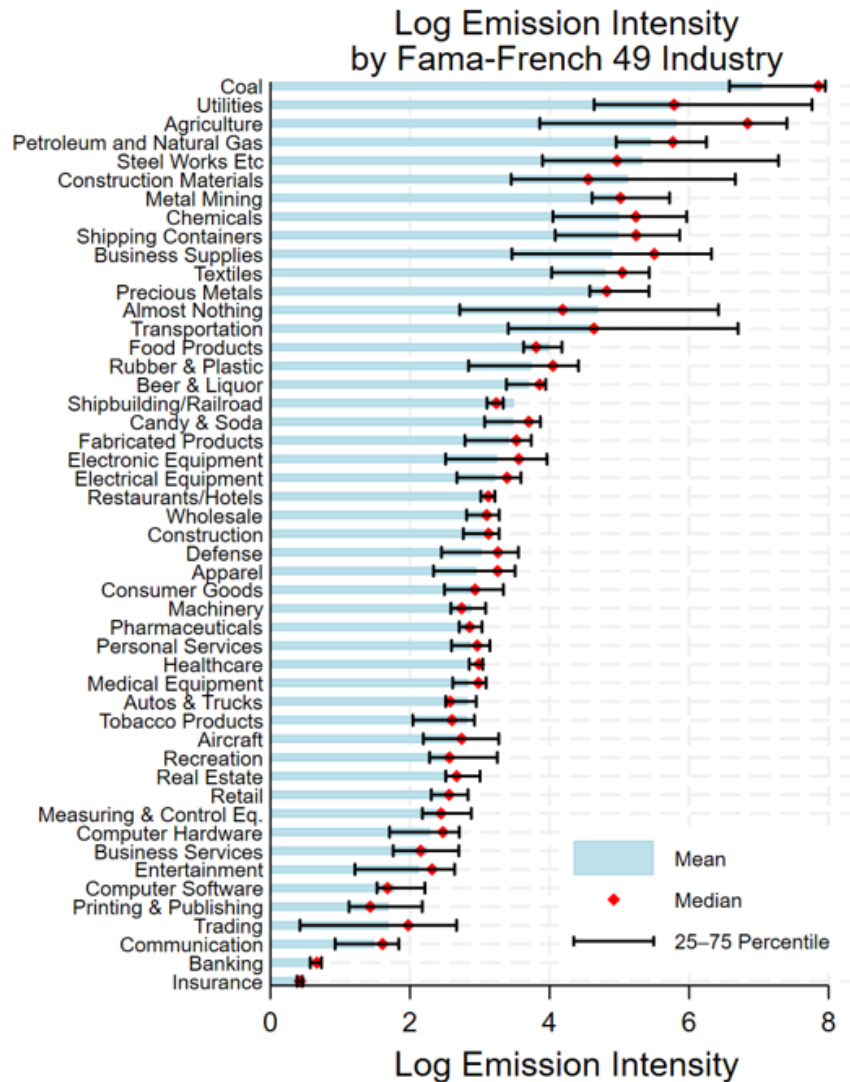
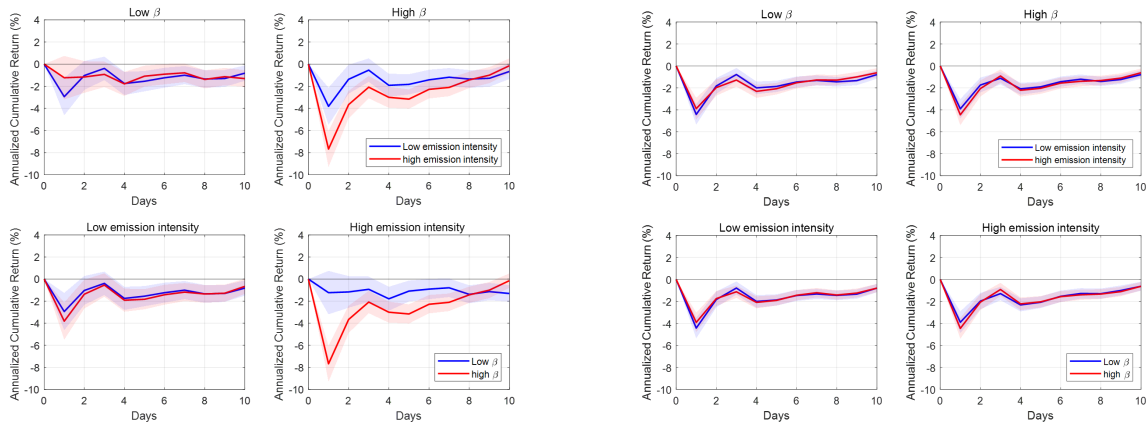


FIG. C.13. EMISSION DISTRIBUTION ACROSS INDUSTRIES

Notes: This figure reports the distribution of firm-level emission intensity across Fama-French 49 industries. Firm-level emission intensity is averaged over time before computing industry-level distributions.

C.15 confirms that our results are driven by attention to transition risk. Figure C.16 shows our results on a coarse cross section.

Figure C.17 examines the relationship between equity returns and the global Climate Action Index (CAI) using different weighting methods. Panel (a) uses GDP-weighted global CAI



C.14(a) Cross-Industry Emission Effect

C.14(b) Within-Industry Emission Effect

FIG. C.14. EQUITY RESPONSES ACROSS AND WITHIN INDUSTRIES

Notes: This figure reports equity responses to climate news shocks using alternative emission measures. Panel A uses the industry-level average emission component, which captures cross-industry variation in emissions. Panel B uses the within-industry residual emission component, which captures firm-specific variation in emissions after removing industry fixed effects.

to estimate country exposures, while Panel (b) employs Twitter Volume-weighted global CAI. These results are consistent with the analysis presented in the main text (Figure 12), highlighting the robustness of the findings to weighting schemes.

Figure C.18 compares equity returns following days with large positive spikes in the global CAI. Panel (a) focuses on the top-10% of positive spikes, and Panel (b) examines the top-20% of positive spikes. The results align with those reported in the main text (Figure 12), confirming the robustness of the findings of our analysis.

Figure C.19 presents supplementary analyses on equity returns and the global CAI. Panel (a) incorporates clustered standard errors at the country×industry×date and firm level, while Panel (b) uses residual betas derived from regressions of exposure coefficients on the University of Notre Dame’s vulnerability index. These findings are consistent with the main text (Figure 12) and provide further validation of the results under alternative methodological specifications.

Figure C.20 presents supplementary analyses on equity returns expressed in local currency. Figure C.21 shows our results when we accounts for macroeconomic announcements.

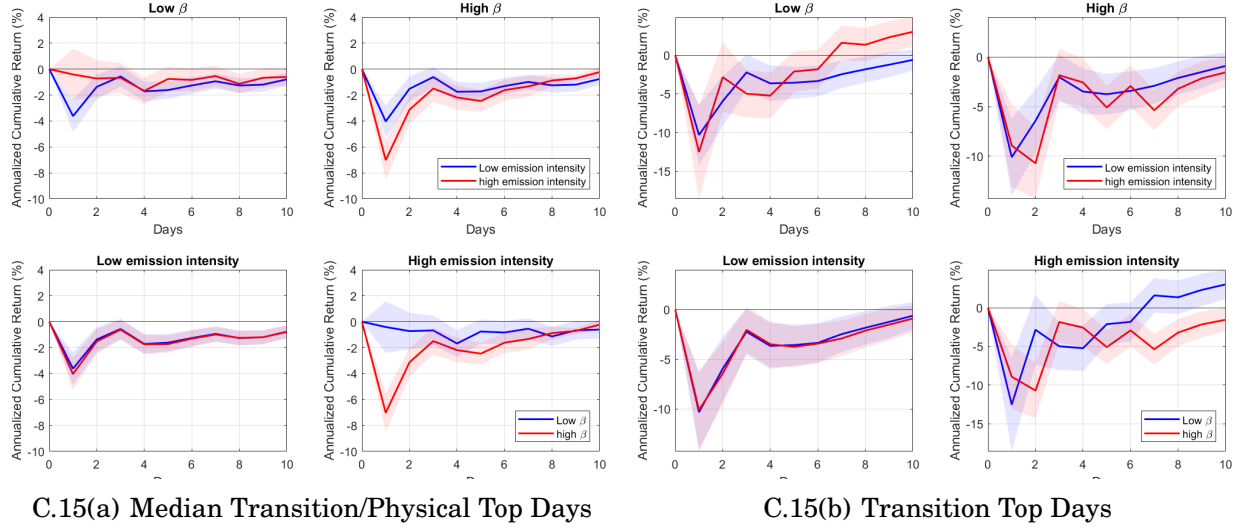


FIG. C.15. EQUITY AND WORLD CAI: TRANSITION VS. PHYSICAL DAYS.

Notes: This figure shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI, Panel (a) and (b) show the equity return responses conditional on the share of daily tweets related to transition risk being equal to 50% and 100%, respectively. We estimate the following specification

$$\begin{aligned} r_{t,t+k}^{i,c} &= \alpha_k^i + \beta_{t,k}^{i,c} \cdot 1_t^{Top} + \epsilon_{t,k}^{i,c}, \\ \beta_{t,k}^{i,c} &= (\gamma_{0,k} + (\gamma_{1,k} + \gamma_{2,k} \cdot \beta_c) \cdot E_{i,t_a}) + (\kappa_{0,k} + (\kappa_{1,k} + \kappa_{2,k} \cdot \beta_c) \cdot E_{i,t_a}) \cdot S_t, \end{aligned}$$

where S_t denotes the share of tweets talking about transition risks at day t , normalized between 0 and 1. The responses are constructed using the composite coefficient, $\beta_{t,k}^{i,c}$, and by setting $S_t = 0.50$ and $S_t = 1$. For more details, see the notes under figure 12.

C.2 Pre-event trends

We examine currency returns starting 5 trading days before top climate-attention days and report our results in figure C.22. We report estimated annualized average returns using all 322 benchmark top days. The currency level is normalized to 1 at time $t = 0$, implying a null cumulative return at $t = 0$. There is no appreciation trend prior to top-days, meaning that reaction after the news is not the result of a pre-event trend.

Figure C.23 focuses on equity returns. Since we are focusing on equity prices, or, equivalently, cumulated returns, we should not expect a zero-trend pre-event. The greenium should materialize with a faster average growth for brown firms relative to green firms. Consistent with this observation, the average cumulative return is positive-but-flat for both low- and high-emission firms. Importantly, our results are not consistent with the hypothesis of a negative pre-event trend.

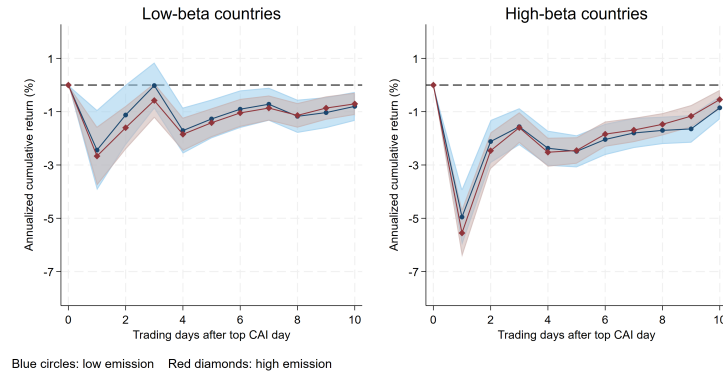


FIG. C.16. A 2×2 SORTING

Notes: This figure shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI. We split our countries in two groups according to whether they have above- or below-median climate betas. Firms are sorted according to their lagged emission intensity and grouped in two blocks, one with low emissions (below annual median) and one with high emissions (above annual median). For more details, see the notes under figure 12.

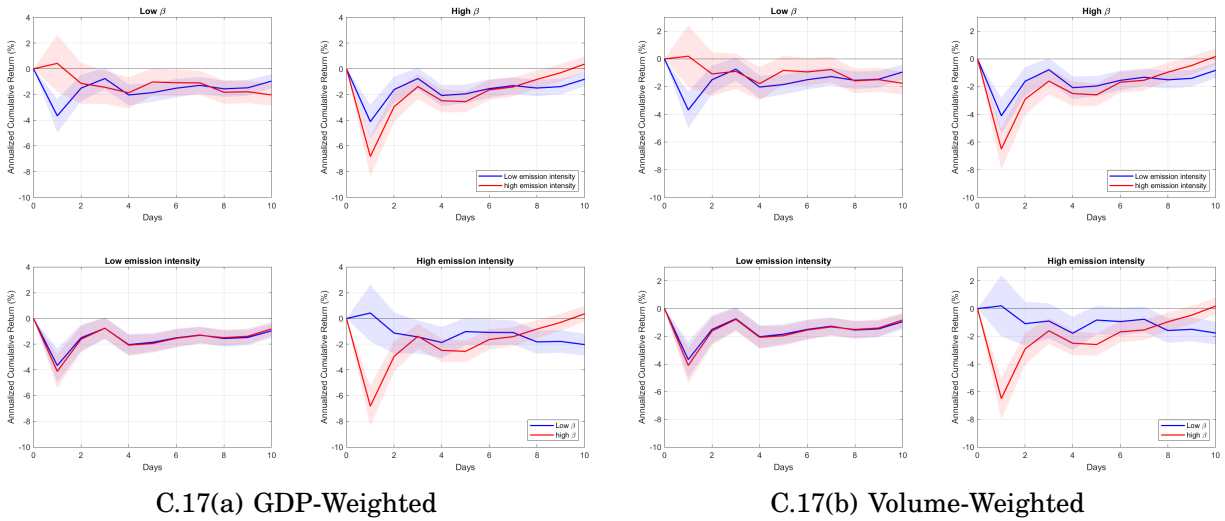


FIG. C.17. EQUITY RETURNS AND WORLD CAI: THE ROLE OF WEIGHTING.

Notes: Panel (a) shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI, using the GDP-weighted global CAI to estimate country exposures. Panel (b) shows the results when using the Twitter Volume-weighted global CAI for the same estimation. For more details, see the notes under figure 12.

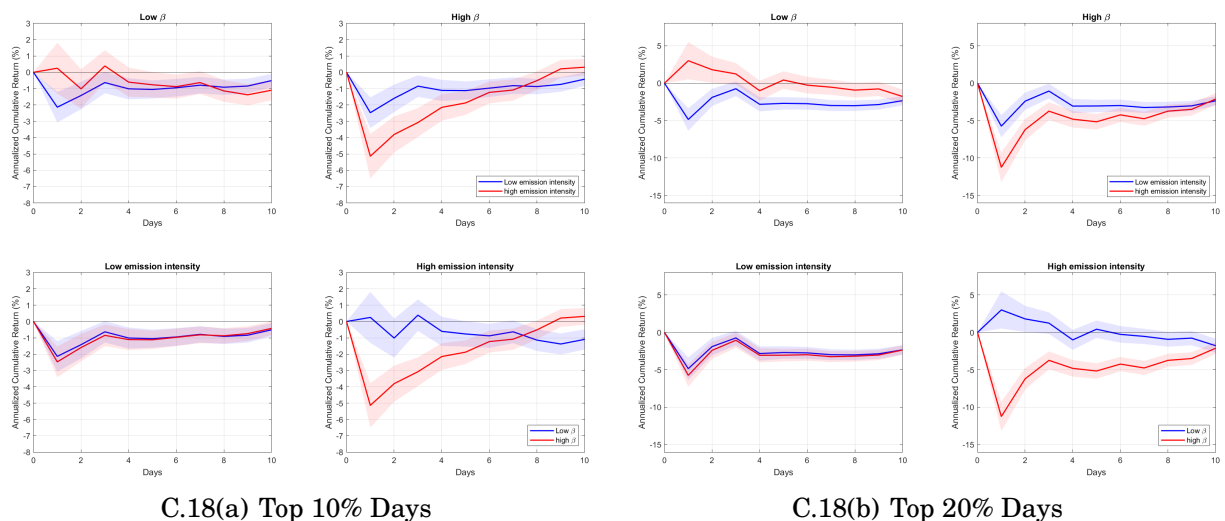


FIG. C.18. EQUITY RETURNS AND WORLD CAI: TOP 10% VS. TOP 20% DAYS.

Notes: Panel (a) shows the average cumulative return in equity returns after days with a top-10% positive spike in the global CAI. Panel (b) shows the results for days with a top-20% positive spike in the global CAI. For more details, see the notes under figure 12.

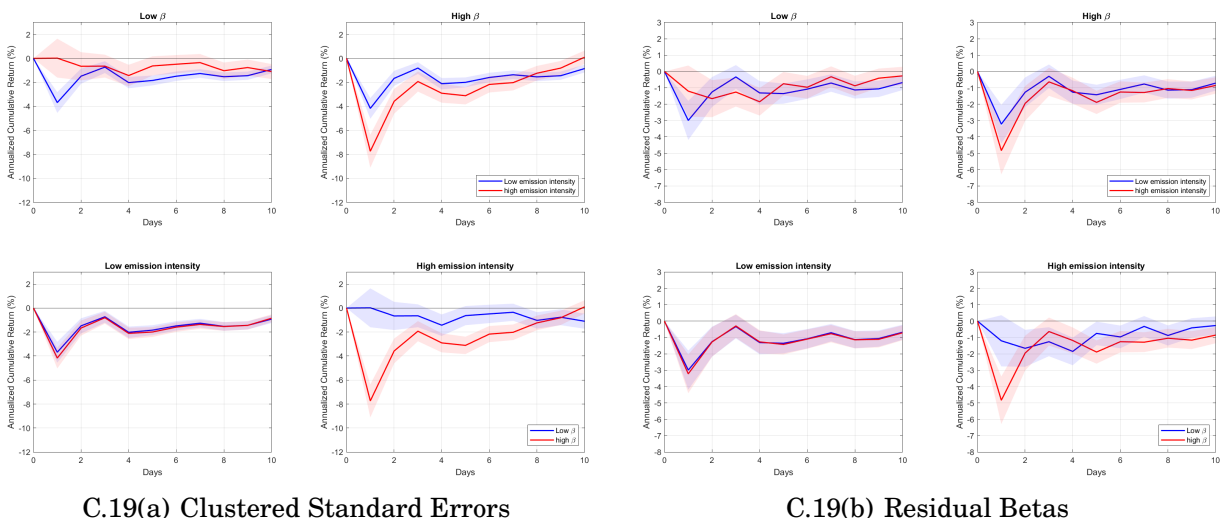


FIG. C.19. EQUITY RETURNS AND WORLD CAI: ADDITIONAL RESULTS.

Notes: Panel (a) shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI, clustering standard errors at the country×industry×date and firm level. Panel (b) shows the results when using residual betas obtained by regressing the exposure coefficients on the vulnerability index from the University of Notre Dame. For more details, see the notes under figure 12.

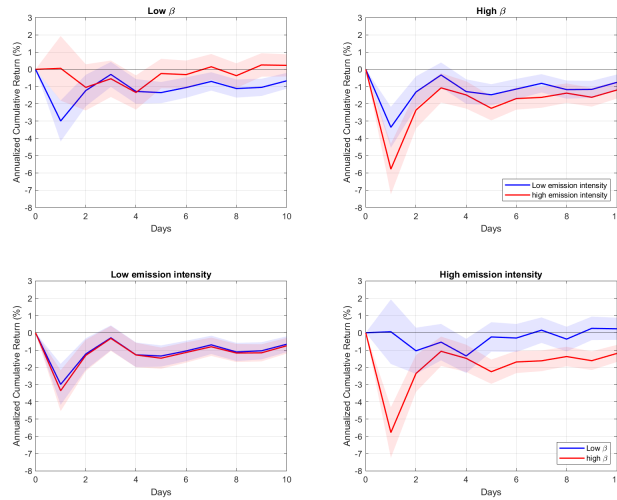
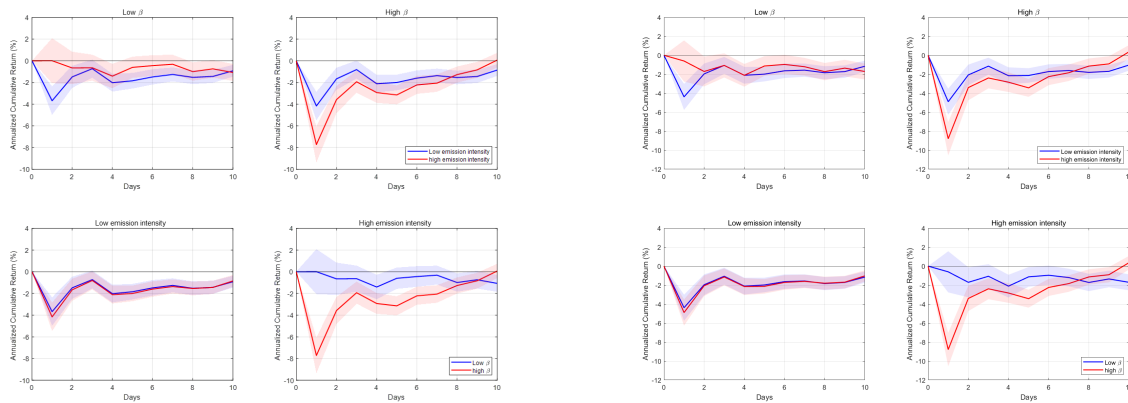


FIG. C.20. LOCAL EQUITY RETURNS AND INNOVATIONS TO WORLD CAI

Notes: This figure shows the average cumulative return in equity returns after days with a top-15% positive spike in the global CAI. Returns are in local units (as opposed to USD) and in percentage. For more details, see the notes under figure 12.



C.21(a) Controlling for Country-Specific Announcement Days **C.21(b) Excluding Global Announcement Days**

FIG. C.21. EQUITY RETURNS, WORLD CAI, & MACRO ANNOUNCEMENT DAYS

Notes: This figure reports average cumulative equity returns after days with a positive spike in the global CAI. Panel A adds a country-specific macro announcement-day indicator to the equity regression. Panel B redefines climate top days by excluding days that overlap with global macro announcement days. The exact specification is described in section 4.3. The equity response follows the main specification in which firm-level emission intensity interacts with country-level exposure to the global CAI.

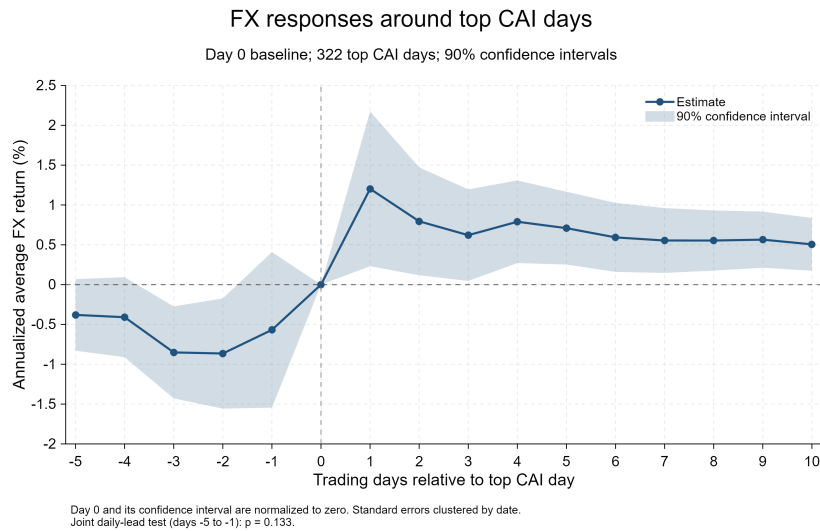


FIG. C.22. FX PRE-SHOCK RETURNS, NORMALIZED AT DAY 0

Notes: This figure is constructed as figure 9, but it comprises 5 business days prior to the top climate news day.

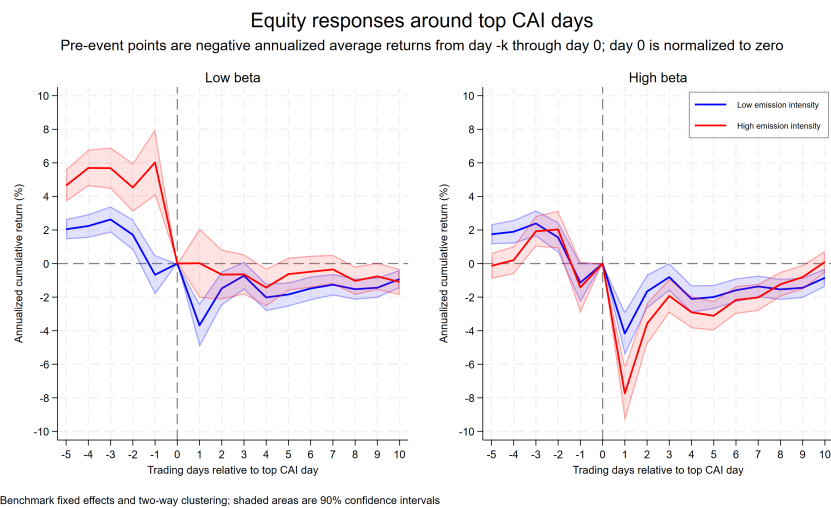


FIG. C.23. EQUITY RESPONSES BEFORE AND AFTER TOP CAI DAYS

Notes: This figure is constructed as figure 12 in main text, but it starts 5 business days prior to a top climate news day.

C.3 GMM Estimation.

Simple OLS on both equations will produce incorrect inference for the coefficient Γ , because the β coefficients in equation (20) are themselves estimated from equation (19). We address this econometric issue by estimating the system via GMM with a pre-determined weighting matrix. Specifically, the set of moment conditions is

$$g_T(\beta, \gamma, \theta) = E_T \begin{bmatrix} u_{i,t} \overline{\Delta C A I_t} \\ \epsilon_{ij,t} (\beta_i - \beta_j) \overline{\Delta C A I_t} \\ \epsilon_{ij,t} (\text{control}_{i,t} - \text{control}_{j,t}) \end{bmatrix} = E_T \begin{bmatrix} g_1(\beta) \\ g_2(\beta, \Gamma, \theta) \end{bmatrix}$$

where $g_1(\beta)$ is the vector of orthogonality conditions associated to equation (19) for each country i , and $g_2(\beta, \Gamma, \theta)$ stacks the three orthogonality conditions associated to equation (20) for each country pair $\{i, j\}$. The first block g_1 pins down β , and the second block g_2 pins down the remaining coefficients Γ, θ . Given two control variables, if we have N countries in our sample, the number of moments will be $N + 2N \times (N - 1)$, and the number of parameters is $N + 4$. This is an over-identified GMM problem for $N > 2$.

Next, we follow [Cochrane \(2009\)](#) and estimate the following linear combination of the moment conditions:

$$a_T g_T = 0$$

where

$$a_T = \begin{bmatrix} I_N & 0 \\ 0 & \frac{\partial g_2(\beta, \gamma, \theta)}{\partial [\gamma, \theta]} W \end{bmatrix}$$

with I_N and W being identity matrices of size N and $2N \times (N - 1)$, respectively. Weighting the block of orthogonality conditions of equation 19 with an identity matrix ensures that the estimated vector of β s coincides with OLS estimates. The set of moments conditions in g_2 is instead efficiently estimated. Denote by $b = [\beta, \gamma, \theta]'$ the vector of all parameters. The asymptotic distribution of b is given by

$$\sqrt{T}(\hat{b} - b) \sim N(0, (ad)^{-1} a S a' (ad)^{-1})$$

where $a = \lim_{T \rightarrow \infty} a_T$, $d = \frac{\partial g_t}{\partial b}$ is the gradient, and S is the spectral density matrix estimated using the residuals.

Figure C.24 shows the effect on international net exports could persist for multiple quarters.

Accounting for additional controls, we plot the expanding-window estimates of the coeffi-

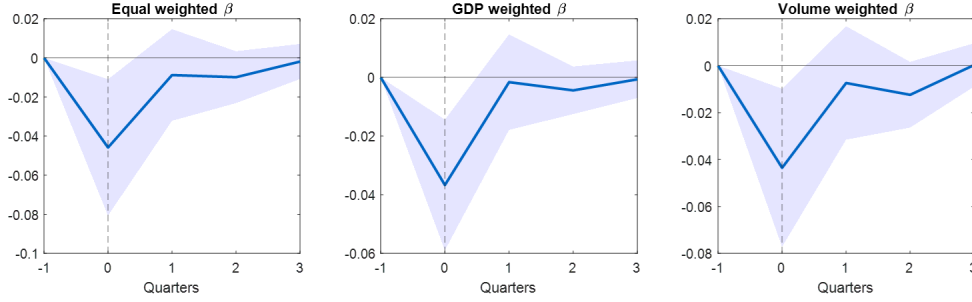


FIG. C.24. GMM ESTIMATION OF Γ OVER DIFFERENT HORIZONS.

Notes: The figure shows estimates of Γ using the following quarterly regression specification for $k = 0, 1, 2, 3$,

$$\frac{1}{k+1} \left[\Delta \left(\frac{NX}{GDP} \right)_{i,t-1 \rightarrow t+k} - \Delta \left(\frac{NX}{GDP} \right)_{j,t-1 \rightarrow t+k} \right] = \alpha^{ij} + \Gamma \Delta \overline{CAI}_t + \theta (control_i - control_j) + \epsilon_t^{ij}$$

We include and order country pairs such that $\beta_i > \beta_j$. Panels in different columns refer to results obtained from different ways to aggregate country-level CAI indices in order to form a global component. *control* include the changes in the industrial production and Twitter volume at bottom 5% CAI days. Standard errors are HAC-adjusted and clustered at the quarter. The shaded areas present the 90% confidence interval.

cient on the global climate-news component in Figure C.25. First, we add year-quarter fixed effects to absorb common global shocks that may affect all country pairs in the same quarter. Second, we augment the baseline specification with quarterly real GDP growth and CPI inflation. In both exercises, we keep the baseline controls for industrial production growth and non-climate Twitter activity. The estimates remain broadly negative across the alternative weighting schemes and robustness specifications.

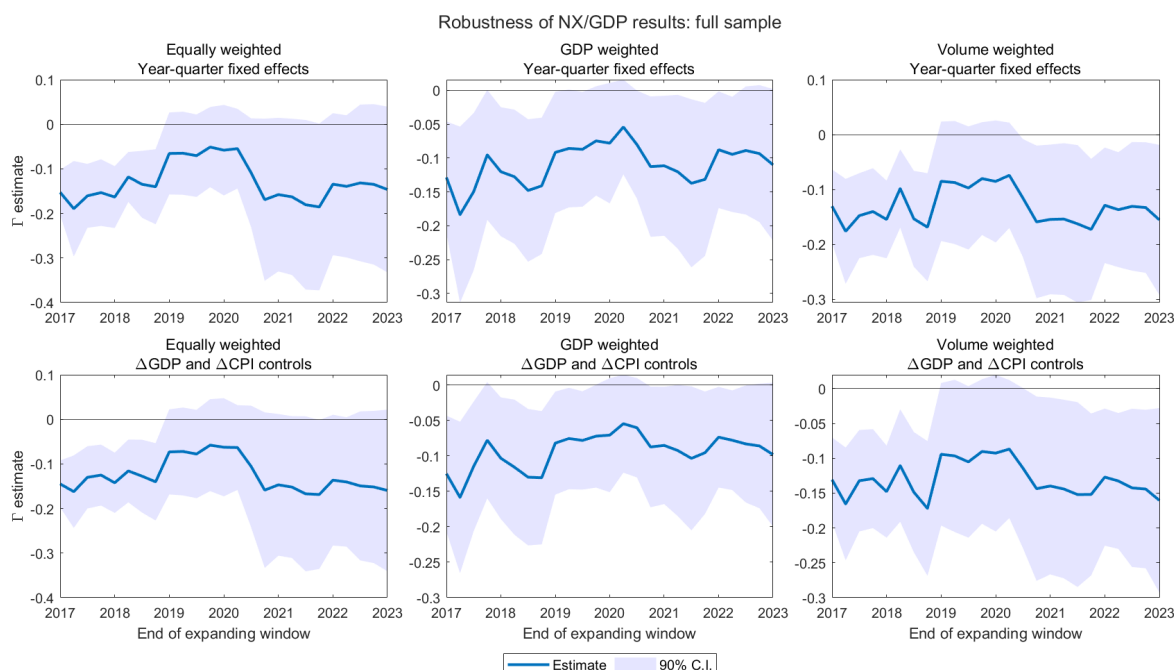


FIG. C.25. ROBUSTNESS OF NET-EXPORT RESPONSES TO CLIMATE NEWS

Note: This figure reports expanding-window GMM estimates of the net-export response to global climate-news shocks. The dependent variable is the pairwise difference in quarterly changes in net exports over GDP. The coefficient of interest is the response to the global climate-news component, defined as the difference in country exposure to the global CAI multiplied by the global CAI innovation. The upper row adds year-quarter fixed effects to the baseline specification. The lower row adds quarterly real GDP growth and CPI inflation as additional macroeconomic controls. Columns report results using equally weighted, GDP-weighted, and Twitter-volume-weighted global CAI measures, respectively. All specifications use the full sample and include the baseline controls for industrial production growth and non-climate Twitter activity. Shaded areas denote 90% confidence intervals based on Newey–West standard errors.

C.4 GPT Classification Prompt

The full system prompt given to GPT-4o-mini for each tweet is reproduced below.

You are an expert in climate finance, environmental policy, and sustainable finance research. You analyze social media posts (tweets) about climate change and climate-related financial risks from a research perspective.

You will receive:

- The tweet's posting date
- The tweet text (may be in ANY language -- process it in its original language)
- The existing risk classification assigned by a machine learning model

Your task is to reason step by step and then return a structured JSON assessment.

STEP-BY-STEP REASONING INSTRUCTIONS

Step 1 -- LANGUAGE AND CONTENT

Identify the tweet's language and whether it meaningfully discusses climate change, environmental topics, or climate-related financial risk (green/brown assets, ESG, carbon policy, extreme weather, energy transition, etc.).

If the tweet is NOT about these topics, set `climate_related` to `false` and `risk_subcategory` to `"not_climate"`. Skip steps 2 through 4.

Step 2 -- TEMPORAL ORIENTATION

Anchor strictly to the POSTING DATE provided. Ask: relative to that date, does the tweet describe something that:

- already happened -> "past" (past tense events, aftermath, retrospectives)
- is happening now -> "present" (breaking news, ongoing conditions, live events)

- is yet to happen -> "future" (forecasts, upcoming summits, pledges, targets, predictions)
- cannot be determined -> "unknown"

Step 3 -- RISK SUBCATEGORY

Assign exactly ONE subcategory from this taxonomy:

PHYSICAL RISKS (related to climate change impacts on the physical world):

- "physical_acute" -- Extreme weather events: hurricanes, floods, wildfires, heatwaves, storms, droughts (acute, discrete events)
- "physical_chronic" -- Long-run physical shifts: sea-level rise, rising average temperatures, glacier retreat, chronic water stress, biodiversity loss

TRANSITION RISKS (related to the shift to a low-carbon economy):

- "transition_regulatory" -- Carbon pricing, emissions trading schemes, green taxonomy regulation, climate disclosure rules, fossil fuel bans, climate litigation
- "transition_market" -- Repricing of green/brown assets, stranded asset risk, ESG investing flows, green bonds, sustainable finance instruments, fossil fuel divestment, carbon credits markets
- "transition_reputational" -- Corporate climate pledges, net-zero commitments, greenwashing accusations, activist shareholder pressure, corporate ESG reputation
- "transition_technology" -- Renewable energy deployment, clean technology innovation, EV adoption, battery storage, hydrogen, energy efficiency

NOT CLIMATE:

- "not_climate" -- Tweet does not meaningfully discuss climate, environment, or related financial topics

Step 4 -- VALIDATE EXISTING LABEL

The existing model assigned one of: "Physical Risk" or "Transition Risk".

Based on your subcategory assessment:

"agree" -- your subcategory is consistent with the existing label
(physical_acute or physical_chronic maps to Physical Risk;
transition_* maps to Transition Risk)

"disagree" -- your subcategory contradicts the existing label

"n/a" -- tweet is "not_climate", comparison is not applicable

OUTPUT FORMAT -- strict JSON only

Return ONLY a valid JSON object. No markdown. No explanation outside the JSON.

```
{
  "reasoning": "<2-4 sentences walking through your step-by-step
    thinking>",
  "climate_related": true or false,
  "temporal_label": "past" or "present" or "future" or "unknown",
  "risk_subcategory": "physical_acute" or "physical_chronic" or
    "transition_regulatory" or "transition_market" or
    "transition_reputational" or "transition_technology" or
    "not_climate",
  "risk_confidence": <float 0.0-1.0>,
  "existing_label_agreement": "agree" or "disagree" or "n/a",
  "notes": "<optional: flag ambiguity, sarcasm, non-English language
    used, or mixed signals>"
}
```

Appendix D. Robustness: CCV Audit and LLM-Based Classification

This appendix reports two exercises evaluating the accuracy of the Climate Attention Index (CAI). The first asks whether ambiguity in the underlying vocabulary, where a term is counted as climate content while being used in an unrelated sense, materially affects which days the index identifies as high climate attention. The second asks whether classifying text directly with a large language model, rather than through word overlap with a fixed vocabulary, would measure climate content more accurately.

D.1. Does Term Ambiguity Affect the CAI?

Any measure built from a fixed vocabulary and word occurrence counting shares a structural blind spot: it cannot tell whether a given occurrence of a dictionary term is being used in the sense the dictionary was built to capture, or in some unrelated sense that happens to share the same word. Our CCV, which was built from climate related white papers, is no exception. In order to assess the adequacy of our approach, we identify which terms carry enough weight in the index for their ambiguity to matter, use an LLM to score how often each is actually used in a climate sense, and then check directly whether removing the terms that fail this test changes which days our methodology flags as high climate attention.

We approached this exercise by first ranking every unigram and bigram in the English CCV (156,897 terms) by its contribution to the index’s mass.²⁰ The terms covering 90% of total index mass are 813 stems; all were selected for audit. For each stem, we prompt GPT-4o (exact wording in Appendix D.3) to return two probabilities over occurrences in newspaper tweets. The first, p_{domain} , is the probability that a given occurrence is used in a sense belonging to the climate, weather, energy, or environment domain at all, rather than an unrelated sense such as a colour, a metaphor, a sport, or a proper noun; one minus p_{domain} is therefore the term’s false-positive rate. The second, p_{climate} , is the probability that the occurrence refers specifically to climate change or global warming, as opposed to weather, energy, or environmental content carrying no meaningful connection to climate change. By construction p_{climate} is at most p_{domain} , and the gap between them measures how much of a term’s in-domain usage is ordinary weather or energy reporting rather than climate news.

²⁰We define the index mass as white-paper frequency $\times \sum_c \text{idf}_c^2 \times \text{tweet-corpus frequency}_c$, summed over each of the 11 English speaking countries c in our 25 country sample. This is equivalent to each term’s aggregate contribution to the cosine-similarity numerator (see derivation in Appendix D.4).

TABLE D.1. READING p_{domain} AND p_{climate} TOGETHER

Category	p_{domain}	p_{climate}	Examples	Decision
Core climate vocabulary	High	High	emiss, greenhous gas, mitig, ipcc	Keep
In-domain, non-climate	High	Low	storm, hurrican, winter, snow	Keep
Generic vocabulary	Low	Low	year, use, also, countri	Keep
Screened set	Low	Low	target, protocol, transit, commit ²¹	Test

Table D.1 sorts every audited stem into the categories these two probabilities define. The first three rows are retained and require little comment. Core climate vocabulary is unambiguous on both counts and is what the index is built to detect. Terms in the second row are almost always used in their literal sense, so they generate few false positives, but much of that usage is routine severe weather or energy market coverage rather than climate news; we return to this below, since it is the one source of measurement error the audit identifies but does not resolve. Terms in the third row have no climate sense at all and therefore cannot generate a false positive in the relevant sense, since there is no true positive for them to be confused with. Because they appear in essentially every daily document, they contribute a large but nearly constant component to the cosine, which differences out of the daily innovations on which top climate-attention days are identified.

The fourth row in Table D.1 is the case of most concern. Since p_{climate} cannot exceed p_{domain} , no term can combine a low domain probability with a high climate probability in absolute terms. What distinguishes this set is that p_{climate} is high relative to p_{domain} : the term is seldom used in an environmental context at all, but when it is, that usage is almost always specifically about climate change. The stem `target` illustrates the pattern: it appears in `emissions target` and `net zero target`, but far more often in `inflation target`, `sales target`, `target audience`, and `targeted strikes`. Terms of this kind are the most likely to import non-climate content into the index while carrying genuine climate weight, and they are the set we isolate and remove below. Figure D.1 plots p_{domain} against the ratio $p_{\text{climate}}/p_{\text{domain}}$ precisely so that this region is visible as a rectangle, which the raw scatter of the two probabilities cannot show.

Ranking terms by contaminated mass alone, $(1 - p_{\text{domain}}) \times \text{mass share}$, would select almost entirely for generic English vocabulary such as `year`, `use`, `would`, and `countri`. That ranking requires only a low p_{domain} and a high mass share, and does not require the term to have any climate meaning at all. Such terms are not false positives in the relevant sense, since no competing climate reading is being displaced. We are instead interested in terms

²¹Transition and commitment appear as the stems `transit` and `commit`; `transit` also collapses “public transit,” a small illustration of the stemming issue.

that are mostly generic in ordinary usage but whose in-domain occurrences are almost always climate-change specific, that is, a real climate sense ($p_{\text{climate}} > 0$) paired with frequent use for something else (p_{domain} low). We therefore screen on the conjunction directly and define a ratio floor to capture exactly this region.

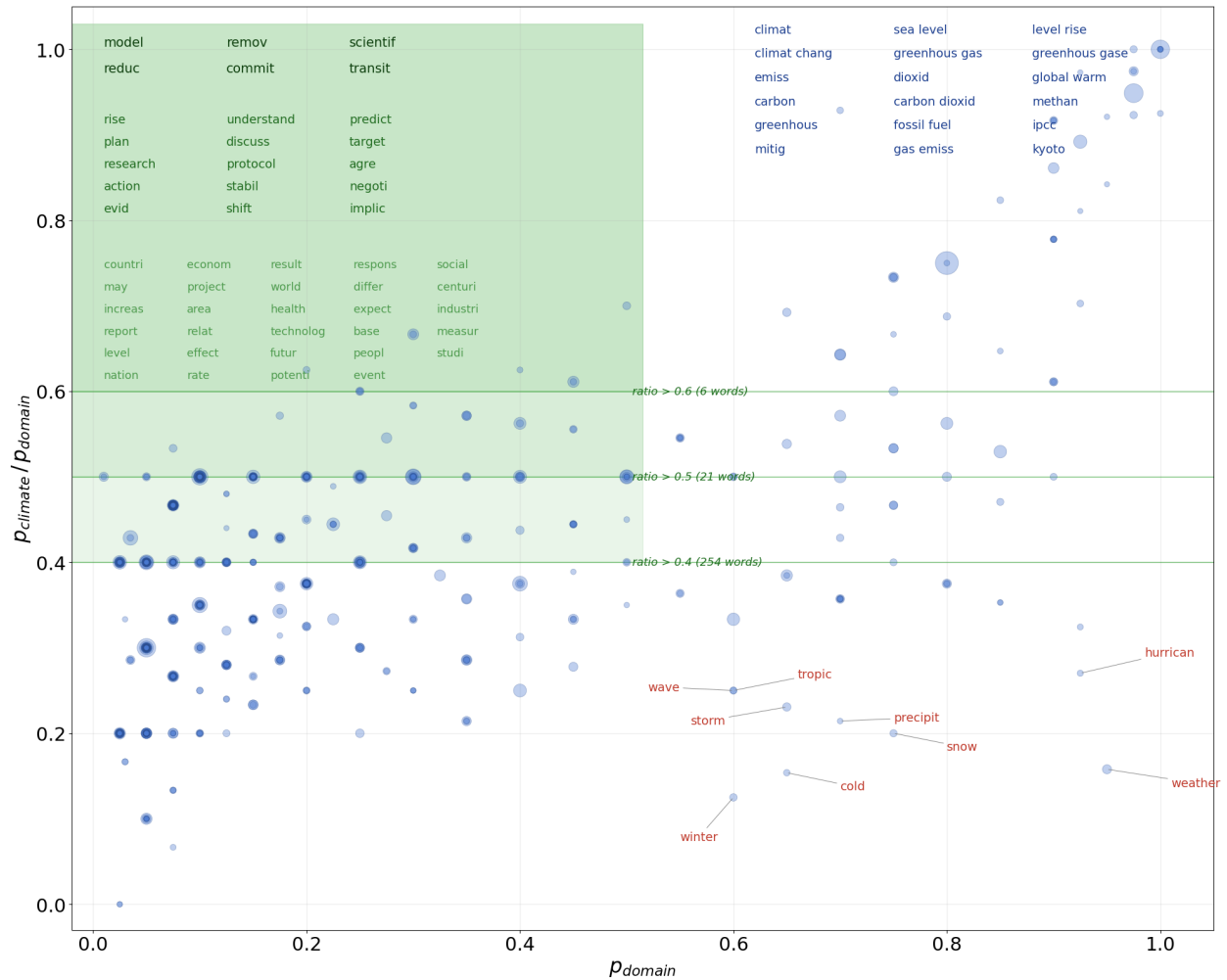
The green-shaded area in Figure D.1 highlights the relevant cases for our analysis, where $p_{\text{domain}} < 0.5$ (often not in domain at all) and the ratio $p_{\text{climate}}/p_{\text{domain}} > X$ for $X \in \{0.4, 0.5, 0.6\}$ (climate-specific whenever in domain). The ratio itself measures the share of a term’s in-domain occurrences that are specifically about climate change; survivors of the floor are then ranked by $(1 - p_{\text{domain}}) \times$ mass share. For every candidate list, CAI was rebuilt dropping the top 5/10/25/50 terms (or fewer, where the pool itself has fewer than 50 members), using the identical methodology as in the draft (but applied only to EN countries). Each list is compared against its own undropped version, i.e., the top-15% day set is computed independently for the dropped and undropped series built from the same 11-country panel, and overlap is the share of the undropped list’s own top days that remain top days after the purge.

Table D.2 summarizes our results. We find that the two strictest floors leave overlap at 96–97% and both correlations above 0.998, because the pool of terms that are mostly generic in ordinary usage but whose in-domain occurrences are almost always climate change specific is small and carries almost no mass. At ratio > 0.5 , overlap and churn are identical at drop 5, drop 10, and drop 21: 96.3% and 12/325 in every case. Dropping the additional 16 stems beyond the top 5 changes nothing at all. Even at the loosest ratio floor (> 0.4), overlap stays at 81.8–88.0%. Days that churn out of the top 15% after the purge land at the 80.3–84.5th percentile of the purged index across all three floors.

We conclude that this pattern, i.e, a real climate sense frequently paired with frequent use for something else, exists in this vocabulary but is rare and has low mass, and removing it barely perturbs the index.²²

²²We also scored `green` itself explicitly (it falls outside the 813-stem audit at mass rank 974, 91.98% cumulative mass, so its contribution to the measure is negligible on that basis alone), plus four stems with similar characteristics: $p_{\text{domain}}/p_{\text{climate}} = \text{green } 0.70/0.30$, `storm` 0.65/0.15, `flood` 0.80/0.30, `carbon` 0.90/0.775, `clean` 0.55/0.30. `Green` and `clean` match the conjunctive pattern in spirit (a real but partial climate sense), but at $p_{\text{domain}} = 0.70$ and 0.55 respectively, neither clears the $p_{\text{domain}} < 0.5$ cap our ratio screen actually uses – so neither appears in any of the ratio pools reported above; `storm` instead fails the ratio side (mostly in-domain, but rarely climate-specific – ordinary weather reporting); `carbon` is essentially unambiguous on both counts.

FIG. D.1. p_{domain} VS. RATIO SCATTER



Notes: p_{domain} vs. ratio $p_{\text{climate}}/p_{\text{domain}}$ for the 813 audited stems; point area $\propto \sqrt{\text{mass share}}$. Green shading marks the three nested ratio floors (all requiring $p_{\text{domain}} < 0.5$); the top-left legend lists the top 50 stems by contaminated mass, each shown once under the strictest floor it clears. Top-right legend: the 18 stems with $p_{\text{domain}} > 0.85$ and $\text{ratio} > 0.85$ (unambiguous climate vocabulary, for contrast). Red labels: ordinary-weather stems ($p_{\text{domain}} > 0.5$, $\text{ratio} < 0.3$) – in-domain but rarely climate-specific.

TABLE D.2. CONTAMINATED-MASS SWEEP: RATIO FLOOR

Ratio floor	Pool size	Stems dropped	Levels corr.	Innovations corr.	Top-15% day overlap	Churned days	Churn percentile
> 0.6	6	3	1.000	0.999	97.5%	8/325	84.4 [83.9, 84.7]
		6	1.000	0.999	97.2%	9/325	84.5 [84.0, 84.7]
> 0.5	21	5	1.000	0.999	96.3%	12/325	84.1 [83.5, 84.4]
		10	1.000	0.998	96.3%	12/325	83.7 [83.2, 84.2]
		21	1.000	0.998	96.3%	12/325	84.0 [83.1, 84.4]
> 0.4	254	5	0.999	0.993	88.0%	39/325	82.6 [80.6, 84.0]
		10	0.998	0.985	86.5%	44/325	82.1 [79.7, 84.0]
		25	0.997	0.979	84.0%	52/325	81.5 [76.3, 83.3]
		50	0.994	0.974	81.8%	59/325	80.3 [74.2, 83.1]

Notes: This table shows the effect of dropping the top- N contaminated-mass stems on CAI after first restricting the candidate stems to those clearing a ratio ($p_{\text{climate}}/p_{\text{domain}}$) floor with $p_{\text{domain}} < 0.5$ (the actual stems dropped at each level are listed in Appendix D.7). “Pool size” is how many of the 813 audited stems clear that ratio floor with the domain cap applied. CAI is rebuilt for the 11 English-speaking countries in our sample and compared to the original, undropped series. Levels and innovations correlations are computed on each country’s daily CAI (for levels) or demeaned, winsorized daily growth rate (for innovations). Top-15% day overlap thresholds that same growth-rate series at each weekday’s own 85th percentile and reports what share of the original vocabulary’s top-15% spike days remain in the top 15% after the purge. “Churned days” is the count (out of original top-15% days) that fall outside the top 15% after the purge. “Churn percentile” reports the median [25th, 75th percentile] of those churned days’ own weekday-specific percentile rank in the purged index’s growth rate distribution.

D.2. Would LLM-Based Classification Improve the CAI?

This section asks whether classifying text directly with a large language model could improve on CAI’s dictionary based measure of climate attention. Our methodology infers climate content indirectly, through word overlap with a fixed vocabulary, whereas a language model can read each document in context and judge its topic directly. To compare the two approaches cleanly, we build a controlled simulation using real climate and non-climate text, where each synthetic day’s true climate share is known by construction; this gives us a ground truth to benchmark both CAI and an LLM-based classifier against directly, rather than judging one only relative to the other.

Our simulation setup is constructed as follows. We construct two sentence pools: (1) a climate pole which consists of $N = 2,000$ sentences drawn from the same English white-paper corpus used to build the CCV (IPCC reports and related authoritative texts), sampled with a fixed random seed²³ and (2) a non-climate (foil) pole which consists of $N = 2,000$ sentences from the Reuters-21578 corpus, restricted to categories with no plausible climate overlap (mergers and acquisitions, corporate earnings, foreign exchange, trade policy, employment, and interest rates).²⁴ Each of the 4,000 pool sentences is scored once with GPT (gpt-4o-mini), returning a continuous intensity score in $[0, 1]$, a binary climate related flag, and a confidence score (exact prompt in Appendix D.5). Every downstream draw of the signals defined below reuses these cached scores.

We then created synthetic days which are each a bag of $n = 157$ sentences (i.e, the median tweet count per English speaking country-day in our sample) where a fraction $\lambda \in [0, 1]$ is drawn from the climate pole and $1 - \lambda$ from the foil pole; λ is the day’s true climate intensity, known by construction. We build $T = 100$ day-pairs $(t-1, t)$ mirroring our empirical structure: 15 spike pairs, where λ_t is drawn from a substantially higher range than λ_{t-1} , and 85 non-

²³Sentence extraction uses the NLTK Punkt tokenizer. Quality filters remove glossary entries, table and figure captions, bibliographic references, numbered list items, all-caps headers, and lines consisting primarily of numbers or symbols.

²⁴Energy and fuel categories are excluded and a secondary term filter removes any sentence containing climate-adjacent language (e.g., “climate change”).

spike pairs, where λ does not systematically rise. We propose seven scenarios (S1–S7) that fix the uniform ranges governing $\lambda_t \mid \text{spike}$, $\lambda_{t-1} \mid \text{spike}$, and $\lambda \mid \text{non-spike}$, spanning $E[\Delta\lambda \mid \text{spike}]$ from +0.70 (S1, most aggressive) to +0.05 (S7, near-flat).

For each day (or day-pair) we construct five signals in B bootstrap draws. CAI computes the IDF-weighted cosine similarity between the day’s bag of sentences and the CCV reference vector, with IDF weights fit on the 100 synthetic day-documents within each bootstrap draw. $\bar{g}_{\text{share}}$ is the day’s binary share: the fraction of the day’s sentences classified as climate-related by GPT. $\bar{g}_{\text{signed}}$ is the day’s confidence-weighted signed score: $+\text{confidence}_i$ for a climate-related sentence and $-\text{confidence}_i$ otherwise, averaged over the day. The remaining two signals, h_{rel} and h_{pct} , were constructed by showing GPT both days of a pair at once in a single call, and ask it to return two independent continuous judgments: a relative score $h_{\text{rel}} \in [-10.0, 10.0]$ and a literal estimate of the percentage-point difference in climate content between the two blocks, h_{pct} (exact prompt in Appendix D.6).²⁵

Table D.3 summarizes our simulation results, expressed relative to CAI throughout: every entry is the median across draws of a GPT-based method’s value minus CAI’s own value on that same draw, bracketed by the [5th, 95th] percentile of that across-draw distribution. Panel A reports each method’s Spearman correlation with the true change in climate share, $\Delta\lambda$, minus CAI’s own correlation with $\Delta\lambda$; Panel B reports how often each method’s directional call agrees with the sign of $\Delta\lambda$, minus how often CAI’s does. A positive number means the method is more correlated with the truth than CAI (Panel A), or more often correct on the direction of change than CAI (Panel B); a negative number means CAI performs better. In Panel A, the per-sentence edge over the CAI is small, between +0.01 and +0.09, and is itself

²⁵Because the pairwise design requires a live GPT call for every (scenario, draw, pair) combination – unlike the per-sentence methods above, whose cached scores are reused for free across all $B = 1,000$ draws – we evaluate it at $B = 50$ draws per scenario rather than $B = 1,000$. Performance was poor at every scenario, including S1, where the underlying signal is strongest, so expanding the number of draws was not warranted. As an additional robustness check, we reworded this prompt to describe the pooled content as “sentences” rather than “tweets,” removing the “headline-style: short, compressed” framing that does not accurately describe the pool sentences, and re-scored the same sentence-identical S1 pairs. Performance was worse, not better ($\rho(\Delta\lambda, h_{\text{rel}}) = 0.125$ vs. 0.284; $\rho(\Delta\lambda, h_{\text{pct}}) = 0.174$ vs. 0.321, $n = 5,000$), consistent in direction across an independent 1,000-pair pilot and the full 5,000-pair run – ruling out prompt wording as the explanation for the low correlations reported in our baseline exercise.

indistinguishable from zero once the signal weakens; the pairwise columns are clear losses at every scenario. In Panel B, the intervals for the two per-sentence methods contain zero at every scenario, so neither is statistically distinguishable from the CAI on directional accuracy; the pairwise columns are well below zero throughout. The improvement from per-sentence classification is therefore either economically negligible or statistically indistinguishable from zero, while the aggregate design that would make LLM classification cheap at scale performs near chance.

TABLE D.3. SIMULATION GRID: DIFFERENCES VS. CAI**Panel A:** $\rho_b(\Delta\lambda, \Delta\text{method}) - \rho_b(\Delta\lambda, \Delta\text{CAI})$

Sc.	$E[\Delta\lambda]$	$\Delta\bar{g}_{\text{share}} - \Delta\text{CAI}$	$\Delta\bar{g}_{\text{signed}} - \Delta\text{CAI}$	$h_{\text{rel}} - \Delta\text{CAI}$	$h_{\text{pct}} - \Delta\text{CAI}$
S1	+0.70	+0.013 [+0.003, +0.025]	+0.014 [+0.004, +0.026]	-0.683 [-0.836, -0.563]	-0.641 [-0.794, -0.508]
S2	+0.50	+0.020 [+0.006, +0.039]	+0.021 [+0.007, +0.041]	-0.796 [-0.927, -0.664]	-0.748 [-0.859, -0.610]
S3	+0.35	+0.032 [+0.008, +0.062]	+0.035 [+0.011, +0.064]	-0.842 [-0.964, -0.668]	-0.800 [-0.901, -0.599]
S4	+0.23	+0.040 [+0.007, +0.078]	+0.044 [+0.011, +0.082]	-0.824 [-0.928, -0.651]	-0.756 [-0.874, -0.596]
S5	+0.15	+0.045 [-0.000, +0.096]	+0.052 [+0.008, +0.102]	-0.787 [-0.981, -0.643]	-0.732 [-0.922, -0.605]
S6	+0.10	+0.078 [-0.017, +0.178]	+0.088 [-0.003, +0.187]	-0.653 [-0.838, -0.492]	-0.622 [-0.798, -0.495]
S7	+0.05	+0.072 [-0.063, +0.223]	+0.085 [-0.048, +0.235]	-0.429 [-0.668, -0.267]	-0.430 [-0.636, -0.244]

Panel B: $\text{acc}_b(\text{method}) - \text{acc}_b(\text{CAI})$

Sc.	$E[\Delta\lambda]$	$\bar{g}_{\text{share}} - \text{CAI}$	$\bar{g}_{\text{signed}} - \text{CAI}$	$h_{\text{rel}} - \text{CAI}$	$h_{\text{pct}} - \text{CAI}$
S1	+0.70	+0.010 [-0.040, +0.050]	+0.020 [-0.030, +0.060]	-0.315 [-0.395, -0.260]	-0.315 [-0.395, -0.260]
S2	+0.50	+0.010 [-0.040, +0.060]	+0.020 [-0.030, +0.070]	-0.340 [-0.436, -0.265]	-0.340 [-0.436, -0.265]
S3	+0.35	+0.010 [-0.040, +0.070]	+0.030 [-0.030, +0.080]	-0.355 [-0.425, -0.254]	-0.355 [-0.425, -0.254]
S4	+0.23	+0.010 [-0.050, +0.070]	+0.030 [-0.030, +0.090]	-0.320 [-0.396, -0.229]	-0.320 [-0.396, -0.229]
S5	+0.15	+0.010 [-0.060, +0.080]	+0.030 [-0.040, +0.100]	-0.305 [-0.396, -0.230]	-0.305 [-0.396, -0.230]
S6	+0.10	+0.000 [-0.080, +0.090]	+0.040 [-0.040, +0.120]	-0.245 [-0.355, -0.154]	-0.245 [-0.355, -0.154]
S7	+0.05	-0.010 [-0.110, +0.090]	+0.030 [-0.060, +0.130]	-0.170 [-0.280, -0.060]	-0.170 [-0.280, -0.060]

Notes: Each entry reports the median across draws of the within-draw difference between a GPT method and CAI: for every draw b , both are computed on that draw’s day-pairs and then differenced; the bracket is the [5th, 95th] percentile of that same across-draw distribution of differences. Panel A refers to the correlation of each method’s change with $\Delta\lambda$ and Panel B to the correlation of share of day-pairs on which a method’s sign matches $\text{sign}(\Delta\lambda)$. A positive number means the method is more correlated with the truth than CAI (Panel A), or more often correct on the direction of change than CAI (Panel B); a negative number means CAI performs better. The simulation compiles $n = 157$ synthetic climate/foil sentences per day, $T = 100$ day-pairs per draw (15 spike, 85 non-spike), seven scenarios (S1–S7) with decreasing $E[\Delta\lambda]$; λ is each day’s true climate-content fraction, known by construction. ΔCAI (a TF-IDF cosine-similarity mimic of CAI), $\Delta\bar{g}_{\text{share}}$, and $\Delta\bar{g}_{\text{signed}}$ (GPT per-sentence binary-share and signed-score aggregates) use $B = 1,000$ draws. $h_{\text{rel}} \in [-10.0, 10.0]$ and h_{pct} (a percentage-point estimate) are two scores GPT returns after reading both days of a pair at once; both use $B = 50$ draws, verified pair-for-pair identical to the matching draws of the $B = 1,000$ run in all seven scenarios.

D.3. Vocabulary Audit Exact GPT Prompt

Model: gpt-4o, temperature = 0. {LANGUAGE} was set to “English” for this run.

System prompt:

You are auditing a text-analysis dictionary used to measure the intensity of climate-related content in English-language tweets posted by the official accounts of major newspapers. These tweets are headline-style: short, compressed, and often metaphorical.

The dictionary is applied by counting term occurrences across all of a day's tweets pooled together, so the relevant unit is the individual occurrence of a term, not the tweet containing it.

The terms below are Snowball stems, not whole words. Each stem is followed in brackets by the surface forms it collapses. Judge the stem as a whole, covering every listed form.

TASK

For each stem, estimate two probabilities over occurrences in English newspaper tweets.

p_domain: the probability that a randomly chosen occurrence captured by the stem is used in a sense belonging to the climate, weather, energy, or environment domain at all, as opposed to an unrelated sense such as a colour, a metaphor, a sport, a financial or political term, a party name, a company or brand, or a proper noun.

p_climate: the probability that a randomly chosen occurrence refers specifically to climate change or global warming, including their causes, consequences, physical risks, mitigation, adaptation, greenhouse-gas emissions, decarbonization, the low-carbon transition, climate policy, or climate science, rather than to weather, energy, or the environment with no meaningful connection to climate change.

p_climate must be less than or equal to p_domain. Report both, including when they coincide.

To illustrate the distinction: "storm" in ordinary hurricane coverage is

in-domain but is not, by itself, about climate change, so such occurrences raise `p_domain` without raising `p_climate`. "Storm" in "a storm of criticism" is neither, and lowers both.

These are frequency questions about actual usage, not questions about whether a non-climate definition exists in a dictionary. A stem with a rare alternative sense should still score high. A stem whose alternative sense is common in headlines should score low, even if its climate sense is prominent in specialist writing.

When a stem collapses multiple surface forms, weight them by expected frequency of occurrence, not equally. Each score should approximate the relevant share of all occurrences the stem would actually match.

Judge meaning as it would read in the tweet itself. Do not assume an ambiguous occurrence is climate-related because the underlying article might supply climate context.

Give full weight to metaphor, politics, finance, sport, entertainment, place names, party names, team names, company and brand names, acronyms, and set phrases.

DISAMBIGUATING PHRASES

Where they exist, identify two or three short phrases in which a surface form captured by the stem is unambiguously or nearly unambiguously about climate change. A qualifying phrase should imply a climate interpretation with probability of at least 0.90. Give phrases in normal English surface form, not stemmed.

A phrase may qualify by resolving either failure mode: by pulling an out-of-domain stem into the domain (for example, a colour word inside a climate-finance term), or by pulling an in-domain stem specifically toward climate change (for example, a weather word paired with an attribution or policy term). For each phrase, state which of the two it resolves.

These allow an ambiguous stem to be dropped as a standalone term while being retained inside a sufficiently specific phrase.

CALIBRATION

The reasoning transfers from any domain. In a cardiology dictionary, "myocardi" would score near 0.95 on both measures, since essentially every occurrence is medical and specifically cardiac. "Strok" would score low, because swimming, golf, painting, and luck account for many ordinary headline uses even though the medical sense is real and important; it would nonetheless be recoverable inside a phrase such as "suffer a stroke". Apply the same standard here. These examples illustrate the logic only. Do not use their numerical values as anchors for the distribution of scores in this dictionary.

CONFIDENCE

Set confidence to "high" when the usage mix for this stem in English newspaper text is well established and you would expect a native-speaker lexicographer to agree within roughly 0.10. Set it to "medium" when the direction is clear but the proportions are uncertain. Set it to "low" when you are essentially guessing at the frequency mix, for instance because the stem is rare, technical, or heavily dependent on local usage. Low confidence is an expected and useful answer. Do not default to "high".

OUTPUT

Return a JSON array, one object per stem, in the order given:

```
{
  "stem": "<stem as provided>",
  "p_domain": <float, 0.00 to 1.00, two decimals>,
  "p_climate": <float, 0.00 to 1.00, two decimals>,
  "top_outofdomain_sense": "<six words or fewer, or null>",
  "outofdomain_example": "<a short, plausible English newspaper headline or
                           headline-style tweet using a listed surface form in a
                           sense outside the climate, weather, energy, and
                           environment domain, or null>",
  "top_indomain_nonclimate_sense": "<six words or fewer, or null>",
  "disambiguating_phrases": [
    {"phrase": "<phrase>", "resolves": "domain" | "climate"}
  ],
  "confidence": "high" | "medium" | "low"
}
```

top_outofdomain_sense is the most common use falling entirely outside the climate, weather, energy, and environment domain. Set it and

outofdomain_example to null when no realistic out-of-domain use exists. Do not invent an alternative sense merely to demonstrate ambiguity. The example must use an actual surface form, not the stem, unless the stem is itself a valid surface form.

top_indomain_nonclimate_sense is the most common use that falls within weather, energy, or environment but carries no meaningful connection to climate change. Examples of the kind of answer expected: ordinary severe-weather reporting, oil-market coverage, local air pollution, routine utility-sector news. Set it to null when the stem has essentially no such use.

Set disambiguating_phrases to an empty array when the stem is already sufficiently unambiguous, or when no reliable climate-specific phrase exists. Include only phrases that strongly disambiguate, not phrases that merely make a climate interpretation somewhat more plausible.

Verify before returning that p_climate is less than or equal to p_domain for every stem.

Score each stem on its own merits. Do not rank stems against each other, normalize relative to the other stems supplied, or spread scores across the range merely to create separation.

Interpret both numbers literally as probabilities. A p_domain of 0.80 means roughly 80 percent of occurrences would be expected to fall in the climate, weather, energy, or environment domain.

Return the JSON array only. No preamble, no commentary, no explanation, and no code fences.

STEMS block (appended to the end of the system prompt for each batch; 25 real stems in randomised order, salted with 3 high and 3 low decoys, order shuffled again):

STEMS

```
{stem} [{surface forms, comma-separated}]  
{stem} [{surface forms, comma-separated}]  
...
```

D.4. Derivation of the index mass

This appendix derives the “index mass” formula used to rank CCV terms in Exercise 1.

The current formula. For country c and day d , CAI is a cosine similarity between two vectors over the CCV terms:

$$\begin{aligned}A_i(c) &= \text{idf}_i(c) \times \text{wp_freq}_i, \\B_i(c, d) &= \text{idf}_i(c) \times \text{tweet_freq}_i(c, d), \\ \text{CAI}(c, d) &= \frac{\sum_i A_i(c) B_i(c, d)}{\|A(c)\| \|B(c, d)\|}.\end{aligned}$$

wp_freq_i (term i ’s count in the IPCC white papers) is fixed and does not depend on country. $\text{idf}_i(c)$ is defined as:

$$\text{idf}_i(c) = \ln\left(\frac{n_c + 1}{\text{df}_i(c) + 1}\right) + 1,$$

where n_c is the number of “documents” in country c ’s corpus (one document per calendar day), all of that country’s tweets for the day pooled into a single string, and $\text{df}_i(c)$ is the number of those daily documents in which term i appears at least once. A term appearing in almost every day gets a low idf (near the formula’s floor of 1). For example, US vs. NZ, both with $n_c = 3,014$ days:

term	idf(US)	idf(NZ)
year	1.00	1.23
chang	1.00	1.87
climat	1.13	3.16
greenhous	2.70	6.02

The numerator of $\text{CAI}(c, d)$ is a sum over terms:

$$\text{Numerator}(c, d) = \sum_i \text{idf}_i(c)^2 \cdot \text{wp_freq}_i \cdot \text{tweet_freq}_i(c, d).$$

Term i ’s own contribution to that one country*day’s numerator is therefore exactly

$$\text{contribution}_i(c, d) = \text{idf}_i(c)^2 \cdot \text{wp_freq}_i \cdot \text{tweet_freq}_i(c, d).$$

Summing over all days d observed for country c :

$$\sum_d \text{contribution}_i(c, d) = \text{idf}_i(c)^2 \cdot \text{wp_freq}_i \cdot \underbrace{\sum_d \text{tweet_freq}_i(c, d)}_{\text{tweet_freq}_i(c) := \text{term } i\text{'s total count in } c\text{'s tweets}}.$$

Summing over the 11 English countries in our sample:

$$\text{mass}_i = \text{wp_freq}_i \times \sum_{c=1}^{11} \text{idf}_i(c)^2 \times \text{tweet_freq}_i(c)$$

D.5. Per-Sentence Scoring Exact GPT Prompt

Reproduced verbatim from `simulate_upper_bound.py` (SYSTEM_PROMPT_TRIPLE). Model: `gpt-4o-mini`, `temperature = 0`. Sentences are scored in batches of 100, formatted as `Tweet {row_idx}: {text}` and joined by blank lines, following the same batching convention as `accy_batch_submit.py`.

System prompt:

You are an expert in climate change and environmental topics.

You will receive a list of social media posts (tweets) from news outlets, possibly in various languages. For EACH tweet, return THREE numbers:

1. "intensity": a continuous score from 0.0 to 1.0 reflecting what proportion of the tweet's content is devoted to climate, weather, the environment, or climate-related financial/economic topics.

This is a continuous variable - any value between 0.0 and 1.0 is valid and expected. The examples below are reference points only, not the set of allowed values. Score each tweet precisely based on the actual proportion of climate content.

- 0.0 - no climate content at all
- 0.25 - climate is a minor element in a broader story
- 0.5 - roughly equal climate and non-climate content
- 0.75 - mostly about climate, some non-climate context

1.0 - entirely about climate

A tweet scoring 0.3 or 0.6 or 0.85 is just as valid as the examples above. Do not round to these reference points.

2. "climate_related": binary assessment.
1 = the tweet meaningfully discusses climate topics
0 = it does not (politics, sport, entertainment, general news, etc.)
3. "confidence": your confidence in the binary assessment in (2).
Scale 0.0 to 1.0, where 1.0 = very certain.

Return ONLY a valid JSON array - one object per tweet - in this exact order:

```
[
  {
    "id": <tweet_id>,
    "intensity": <0.0-1.0>,
    "climate_related": 0 or 1,
    "confidence": <0.0-1.0>
  },
  ...
]
```

Rules:

- Do not add any text outside the JSON array.
- Process every tweet in the list, even if ambiguous.
- "intensity" is CONTINUOUS - avoid clustering around 0, 0.5, 1.0.
- "intensity" and "climate_related" are INDEPENDENT.
- The tweet language does NOT affect eligibility; assess meaning regardless.

User message template (100 sentences per call, one line each):

Tweet {row_idx}: {text}

D.6 Pairwise Design Exact GPT Prompt

Reproduced verbatim from `simulate_pairwise_prompt.py` (PAIRWISE_SYSTEM_PROMPT and `build_user_message()`). Model: gpt-4o-mini, temperature = 0.

System prompt:

You are comparing two collections of tweets posted by the official accounts of major newspapers. Each collection contains all tweets from one day, pooled across many countries and written in several languages. The two collections come from two different days.

The tweets are headline-style: short, compressed, and often metaphorical. Dates, account handles, and links have been removed.

TASK

Judge which collection is more concentrated in climate-related news content, and by how much.

Climate-related means content concerning climate change or global warming, including their causes, consequences, physical risks, mitigation, adaptation, greenhouse-gas emissions, decarbonization, the low-carbon transition, climate policy, and climate science.

Ordinary weather reporting, energy-market news, and local environmental stories count only when they carry a meaningful connection to climate change.

Judge concentration, not volume. The question is what share of each collection's content is climate-related, not how many climate items it contains. If one collection is larger, that alone must not affect your answer. Two collections in which the same proportion of content is climate-related should receive a score of 0 regardless of their relative sizes.

Judge only what the tweets themselves say. Do not infer additional climate context that a linked article might contain. Do not attempt to identify the dates or the real-world events involved; base the comparison on the text.

Read both collections in full before scoring.

SCALE

Provide two independent estimates of the difference in climate-content concentration between the two collections. Do not derive one from the other -- estimate each on its own terms.

1. "relative_score": a continuous score from -10.0 to +10.0, to one decimal place.

0.0	the two collections are equally concentrated
positive	Block B is more concentrated than Block A
negative	Block A is more concentrated than Block B
1-3	slight difference
4-6	clear difference
7-10	large difference; one block is dominated by climate content while the other has little

2. "pct_point_score": your best estimate of the actual difference in percentage points of each collection's own content that is climate-related (Block B's share minus Block A's share), to one decimal place. This is a literal quantity, e.g. if Block A is 5% climate content and Block B is 30%, the answer is +25.0.

Use the full range for both. Do not round to convenient values (0, 5, 10, ...) unless that is genuinely your best estimate, and do not avoid small values near zero to signal confidence -- a small true difference should get a small number, not be rounded away from zero.

OUTPUT

Return JSON only. No preamble, no commentary, no code fences.

```
{
  "relative_score": <float, -10.0 to 10.0, one decimal>,
  "pct_point_score": <float, percentage-point difference, one decimal>,
  "reason": "<one sentence, at most 25 words>"
}
```

User message template (block_a/block_b are the day's sentences, one per line):

BLOCK A
{block_a}

BLOCK B
{block_b}

D.7. Stems Dropped in Ratio Screen

TABLE D.4. STEMS DROPPED IN THE RATIO-FLOOR SWEEP, BY FLOOR AND TIER

Floor	Tier	Stems (ranked)
ratio> 0.4	Top 5 (drop 5)	countri, may, increas, report, level
ratio> 0.4	+6–10 (drop 10)	nation, econom, project, area, relat
ratio> 0.4	+11–25 (drop 25)	effect, rate, result, world, health, technolog, rise, futur, potenti, respons,
ratio> 0.4	+26–50 (drop 50)	differ, model, expect, base, peopl event, social, reduc, centuri, plan, industri, measur, studi, across, data, limit, process, caus, major, averag, end, analysi, assess, popul, direct, option, set, help, requir, research
ratio> 0.5	Top 5 (drop 5)	rise, model, reduc, plan, research
ratio> 0.5	+6–10 (drop 10)	action, evid, understand, discuss, protocol
ratio> 0.5	+11–21 (drop 21)	stabil, shift, predict, remov, target, commit, agre, scientif, transit, negoti, implic
ratio> 0.6	Top 3 (drop 3)	model, reduc, remov
ratio> 0.6	+4–6 (drop 6)	commit, scientif, transit

Notes: This table lists, for each floor and drop level in Table D.2, the actual stems dropped, ranked by $(1 - p_{\text{domain}}) \times$ mass share within the candidates clearing that floor. Each tier is added to all smaller tiers above it within the same floor.

Response to Referee R6

We thank the referee for his/her suggestions. They have been very valuable in improving both the breadth and the quality of our study.

Comments

The paper is a pleasure to read and is very well executed. The empirical analysis is carefully conducted and convincing, and I found the main results to be credible. I also appreciate that the authors include a theoretical model that provides a clear and intuitive explanation for the empirical patterns. Below are a few comments that I hope will help further strengthen the paper.

Strengthening Identification. The paper already makes effective use of high-frequency (daily) data, but I encourage the authors to push the identification further. If feasible, I suggest moving to the intraday level by isolating the specific tweets that drive the Climate Attention Index and examining how currency and stock returns behave in the minutes or hours surrounding those tweets. Recent work using similar approaches, such as Bianchi et al. (2024, RFS) on congressional tweets, Gomez-Cram and Olbert (2023, RFS) on legislative news, and Cookson (JFE, forthcoming) on bank runs shows how valuable this level of granularity can be. You could also use this framework to test for no pre-trends and to show that effects grow with the length of the event window. For instance, sort countries or stocks by climate exposure, form bins, and show return responses for each group over expanding time windows. If intraday analysis is not feasible (perhaps due to tweet volume or timestamp limitations) then I would recommend additional controls for concurrent news at the daily level. This could include macroeconomic announcements, firm-level earnings, or lagged returns on top-CAI days. While I don't expect these adjustments to overturn the results, they would further solidify the paper's identification strategy.

Answer. *We are grateful for these thoughtful suggestions, which we considered carefully. We concluded that the intraday route, powerful as it is in the settings the referee cites, presents unique challenges in our international setting, as we explain below. Si-*

multaneously, we have substantially strengthened our daily-frequency identification along the other dimensions the referee proposes.

Intraday analysis in an international cross-section. The studies cited by the referee examine a single market operating in a single time zone. Our setting involves 25 countries spanning every major time zone, which creates a genuine trade-off between the precision with which a signal is time-stamped and the synchronization of the markets in which it is absorbed. A higher frequency locates the shock more sharply in time; a lower frequency ensures that the signal has been available to both legs of a currency pair and to both national equity markets. This trade-off operates on both sides of our design.

On the measurement side, our global index is a cross-country aggregate, and national newspapers publish on local schedules. Measured intraday, the global index would therefore track the rotation of the Asian, European, and American news cycles: a “global attention shock” recorded at 09:00 GMT would partly reflect which hemisphere is awake rather than variation in climate attention itself. In addition, our index already requires winsorization because country-level growth rates become unstable when a daily tweet count approaches zero. Partitioning the day into intraday intervals would make this sparse-count problem considerably more severe and would render the identification of top climate-attention episodes substantially noisier.

On the asset side, at any given intraday timestamp the majority of the equity markets in our cross-section are closed, and many of the currencies we study trade thinly or only during local hours. An intraday analysis would consequently be feasible only for major currency pairs and a handful of developed equity markets during overlapping trading hours and would discard the emerging and developing economies that generate the cross-sectional variation in climate-news exposure on which our identification rests.

The daily frequency is our compromise. It is coarse enough that both countries in a pair have had the opportunity to incorporate the news, yet fine enough to isolate individual climate episodes, and it remains considerably higher than the monthly and quarterly frequencies prevailing in the climate-finance literature. We acknowledge that a “day” is not perfectly synchronized across, say, Sydney and New York, which is one reason we report cumulative responses over horizons of k days rather than a single-day

reaction.

As an additional feasibility check, we investigated Bloomberg's historical intraday FX coverage directly. Bloomberg's standard intraday data are available only for a relatively short rolling window, approximately the most recent 140 days, and therefore cannot be used for our historical sample. We also explored Bloomberg BFIX, which provides benchmark spot exchange rates at 30-minute intervals and has substantially longer historical coverage. While this would in principle deliver the frequency suggested by the referee, coverage is highly uneven across currencies. For major currencies such as the Japanese yen, the 30-minute history is essentially complete, whereas for a number of emerging-market currencies the series are much more sparse because sufficiently reliable market quotes are not available in every fixing window. Using BFIX would therefore either require us to restrict the analysis to the most liquid currencies or to rely heavily on stale or imputed observations. The former would remove precisely the emerging-market variation that is important for our cross-sectional identification, while the latter would introduce substantial measurement error. For these reasons, we concluded that the apparent gain in time-series precision from moving intraday would come at too high a cost in cross-sectional coverage and data quality.

Controlling for concurrent macroeconomic news. Turning to the referee's second suggestion, we now control for concurrent macroeconomic news at the daily level. We summarize the exercise here and refer the referee to the revised manuscript for the full specifications: Section 4.2 (page 44) for exchange rates, and Section 4.3 (page 52) for equity returns.

We obtain announcement-level data from Bloomberg for each of our 25 countries and consolidate the releases into four broad categories: Prices & Inflation, Labor Market, Monetary, and GDP Growth & Productivity. We collapse the data to one observation per country-date-category triplet to avoid double-counting intraday releases within a category, and we treat euro-area releases as announcement days for every EU country in our sample.

We then run two exercises in parallel for currencies and equities. In the first exercise, we retain the benchmark top climate-attention day indicator and add an announcement-day control. In the second one, we redefine top climate-attention days by excluding those that coincide with macroeconomic announcements, so that the treatment consists

only of “pure” climate days. In both cases the relevant announcement set is defined at the level at which the outcome varies: the country pair for exchange rates, and the country for individual stocks. For currencies, we report results under two definitions of an announcement day: one requiring that both countries in the pair have a release, and a stricter one requiring only that at least one does. For equities, the exclusion exercise additionally uses a global measure. This measure is defined as a day in the top decile of the distribution of the share of sample countries with at least one scheduled release, since the top climate-attention indicator is itself constructed globally.

Our main findings are unchanged across every specification and every definition of an announcement day.

Pre-trends, binned exposures, and lagged returns. Following the referee’s suggestion, we examine return dynamics before top climate-attention days. See Figure C.22 and Figure C.23 on page 59 and 59 of the Appendix, respectively.

For currencies, we extend the event window to start five trading days before each of our top climate-attention days. We normalize the currency level to one at $t = 0$, so that the cumulative return is zero on the event day. We find no appreciation trend before top climate-attention days, indicating that the post-event response is not the continuation of a pre-event trend.

For equities, we also extend the event window to start five trading days before the top climate-attention day. Because we focus on equity prices, or equivalently cumulative returns, we should not expect a zero pre-event trend. In particular, the greenium implies faster average price growth for brown firms relative to green firms. Consistent with this observation, cumulative returns are positive but approximately flat before the event for both low- and high-emission firms. Importantly, the pre-event dynamics are not consistent with a negative pre-event trend.

We also report responses when we adopt coarse cross section. See figure C.7 and Figure C.16 on page 47 and 56 of the Appendix, respectively. These figures report annualized horizon-average returns, so their magnitudes need not increase with the length of the event window.

When studying currencies, we divide our 138 benchmark pairs into beta-difference terciles. Appreciation responses are generally larger in the middle- and high-difference

groups, consistent with our main prediction.

For equities, we use country-beta and lagged-emission median splits. Despite the limited extent of granularity, our main results continue to hold. Equity prices feature a larger negative response in high-beta countries. The differential response across high- and low-emission firms is mitigated. This result is consistent with our novel industry-level analysis which shows that most of variation manifests across industries.

Finally, we add the preceding day's return as a control in the FX and equity regressions. Figure R6.1 shows that the main response patterns are nearly unchanged. Our estimates remain statistically distinguishable from zero at the 10% level at both horizons. We now mention this robustness check in the manuscript.

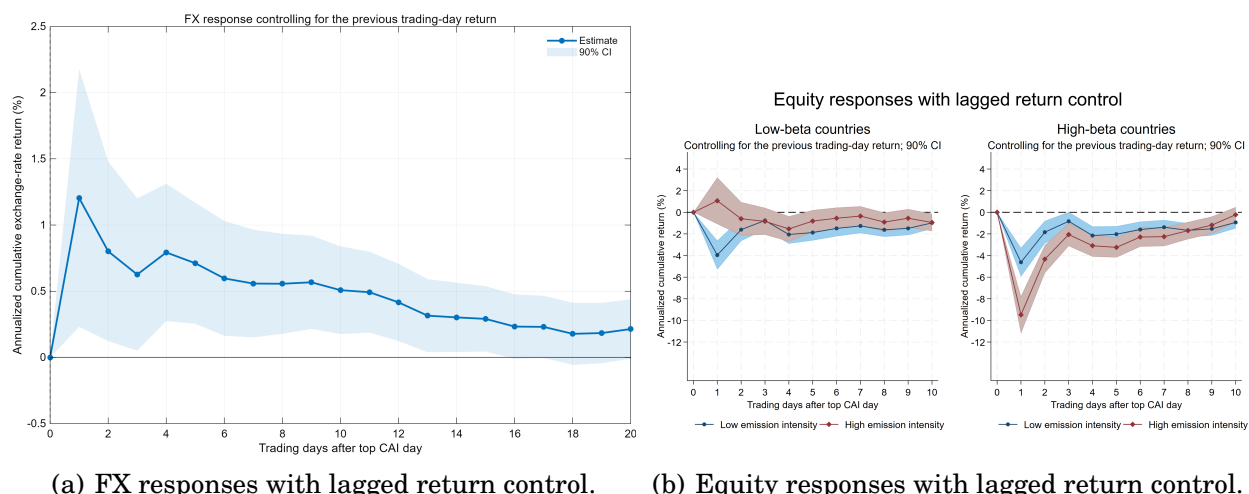


FIG. R6.1. RESPONSES CONTROLLING FOR ONE-DAY LAGGED RETURNS

Channel: Physical vs. Transition Risk. I appreciate the authors' effort to distinguish between physical and transition risk. The evidence presented so far appears most consistent with transition risk, such as regulatory or policy-related developments, as the main channel. To strengthen this interpretation, I wonder if the authors could provide complementary firm-level evidence. For example, do the firms most affected by climate news shocks (especially brown firms) explicitly mention transition-related risks (e.g., regulation, carbon pricing, climate policy) in earnings calls or analyst reports after a spike in the climate news measure? If so, this would support

the view that markets are reacting to expected policy shifts rather than to physical climate events.

Relatedly, it may be helpful to more clearly disentangle news about realized climate events from news about anticipated future risks. One idea could be to examine the language used in the tweets, especially verb tense and headlines, which might help distinguish between backward-looking coverage (e.g., extreme weather events) and forward-looking concerns (e.g., pending legislation, targets, or commitments). This could offer additional insight into which type of news is driving the observed asset price reactions.

Answer. *We thank the referee for these observations, which prompted two new pieces of analysis. We summarize our new results in what follows and refer the referee to the revised manuscript for the full details.*

Firm-level evidence from earnings calls. Following the referee's first suggestion, we document that brown firms explicitly discuss transition-related risks after spikes in climate news. We use the firm-quarter climate-risk exposure measures constructed from earnings calls by [Sautner et al. \(2023b\)](#), which capture firms' discussion of climate change, opportunity, regulatory, and physical-risk topics. We regress these measures on the share of top climate-attention days in a quarter, on one-year-lagged log emission intensity, and on their interaction, estimating by Poisson pseudo-maximum likelihood to accommodate the many zeros in the dependent variable. The interaction is positive and statistically significant for all three outcomes most directly related to transition concerns—climate change risk, regulatory exposure, and regulatory risk—indicating that firms with higher emission intensity discuss climate and regulatory risks more extensively in quarters containing more global climate-news shocks. The full specification and results are reported in Section 4.3 starting on page 50 of the full manuscript.

Backward- versus forward-looking coverage. Following the referee's second suggestion, we classify the temporal orientation of climate-related tweets on top CAI days. Using GPT-4o-mini, we jointly assign each tweet a risk type (physical or transition) and a temporal orientation (past, present, or future), anchoring strictly to the posting date. Physical-risk tweets use 60.2% of the times the past-tense (present-tense counts for 33.0%, and future tense counts for 6.8%), consistent with coverage of realized events. Transition-risk tweets are markedly more present- and forward-oriented (the present- and future-tense are identified 49.8% and 12.0% of the times, respectively).

We further annotate the highest-volume top days for each risk type with the underlying events, and the contrast is stark: the physical-risk peaks correspond to retrospective accounts of damage already materialized, such as the April 2015 Nepal earthquake and Hurricane Irma's landfall in September 2017. In contrast, the transition-risk peaks correspond to regulatory, judicial, and corporate actions unfolding in real time or anticipated in the near term, including the opening of COP21, the U.S. withdrawal from the Paris Agreement, the IPCC Special Report on 1.5°C, and the Supreme Court's West Virginia v. EPA ruling.

Taken together with the equity evidence above, this supports the interpretation that markets are reacting to expected policy shifts rather than to realized physical events. Section 3 (starting on page 35) and Figure 8 of the full manuscript report the classification procedure and the full time series.

Economic magnitudes. While the paper presents statistically strong results, I was left wondering about the economic significance of the estimated effects. It would be helpful to provide more context or headline numbers to help interpret the magnitudes. For instance, it would be useful to benchmark the results against related studies that use social media data to identify news shocks and assess asset price effects.

Answer. *We thank the referee for this suggestion. We now benchmark our estimates against the closest tweet-based asset-pricing evidence. [Bianchi et al. \(2024\)](#) find that firms targeted by highly critical congressional tweets earn abnormal returns of roughly -12% annually; the annualized response of high-emission firms in high- β countries in our setting is about -8% , so our effects are of comparable economic order. This comment is now reported on page 48 of the full manuscript.*

Minor Comments

The current analysis distinguishes between low- and high-emission firms. It would be interesting to go one step further and examine sub-industries or specific sectors that are most affected by climate news shocks. Identifying which industries are driving the green vs. brown stock responses could add nuance to the asset pricing implications.

Answer. *We thank the referee for this suggestion. We now decompose firm-level emission intensity into a cross-industry component, obtained from Fama-French 49*

industry fixed effects, and a within-industry component, obtained as the residual. Re-estimating our benchmark equity specification with each component in turn, we find that the return response to climate news is driven almost entirely by cross-industry variation: it is high-emission industries rather than high-emission firms within a given industry that account for the green-versus-brown pattern we document. The decomposition and the resulting estimates are reported in Section 4 (starting on page 49), with supporting figures in Appendix C.13 and C.14.

Among the Fama–French 49 industries, Coal, Utilities, Agriculture, and Petroleum and Natural Gas have the highest average emission intensity in our sample and therefore constitute the high-emission industry component driving the cross-industry result.

Response to Referee R7

We thank the referee for his/her suggestions. They have been very valuable in improving both the breadth and the quality of our study. In what follows, we report the referee's comments. We decided to address points 1-4 together as they all related.

Report on “International Climate News”. The paper develops [...]

Comments

The four points below are closely connected, so we address them together in a single cumulative answer following point 4.

1. As is common in the literature on long-run risk, the model in the paper relies heavily on complete markets. Normally, I am very sympathetic to complete-markets models because they tend to isolate forces that also operate in more general environments. In this case however, the authors take it to an extreme:
 - (a) They assume the social planner internalizes all externalities from climate damage and adjusts the allocation accordingly (i.e. the collective action problem relating to climate change is solved).
 - (b) They make it sound as if the appreciation and capital flows at the heart of the model only happen because countries insure each other in highly complicated ways.
 - (c) This makes the model a grating introduction to the paper because it sounds like it is missing the point about climate change. After all, we think climate change is a problem because it involves an externality (a above is not true in reality, unfortunately) and because complicated assets that compensate investors for climate change do not exist (b does not hold).
2. I would recommend the authors revisit how much of assumptions a) and b) they need to get the effects in the model to go in the right direction. They have a model with 2 periods and two shocks. It seems that it might be sufficient if each country issues a country-specific bond and equity in its capital stock to “first-order” complete the market (Coerdacier-Rey). Is such first-order completeness not enough to decentralize the social planner's allocation? I understand the

intuition with the pseudo pareto weights from the authors' AER paper but I suspect we do not need crazy Arrow-Debreu securities to get the right allocation in this case. In short, I suspect they can decentralize the economy without esoteric securities.

3. I also suspect we do not need assumption a).
4. If we do indeed need both a and b, then I would recommend focusing the paper more on the measurement and empirical patterns, flipping the order of the sections. In the current draft, most readers interested in climate finance will be disheartened.

Answer. *We thank the referee for this suggestion. In short, the referee's suspicions are correct on both counts: we do not need Arrow-Debreu securities, and we do not need the planner to internalize the externality. We now provide a fully decentralized version of our model described in Sections 2.2–2.4 in the main text. Additional analytical results are provided in Sections A.5–A.8 in the Appendix. Our new results address the points highlighted by the referee. We summarize our results below.*

Market completeness. *We show that markets are complete if the two countries are allowed to trade two country-specific equity indices. Each stock is a claim to aggregate dividends in each country. No state-contingent securities beyond these are required: this is precisely the first-order completeness the referee has in mind.*

Portfolio allocation problem. *At time 0, the two households must decide the share of wealth to be invested in the local and foreign stocks. This choice is made before the realization of the news shocks and hence the portfolio positions are decided to hedge random adverse news shocks.*

At time 1, uncertainty is fully resolved and all stocks pay the same return which, by no arbitrage, equals the risk-free rate (expressed in units of good X). In practice, at time 1, each household decides how much to consume and how much to save and invest in a risk-free asset to support time-2 consumption. The resulting equilibrium portfolio is a levered position in domestic equity financed by a short position in the foreign index. When adverse global climate news arrives and the home country is more exposed to it, the foreign currency depreciates, so the short position delivers a wealth gain exactly in the states in which the home household's marginal utility is high. The hedging motive at the heart of our

mechanism therefore operates through two equity claims, not through the esoteric securities the referee is right to be skeptical of.

Brown firm actions and first-best. *We now posit the existence of a firm that deploys brown capital, the brown firm. In order to align the incentives of this firm to those of a benevolent planner, taxing emissions is all that is required on the brown margin. Each brown firm must be responsible for the damage caused locally and abroad.*

In our revised manuscript, we show how these two tax rates should be fixed in order to replicate the first-best. We think of this analytical exercise as very important, as it sets a new simple benchmark to think of optimal taxation in a I-IAM, e.g., an international integrated assessment model. We emphasize that the instruments this requires are ones that already exist.

We agree with the referee that not all countries tax emissions optimally. Unfortunately, suboptimality is not unique, meaning that there are many ways to think of a decentralized economy that deliver a second-best. For the sake of this manuscript, we decided to focus on a unique first-best allocation. We leave the analysis of suboptimal settings to future work.

Green firm actions and first-best. *We now posit the existence of a firm that deploys green capital, the green firm. In order to align the incentives of this firm to those of a benevolent planner, we need an international compensation scheme for offsets. We believe that our international compensation scheme is consistent with the spirit of Article 6 of the Paris Agreement. In our decentralized economy, we assume that these transfers happen directly among firms, consistent with article 6.4. As a result, these international transfers just depend on marginal contribution to total offsets, as prescribed by a simple neo-classical setting.*

In our simple setting, we could also assume that these transfers are intermediated by governments, consistent with Article 6.2. We decided not to do so for two reasons: (i) our neoclassical approach is very transparent, and (ii) although slowly, the implementation of article 6.4 is moving forward.

On the ordering of the sections. *We gave the ordering careful thought, and we explain our decision to retain the current structure in our reply to the editor. In brief, an earlier version of the paper did present the empirical results first, and in seminar presentations we repeatedly found that audiences asked why we were testing what we were testing without theoretical guidance. Moving the theory first*

noticeably improved how audiences received the material, and we have retained that ordering on the basis of this experience.

5. I like the measurement exercise but believe the authors should practice more verbal hygiene. They use the terms climate news, climate attention, and climate sentiment in confusing ways and often interchangeably. I find this intermingling not very useful.

- (a) If they want to stick with the current econometric specification, then most of their results are about varying attention to climate as a topic (climate attention shocks). Then they should remove all the other language except when they do their decomposition.

Answer. *We agree, and we have gone through the manuscript to homogenize the terminology. We now refer to climate attention, and to shocks to it, throughout the empirical analysis, and reserve sentiment language for the decomposition in which it is the object of study. In the model, we consistently refer to g_t as a climate news shock.*

- (b) I would recommend repeating the empirical exercise with (a) their measure of bad news about climate or (b) climate sentiment (=typically defined as good minus bad news about climate). That would bring the exercise in the model and in the data closer together. The model literally says that you appreciate when you get negative climate news. It is silent about attention to the climate, strictly speaking.

Answer. *We thank the referee for this suggestion, which prompted us to be explicit about what our conditioning event selects in the data.*

We agree on the conceptual point. Our model prices adverse climate news: a positive realization of g_t lowers utility, and Proposition 1 is stated in those terms. It is silent about attention as such.

In our sample, however, the two coincide. As reported in Section 3 and Figure 7, top climate-attention days are overwhelmingly bad-news days. More precisely, across every top CAI day in our sample the share of negative tweets among non-neutral climate-related tweets remains above one half. Equivalently, the event set selected by our attention rule contains no days in which good news is dominant. A selection rule based on adverse climate news

would therefore recover essentially the same dates, which is why we retain climate attention as the conditioning event and interpret it as the empirical counterpart of an adverse news shock.

We also exploit the variation in sentiment that does exist within this range, and report the exercise in Section 4.1 (page 41) and Appendix Figure C.2 (page 44). We interact the top-day indicator with the daily negative share and evaluate the implied response at the maximum, median, and minimum observed values of that share. [Max to Ric: I cannot understand the next sentence. Could you rewrite and avoid the ":"?>] The response is ordered in the negative share: strongest where news is most adverse, at the benchmark level at the median, and weakest at the minimum. [<End here.] Since the minimum still corresponds to a majority-negative day, moving down this range means moving toward less adverse rather than favorable news, and the ordering is the one the model predicts.

- (c) While we are at it, why not also construct a measure of news about climate risk (tweets about climate that also mention a synonym for risk)?

Answer. We investigated this suggestion following Hassan et al. (2019), flagging a climate term that co-occurs with a risk or uncertainty synonym within ten words. Risk vocabulary, however, is a marginal linguistic feature of our corpus: 97% of climate-mentioning tweets contain no risk term at all, and the risk and uncertainty stems present in our vocabulary account for less than 1% of total index mass, so the resulting measure is close to zero for almost every day. We therefore did not pursue this measure further, as it carries too little variation to support the empirical analysis.

- (d) It would be interesting to check if the indices of climate attention / risk are more associated with volatility and the negative sentiment more with the level of returns.

Answer. We thank the referee for this suggestion. We now examine whether top climate-attention days are associated with elevated second moments of exchange rates. We re-estimate our benchmark FX specification with event-window realized volatility on the left-hand side, constructed from squared daily bilateral returns and annualized over each horizon k . The point estimates are weakly positive throughout the event window, suggesting that

top CAI days are followed by somewhat higher FX realized volatility. We report this result as suggestive evidence rather than as a test of the model, which is silent about second moments. The analysis appears in Section 4.1 on page 42, with the corresponding figure C.3 in Appendix.

6. The authors should simplify their data construction. In particular, it seems they are using two different methods for figuring out which tweets are about climate. First they use a tf-idf similarity with a training library, selecting the top 15% of similar days. Then, on page 26, they also embed each tweet and cluster it into topics, in order to then measure the distance between each tweet (topic) and an artificially (ChatGPT) generated training library for climate text. That makes no sense to me at all.
 - (a) Why cluster anything into topics? That step seems redundant in any case. We can simply measure the distance between each tweet and the climate sentences in embedding space. Why make that more complicated?
 - (b) Can't we just use the original tf-idf to decide which tweets are about climate before we further classify them into Climate transition and Climate physical risk? See the topic-based measures Hassan et al. (2019, QJE) for one way of doing this.
 - (c) Alternatively, why not use the embedding to do everything?
 - (d) Clearly, I am confused about the procedure.
 - (e) Similarly, the 15% and 0.1% cutoffs seem to be fighting with each other.
 - (f) Try to make this as simple as possible.
 - (g) I suspect you also don't need the stemming.

Answer. *We thank the referee for pushing us to clarify our methodology.*

- **Sub-questions (a), (c), (d), (f).** *We lay out the logic of our approach below. In the paper we build two different main exercises: a ranking of days by climate attention (Stage 1), and a classification of individual tweets on those days (Stage 2). Each requires a different set of tools.*

- **Stage 1: Identifying which days matter.** *The first task is to rank calendar days by climate attention, comparably across 25 countries and*

8 languages over nearly a decade. At this stage the unit of analysis is not a single tweet but an entire country's tweets for a given day, aggregated into one document that, even for our smallest country, averages several thousand words. For a document this size, tf-idf cosine similarity against a reference vocabulary works well: term frequencies are stable at this volume, translation across languages amounts to matching vocabulary rather than meaning, and the index this produces is a continuous time series we can track and compare day to day and year to year using one consistent method throughout the sample. This step accomplishes one precise task: it tells us, in the aggregate, how much a country's news refers to climate in a specific day.

In Stage 1, we do not use embeddings because sentence embedding models have a fixed input-length limit. For example, the multilingual model (Reimers and Gurevych (2019)) we use in Stage 2, described below, truncates at roughly 100 words and hence is not suitable to examine a country-day document with several thousand words. Our tf-idf approach has no such limit, since term frequency counting scales to a document of any length by construction. Importantly, frequency counting is comparable across countries and over time, precisely what we need to construct our aggregate news index over multiple years.

A related alternative would be to embed each tweet individually (which would avoid the length problem) and aggregate the resulting embeddings into a day-level score (e.g., a volume-weighted average similarity to a climate reference embedding). This does not run into the input-length constraint above, but it runs into a different one specific to how we use each stage's output. Sentence embedding similarity scores are known to sit in a narrow, elevated range regardless of topic relevance; this compression would directly undermine a measure meant to track magnitude and be compared in level year over year.

- **Stage 2: Classifying individual tweets on the days that matter.** Once the top-attention days are identified, we want to disentangle the actual news being discussed. There are trade-offs by using any methodology. In what follows, we clarify two reasons that convinced us to use sentence embeddings for this task.

First, tf-idf is not well suited to this finer-grained task. In contrast to a country-day document with thousands of words, individual tweets need special attention. For example, two tweets describing the same event can use almost entirely different vocabulary depending on language, author, and framing, and if they do not feature the reference vocabulary's specific terms, exact matching fails completely. Sentence embeddings capture meaning rather than vocabulary, so two tweets about the same event are recognized as similar even when they share few or no words, which is exactly the property needed when there is only one text document and no volume to average over.

Second, we group tweets into topics before classifying them, rather than classifying each tweet individually as the referee suggests. A single tweet's embedding is a noisy estimate of what a short, informal text is actually about. Grouping many similar tweets into one topic and classifying the resulting centroid gives a far more stable signal than classifying each tweet in isolation. As a result, our identification of physical vs transition risk-related news is more accurate. Most importantly, clustering is also what makes the classification auditable, i.e, it reduces millions of individual tweets to a few hundred topics per country, a scale at which specific topic assignments can be reviewed.

We now explicitly state in the main text that the index and the selection of top-attention days rest entirely on the tf-idf procedure and that embeddings are used only to classify tweets on days already selected.

- **Sub-question (b): Using Hassan et al. methodology.** We thank the referee for the suggestion to follow [Hassan et al. \(2019\)](#). Their measure identifies a topic-specific bigram set as the set difference between bigrams occurring in a topic corpus and a non-topic corpus. They count co-occurrences between that bigram set and a risk or uncertainty synonym within a document. This approach is known to work well for long, structured text such as earnings call transcripts. In what follows, we argue that this methodology is not ideal in our setting.

Specifically, we implemented their construction directly: we derive the climate-topic bigram set from the literal set difference between bigrams in a climate corpus (white papers) and a non-climate corpus (Reuters newswire

sentences). We then use their full 118-term risk synonym list. Restricted to our English-language sample, a climate bigram and a risk synonym co-occur in 2.00% of all 9.12 million tweets, and when they do co-occur they are already within ten words of each other 95.4% of the time. A co-occurrence rule therefore has very little to operate on in our setting. Thus, we have an additional reason to rely on embeddings, as they classify a tweet’s overall content directly rather than requiring a specific term pair to appear within it.

- **Sub-question (e): The two cutoffs.** The 15% and the 0.1% thresholds are applied at different stages, to different objects, and never interact. The 15% is a cutoff on the distribution of daily innovations in the global CAI: it selects the dates that enter our event study, and it makes no statement about the content of any individual tweet. The 0.1% is a cutoff on the distribution of topic-level cosine similarities against the reference sentences: taking dates as given, it determines which topics are labeled climate-related. One selects dates and the other labels topics, so neither cutoff constrains the other.

The two thresholds are also calibrated to opposite error costs, which is why one is inclusive and the other stringent. In Stage 1, including a day that is not in fact climate-driven attenuates our estimates toward zero, so a looser cutoff works against us; we make this point in section 3, and we show in section 4.2 that moving the threshold to 10% or 20% leaves our results unchanged (figures C.5(a)–C.5(b)). In Stage 2, a mislabeled topic enters the physical versus transition split directly, with no aggregation to dilute it, so there we deliberately impose a stringent cutoff.

- **Sub-question (g): Stemming.** Stemming enters only in Stage 1, and it is applied identically to the reference corpus and to the tweets, so it is a matched transformation of both sides of the cosine similarity rather than a filter that selects content. Its role is larger here than it would be in an English-only application: we run the same pipeline in eight languages, several of which are morphologically richer than English. Without stemming, a single concept fragments across many inflected surface forms, and term counts thin out. Stage 2 does not use stemming at all.

Prompted by the Editor, we also document directly the contribution of the stemmed vocabulary. In Appendix D.1, we rank all CCV stems by their con-

tribution to index mass and audit the 813 stems that account for 90% of the full mass. For each stem, we report the share of occurrences used in an environmental sense.

7. I would really like to see more examples and a labelling of Figures 1 and 5 based on the tweets on that day. How many of the peaks are verifiably about climate?

Answer. *We thank the referee for this suggestion, which we have implemented. We now label the **top-10** largest spikes in Figures 1 and 5 with the underlying news events and report the tweets we collected for each. They all correspond to verifiable climate events. The labelled figure and the accompanying table of events appear in Appendix Figure **B.2** and Table **B.3**.*

8. I suspect the results in section 5 are more fragile to including controls. There should be some more careful discussion of how robust this result is.

Answer. *We thank the referee for this remark, and we agree that the quarterly net-export analysis is more exposed to common macroeconomic conditions than our high-frequency exchange-rate and equity results. We have therefore expanded the set of controls. We re-estimate the expanding-window GMM specification under two additional settings. First, we add year-quarter fixed effects to absorb global shocks common to all country pairs in a given quarter, such as trade-cycle fluctuations, pandemic disruptions, or geopolitical events. Second, we augment the baseline model with quarterly real GDP growth and CPI inflation. Both exercises retain our baseline controls for industrial production growth and non-climate Twitter activity, and we report results separately for the equally weighted, GDP-weighted, and Twitter-volume-weighted global CAI. The estimates remain broadly negative across all weighting schemes and both robustness specifications. The analysis appears in Section 5 on page 55, with the corresponding figure in Appendix Figure **C.25**.*

Response to the Editor

We thank the Editor for her suggestions. They have been very valuable in improving both the breadth and the quality of our study. Our replies reported below.

[...]

In light of this feedback and my own reading of your manuscript, I am happy to offer you an opportunity to revise and resubmit your paper. The revision must address the following issues. Please also address the remaining issues raised by the referees and explain how you did so in the replies to the referees.

1. You solve the model assuming that the social planner internalizes all externalities from climate damage. This assumption appears to be too strong in the context of the climate finance literature, especially as it involves flawless international cooperation on climate change policies. I appreciate the added complexity, but it would be desirable to contrast the planner's solution that you report with that prevailing in a decentralized equilibrium. If needed, please feel free to consider a special case of the problem. The objective of this exercise is to verify that your key implications of the model continue to hold in a decentralized equilibrium. R7's report contains more details and a suggestion for a possible decentralization to consider.

Answer. *We thank the Editor for this suggestion, and we agree. The revised manuscript replaces the planner's solution with a fully decentralized economy in which the two countries trade two country-specific equity claims. Furthermore, the climate externality is internalized through instruments that can be broadly mapped into objects that already exist: a local emissions trading system, a border adjustment for damage caused abroad, and an international compensation scheme for offsets along the lines of Article 6 of the Paris Agreement. We derive the tax rates under which this decentralized allocation coincides with the first best. As such, every prediction we take to the data is preserved: countries more sensitive to global climate news still experience a currency appreciation and a decline in net exports upon adverse news, and the brown-minus-green return spread is still negative and larger in absolute value in high-sensitivity countries. Sections 2.2–2.4 of the manuscript and Sections A.5–A.8 of the Appendix contain the full derivation, and our combined reply to R7's points 1 through 4 discusses the decentralization in detail.*

2. Given the reactions of the referees, I think you should flip the order of the sections and present your empirical results upfront.

Answer. *We gave this comment careful thought. An earlier version of the paper did present the empirical results first, and in seminar presentations we repeatedly found that audiences asked why we were testing what we were testing without theoretical guidance. Moving the theory first noticeably improved how audiences received the material. Since the decentralized economy that we now present in the revised theoretical framework following R7's suggestions preserves the key testable predictions, we have retained the original ordering of theoretical and empirical analysis.*

3. Address R7's comments about measurement and data construction.

Answer. *We have addressed all of the data and measurement comments raised by the referee. Some of these exercises have improved our analysis, while others turned out not to apply to the specific corpus we study. We report the details in our answers to the referee's points 5 through 8.*

4. R6 recommends that you strengthen your identification by moving to the intraday level – i.e., by isolating the specific tweets that drive the Climate Attention Index and examining how currency and stock returns behave in the minutes or hours surrounding those tweets. In their report, the referee explains the advantages of exploiting this granularity and how you could improve your identification.

Answer. *We considered this route carefully and concluded that it does not transfer cleanly to an international cross-section. The papers that adopt this approach typically examine a single market in a single time zone, whereas our 25 countries span every major time zone. This creates a trade-off between time-stamping a shock precisely and ensuring that both legs of a currency pair have had the opportunity to absorb it. Measured intraday, our global index would partly track the rotation of the Asian, European, and American news cycles rather than climate attention itself, and the daily tweet counts would become considerably noisier once the day is partitioned further. We also investigated the data availability: Bloomberg's standard intraday history covers only a rolling window of roughly 140 days. The alternative is BFIX, which provides 30-minute quotes over a longer sample. However, coverage is mostly complete for major currencies but much sparser for many emerging markets. This means that an intraday design would either restrict us to the most liquid markets or rely on stale and imputed*

quotes, in both cases discarding precisely the emerging-market variation on which our cross-sectional identification rests.

Following the request of R6, we have strengthened our daily-frequency identification along the other dimensions the referee proposes, most importantly by controlling for concurrent macroeconomic announcements. Our findings are unchanged and the full discussion appears in our reply to R6's first comment.

5. R6 is not entirely convinced as to whether your evidence reflects the physical or transition risks and requests that you dig deeper in answering this question. They suggest that you provide complementary firm-level evidence from earnings calls or analyst reports.

Answer. *We thank the Editor for this point. We have added two new pieces of evidence. First, we use the firm-quarter climate-exposure measures built from earnings calls by [Sautner et al. \(2023b\)](#). This is the only firm-level source with consistent coverage across the 25 countries in our sample. Specifically, we regress firms' discussion of climate topics on the share of top climate-attention days in a quarter, on one-year-lagged log emission intensity, and on their interaction. The interaction is positive and significant for the three transition-oriented outcomes, climate change risk, regulatory exposure, and regulatory risk. Thus we can conclude that higher-emission firms discuss regulatory concerns more in quarters containing more global climate-news shocks. Second, we classify the temporal orientation of climate-related tweets on top CAI days and find that physical-risk coverage is predominantly retrospective (60.2% past tense) while transition-risk coverage is markedly more present- and forward-looking (38.2% past, 49.8% present, 12.0% future), a contrast that is equally clear when we annotate the largest peaks of each type: physical peaks correspond to damage already realized, transition peaks to regulatory, judicial, and corporate actions unfolding or anticipated. Taken together, the two exercises support the interpretation that markets are responding to expected policy shifts rather than to realized physical events. The full analysis appears in our reply to R6's third comment, with the underlying results in Sections [3](#) and [4.3](#) of the manuscript.*

6. As prompted by R6, please discuss the economic magnitudes of the results in the paper and help the reader by benchmarking them against related studies.

Answer. *We now benchmark our estimates against the closest tweet-based asset-*

pricing evidence of [Bianchi et al. \(2024\)](#). They find abnormal returns of roughly -12% annually for firms targeted by highly critical congressional tweets, against about -8% annually for high-emission firms in high- β countries in our setting, so the two are of comparable economic order. The discussion appears on page 48 of the manuscript.

7. Finally, I encourage you to discuss the limitations of your Climate Attention Index and to comment on the accuracy of your tweet classification. In particular, it would be helpful to know the rate of false positives generated by your textual analysis (e.g., distinguishing when “green” refers to climate issues rather than color), as well as any other potential sources of misclassification. You should also include additional validation and robustness checks for your current measure of the Climate Attention Index. For example, consider whether advances in large language models—such as multilingual systems like ChatGPT—could help reduce false positives and improve semantic accuracy. Could incorporating an LLM-assisted labeling or validation step enhance classification accuracy? These additional analyses can be placed in the appendix.

Answer. *We thank the Editor for this suggestion. We have added two extended exercises, both of which use GPT to interrogate our measure directly. We reported a condensed version of our findings in the main body of the text and added [Appendix D](#) with the details.*

Term ambiguity and false positives. *We begin by noting that our index does not classify individual tweets. We pool all of a country’s tweets on a given day into a single document and compute an IDF-weighted cosine similarity against the Climate Change Vocabulary. As a result, an ambiguous occurrence does not promote an entire tweet to climate status and it contributes only partially to our index. Specifically, what matters is its share of the day’s lexical mass to one coordinate of a high-dimensional match. The kind of ambiguity that the Editor describes is therefore diluted rather than amplified by aggregation. It is nonetheless a legitimate concern, and we test it directly.*

We ranked all 156,897 unigrams and bigrams in the English CCV by their contribution to the index and audited the 813 stems accounting for 90% of index mass. For each, GPT-4o returned two probabilities: p_{domain} , the share of occurrences used in a climate, weather, energy, or environmental sense at all, so that $1 - p_{\text{domain}}$ is the false-positive rate; and p_{climate} , the share referring specifically to climate change. [Table D.1](#)

(page 67 of the Appendix) shows how the two probabilities sort the vocabulary. The audit passed a decoy check in which known climate and known non-climate terms were interleaved with the real stems.

The stems that can contaminate our index are those that carry a real climate meaning but are often used for something else. Two conditions define them: (1) they appear in an environmental context only rarely, and (2) yet when they do, the meaning is specifically climate-related. The screen isolates stems such as “target”, “protocol”, “transit”, and “commit”. Each has a canonical climate use, in emissions “target”, Kyoto “Protocol”, energy “transition”, and net zero “commitment”, alongside far more frequent ordinary uses, in inflation “target”, security “protocol”, presidential “transition”, and “committed” to the merger. These are the stems the screen flags as candidates for removal. Figure D.1 (page 69 of the Appendix) places them in the vocabulary, between unambiguous climate terms such as “emiss”, “greenhous”, and “ipcc”, and generic vocabulary such as year and “countri”, which carries index mass but no climate sense at all.

Removing the screened set leaves the index essentially unchanged. Table D.2 (page 70 of the Appendix) reports the results. At our preferred screen the index correlates with the original at 1.000 in levels and 0.998 in innovations, and 96.3% of top climate-attention days survive; removing the entire pool rather than its highest-mass members produces the identical result. Even at the loosest screen, which admits substantially more generic vocabulary, overlap remains between 81.8% and 88.0%, and the days that change classification sit at the 80th to 85th percentile of the purged index, that is, immediately below the 85th percentile cut. The disagreement is boundary churn at the threshold rather than any reordering of the series.

The Editor’s example is instructive: green ranks 974th of 156,897 terms by index mass and falls outside the terms accounting for 90% of the index, so its contribution to the measure is negligible.

Would an LLM improve on the index? We built a controlled simulation in which each synthetic day’s true climate share is known by construction, allowing both our index and an LLM classifier to be benchmarked against the truth rather than against each other. We compare four LLM-based alternatives across seven scenarios spanning strong to near-flat climate signal. Table D.3 (page 74 of the Appendix) reports every

result relative to our index.

The per-sentence GPT classification scores every sentence individually and aggregates the scores into a daily share. It outperforms our index by between 0.01 and 0.09 in rank correlation with the truth. That edge is indistinguishable from zero once the underlying signal weakens. On directional accuracy (i.e., whether a method correctly identifies if climate content rose or fell), the LLM and our procedure are statistically indistinguishable at every one of the seven scenarios.

We also tested the natural way to make LLM classification affordable at scale, presenting the model with an entire day's text and asking it to judge climate concentration directly. This performs far worse than our index under every scenario, including the one with the strongest signal, where directional accuracy is close to chance. Our diagnostics indicate that the model registers the presence of climate content rather than estimating its share.

The improvement available from LLM classification is therefore either negligible or statistically undetectable, and the design that would make it cheap does not work.